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Sheet paper processing device

Abstract

There is provided a base machine including a base machine feed path that conveys a paper sheet, and a validator identifying the paper sheet conveyed by the base machine feed path; an expansion unit including an extension feed path extending from the base machine feed path, a branching feed path branching from the extension feed path, and a plurality of stacking portions that are connected to the branching feed path and stack the paper sheet identified by the validator; and a control portion that, when a conveyance abnormality occur, continues conveyance operation of at least one of the extension feed path and the branching feed path without immediately stopping the conveyance operation and conveys the paper sheet to a predetermined stacking portion that is any of the plurality of stacking portions, in accordance with a location where the conveyance abnormality occurred and a type of the conveyance abnormality.

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Background/Summary

(1) This application is the U.S. national phase of International Application No. PCT/JP2020/048843, filed Dec. 25, 2020, which designated the U.S., the entire contents of which is hereby incorporated by reference.

TECHNICAL FIELD

(2) The present invention relates to a paper sheet processing device.

BACKGROUND ART

(3) Some paper sheet processing devices count and classify a paper sheet according to type, and accumulate them in a plurality of accumulation units (see, for example, Patent Document 1).

PRIOR ART DOCUMENTS

Patent Documents

(4) Patent Document 1: Japanese Unexamined Patent Application, First Publication No. 2017-178509

SUMMARY OF THE INVENTION

Problems to be Solved by the Invention

(5) In such a paper sheet processing device, for example, when a paper sheet conveyance abnormality such as a jam occurs on the feed path, the conveyance operation of all the feed paths is temporarily stopped at that point. In this state, the operator removes all the paper sheets that have become stuck on the feed path and all the undetermined paper sheets in the plurality of accumulation units, and sets them back in the loading unit of the paper sheet processing device. Therefore, there is a problem that the amount of paper sheets to be removed is large and the removal work becomes complicated.

(6) An object of the present invention is to provide a paper sheet processing device that facilitates paper sheet removal work when a conveyance abnormality occurs.

Means for Solving the Problem

(7) According to the first aspect of the present invention there is provided a base machine including a base machine feed path that conveys a paper sheet, and a validator that identifies the paper sheet conveyed by the base machine feed path. There is provided an expansion unit including an extension feed path that extends from the base machine feed path, a branching feed path that branches from the extension feed path, and a plurality of stacking portions that are connected to the branching feed path and stack the paper sheet identified by the validator. There is provided a control portion that, when a conveyance abnormality occur, continues conveyance operation of at least one of the extension feed path and the branching feed path without immediately stopping the conveyance operation and conveys the paper sheet to a predetermined stacking portion that is any of the plurality of stacking portions, in accordance with a location where the conveyance

abnormality occurred and a type of the conveyance abnormality.

(8) According to the second aspect of the present invention, in the first aspect, the control portion, when the conveyance abnormality occurs, may switch a conveyance destination so as to convey the paper sheet being conveyed on the extension feed path or the paper sheet being conveyed on the branching feed path to the empty stacking portion where no paper sheet is stacked.

(9) According to the third aspect of the present invention, in the second aspect, the control portion, when the conveyance abnormality occurs, may switch the conveyance destination so as to convey the paper sheet being conveyed on the extension feed path or the paper sheet being conveyed on the branching feed path to the stacking portion with a lowest number of stacked sheets among the stacking portions, when there is no empty stacking portion where no paper sheet is stacked.

(10) According to a fourth aspect of the present invention, in the first aspect, the predetermined stacking portion may be set in advance. In this case, the control portion, when the conveyance abnormality occurs, may switch a conveyance destination so as to convey the paper sheet being conveyed on the extension feed path or the paper sheet being conveyed on the branching feed path to the predetermined stacking portion.

(11) According to the fifth aspect of the present invention, in the first aspect, the expansion unit may include a plurality of expansion units. In this case, the control portion, in accordance with the location where the conveyance abnormality occurred and the type of the conveyance abnormality, may individually switch a conveyance destination of a paper sheet being conveyed downstream of the location where the conveyance abnormality occurred and a paper sheet being conveyed on an upstream side of the location where the conveyance abnormality occurred so that each paper sheet is conveyed to the predetermined stacking portion or to another stacking portion.

(12) According to the sixth aspect of the present invention, any one of the first to fifth aspects may further include a terminal stacking portion that receives a paper sheet from an opening portion at the terminal end of the extension feed path and stacks it. In this case, the control portion, when the location where the conveyance abnormality occurred is not the extension feed path, may switch a conveyance destination so as to convey the paper sheet being conveyed on the extension feed path to the terminal stacking portion.

(13) According to the seventh aspect of the present invention, in any one of the first to sixth aspects, the control portion may switch a conveyance speed of at least any one of the extension feed path and the branching feed path to a lower speed than a conveyance speed during normal counting except when a conveyance abnormality occurs, in accordance with the location where the conveyance abnormality occurred and the type of the conveyance abnormality.

(14) According to the eighth aspect of the present invention, in any one of the first to seventh aspects, the control portion, after having completed conveyance of the paper sheet being conveyed on the extension feed path or the paper sheet being conveyed on the branching feed path to the predetermined stacking portion, may finalize counting of a paper sheet normally conveyed to the stacking portion other than the predetermined stacking portion.

(15) According to the ninth aspect of the present invention, in any one of the first to eighth aspects, the control portion, after having completed conveying paper sheet being conveyed on the extension feed path or paper sheet being conveyed on the branching feed path to the predetermined stacking portion, may finalize the counting of the paper sheet normally conveyed to the stacking portion other than the predetermined stacking portion.

Effect of Invention

(16) The present invention can facilitate paper sheet removal work.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 A schematic block diagram showing a configuration example of a paper sheet processing device according to an embodiment of the present invention.
- (2) FIG. 2 A front view schematically showing a base machine that constitutes the paper sheet processing device according to the embodiment of the present invention.
- (3) FIG. 3 A front view schematically showing a first expansion unit capable of constituting the paper sheet processing device according to the embodiment of the present invention.
- (4) FIG. 4 A front view schematically showing a second expansion unit capable of constituting the paper sheet processing device according to the embodiment of the present invention.
- (5) FIG. 5 A front view schematically showing a third expansion unit capable of constituting the paper sheet processing device according to the embodiment of the present invention.
- (6) FIG. 6 A front view schematically showing a fourth expansion unit capable of constituting the paper sheet processing device according to the embodiment of the present invention.
- (7) FIG. 7 A front view schematically showing a terminal pocket capable of constituting the paper sheet processing device according to the embodiment of the present invention.
- (8) FIG. 8 A front view showing the contents displayed on the operation display unit at the time of a first abnormality in the second counting mode of the paper sheet processing device according to the embodiment of the present invention.
- (9) FIG. 9 A front view showing the contents displayed on the operation display unit at the time of a first abnormality in the second counting mode of the paper sheet processing device according to the embodiment of the present invention.
- (10) FIG. 10 A front view showing the contents displayed on the operation display unit at the time of a first abnormality in the second counting mode of the paper sheet processing device according to the embodiment of the present invention.
- (11) FIG. 11 A front view showing the contents displayed on the operation display unit at the time of a first abnormality in the second counting mode of the paper sheet processing device according to the embodiment of the present invention.

EMBODIMENTS FOR CARRYING OUT THE INVENTION

- (12) A paper sheet processing device according to an embodiment of the present invention will be described below with reference to the drawings.
- (13) As shown in FIG. 1, a paper sheet processing device 1 according to the embodiment includes a combination of a base machine 2, a first expansion unit 3 (expansion unit) to a fourth expansion unit 6 (expansion unit) of a plurality of types, and a terminal pocket 7 (stacking portion, terminal stacking portion). Here, the paper sheet processing device 1 is manufactured by selectively setting at least either of the number and type of the first expansion unit 3 to the fourth expansion unit 6 of a plurality of types for combining with the base machine 2 and the terminal pocket 7. The paper sheet processing device 1 is, for example, a banknote processing device that processes banknotes among various types of paper sheets. The paper sheet processing device 1 is constituted by one being appropriately selected by user operation or the like from the first expansion unit 3 to the fourth expansion unit 6 being connected to the base machine 2 serving as a main body, and the terminal pocket 7 being connected to the position most distant from the base machine 2. Of course, it is also possible to have a configuration in which the terminal pocket 7 itself is not connected regardless of the position farthest from the base machine 2 or the like.
- (14) <Base Machine 2>
- (15) As shown in FIG. 2, the base machine 2 is a paper sheet sorting machine that sorts the paper sheet S, and performs a predetermined classification process on the paper sheet S loaded from outside the machine by the operator. For example, a classification process of identifying the paper sheet S loaded from outside the machine by an operator, counting the paper sheet S by denomination, and classifying the paper sheet S by denomination, or a classification process that

identifies and counts a plurality of batch processing groups of a paper sheet S by each batch processing group, and stacks the paper sheet S separately by batch processing group may be performed.

(16) The base machine **2** includes a loading portion **15** provided on the right side surface **11** of the main body as viewed from the operator side and that continuously opens to the right side **11** and the front face of the machine body on the operator side, and a rejection portion **16** that is provided above the loading portion **15** on the right side surface **11** and that continuously opens to the right side surface **11** and the front face of the main body.

(17) Paper sheets S are loaded into the loading portion **15** from the outside by the operator. At this time, the paper sheet S is vertically stacked and placed on the bottom surface **17** of the loading portion **15** in a state of the long sides of the paper sheet S being aligned along the front-rear direction of the main body and the short sides along the left-right direction of the main body. The loading portion **15** moves the stacked paper sheet S loaded in this way one by one from the bottom so that the short sides thereof are along the conveyance direction, and feeding to the left of the main body. The paper sheet S moves throughout the entire paper sheet processing device **1** shown in FIG. **1** with the long sides thereof being aligned with the front-back direction of the main body.

(18) The base machine **2** has a base machine feed path **20** for receiving and conveying the paper sheet S that has been loaded into the loading unit **15** and fed out from the loading unit **15**. The base machine feed path **20** has an identification conveyance portion **21** that is connected to the input portion **15** and that receives and conveys the paper sheet S fed out from the loading portion **15**. The base machine **2** has a validator **22** that identifies the paper sheet S conveyed by the identification conveyance portion **21** in the identification conveyance portion **21**. The identification conveyance portion **21** horizontally extends from the loading portion **15** toward the left side surface **24** opposite to the right side surface **11** of the main body, and then extends upward. The validator **22** is provided in this horizontally extending portion of the identification conveyance portion **21**.

(19) The base machine feed path **20** has an outward-facing conveyance portion **28** that extends from the upper end of the identification conveyance portion **21** toward the left side surface **24** and opens to the left side surface **24**. The base machine **2** also has a reject conveyance portion **27** that extends from the upper end of the identification conveyance portion **21** toward the right side surface **11** and is connected to the rejection portion **16**. At the upper end position of the identification conveyance portion **21**, a sorting portion **30** for sorting the paper sheet S conveyed by the identification and conveyance portion **21** to the outward-facing conveyance portion **28** and the rejection conveyance portion **27** is provided. In other words, the sorting portion **30** switches the conveyance destination of the paper sheet S from the identification conveyance portion **21** to the outward-facing conveyance portion **28** and the reject conveyance portion **27**. The outward-facing conveyance portion **28** and the rejection conveyance portion **27** receive and convey the paper sheet S from the identification conveyance portion **21**. The outward-facing conveyance portion **28** receives from the identification conveyance portion **21** the paper sheet S that has been successfully identified as an object to be counted by the validator **22**, and conveys it to the outside of the base machine **2**. The rejection conveyance portion **27** receives the paper sheet S that could not be successfully identified as a counting target by the validator **22** from the identification conveyance portion **21** and conveys it to the rejection portion **16**.

(20) The base machine **2** has a plurality of, specifically, three tributary conveyance portions **31** to **33** that branch off from the intermediate portion of the rejection conveyance portion **27**. The base machine **2** has a sorting portion **34** at the position of the tributary conveyance portion **31** that sorts the paper sheet S of the rejection conveyance portion **27** to the tributary conveyance portion **31** and more to the rejection portion **16** side than the tributary conveyance portion **31** of the rejection conveyance portion **27**. The base machine **2** has a sorting portion **35** at the position of the tributary conveyance portion **32** that sorts the paper sheet S of the rejection conveyance portion **27** to the tributary conveyance portion **32** and more to the rejection portion **16** side than the tributary

conveyance portion **32** of the rejection conveyance portion **27**. The base machine **2** has a sorting portion **36** at the position of the tributary conveyance portion **33** that sorts the paper sheet **S** of the rejection conveyance portion **27** to the tributary conveyance portion **33** and more to the rejection portion **16** side than the tributary conveyance portion **33** of the rejection conveyance portion **27**, that is, to the rejection portion **16**.

(21) The paper sheet **S** is fed out from the end of the rejection conveyance portion **27**, and rejection portion **16** stacks the paper sheet **S** thus fed out so as to be removable to the outside. In other words, the rejection portion **16** ejects the paper sheet **S** conveyed by the rejection conveyance portion **27** to the outside of the machine. Specifically, the rejection portion **16** stacks the paper sheet **S** other than the paper sheet **S** successfully identified as the counting target by the validator **22**. Therefore, the rejection conveyance portion **27** receives and conveys the paper sheet **S** other than the paper sheet **S** successfully identified as the counting target by the validator **22** from the identification conveyance portion **21**. Depending on the counting mode, the rejection conveyance portion **27** also receives and conveys the paper sheet **S** that has been successfully identified as a counting target by the validator **22** from the identification conveyance portion **21**.

(22) The rejection portion **16** has a bottom surface **37**, a wall surface **38**, and an impeller **39** that causes the paper sheet **S** fed out from the rejection conveyance portion **27** to abut the wall surface **38** and fall toward the bottom surface **37**. Therefore, the rejection portion **16** stacks the fed out paper sheet **S** vertically to be placed on the bottom surface **37**.

(23) Each of the tributary conveyance portions **31** to **33** branches downward from the rejection conveyance portion **27**, and in conjunction therewith, a plurality of, specifically, three identically shaped stacking portions **41** to **43** are provided on the underside thereof, being arranged side by side. The stacking portions **41** to **43** open to the front surface of the main body of the base machine **2**. The leftmost stacking portion **41** among the stacking portions **41** to **43** is connected to the leftmost branch conveyance portion **31** among the branch conveyance portions **31** to **33**, the middle stacking portion **42** to the middle branch conveyance portion **32**, and the rightmost stacking portion **43** to the rightmost tributary conveyance portion **33**. The stacking portions **41** to **43** are provided above validator **22**.

(24) The stacking portion **41** stacks the paper sheet **S** fed out from the tributary conveyance unit **31** so that they can be taken out from the front surface of the main body of the base machine **2**. The stacking portion **41** includes an elevatable bottom plate **45** that slopes slightly downward to the right with respect to the horizontal, a wall surface **46** that extends upward on the right side of the bottom plate **45**, and an impeller **47** that is arranged at the upper portion of the wall surface **46** and that causes the paper sheet **S** fed out from the tributary conveyance portion **31** to abut the wall surface **46** and fall toward the bottom plate **45**. Therefore, the paper sheet **S** fed out to the stacking portion **41** is stacked vertically on the bottom plate **45**. The bottom plate **45** moves up and down according to the stacking amount of the paper sheet **S** in the stacking portion **41**.

(25) The stacking portions **42** and **43** also have the same structure as the stacking portion **41**, with the stacking portion **42** stacking the paper sheet **S** fed out from the branch conveyance portion **32** so as to be removable to the outside, and the stacking portion **43** stacking the paper sheet **S** fed out from the branch conveyance portion **33** so as to be removable to the outside. The stacking portions **41** to **43** selectively stack the paper sheet **S** that has been successfully identified as objects to be counted by the validator **22**.

(26) Independently provided in the base machine **2** are a loading portion drive portion (not shown) for driving the loading portion **15** and a base machine feed path drive portion (not shown) for driving the base machine transport path **20**, whereby the loading portion **15** and the base machine feed path **20** are individually and independently operable.

(27) The base machine **2** has a touch panel type operation display portion **55** on the front surface of the main body, for inputting operations by the operator and for displaying information to the operator. Further, the base machine **2** is provided with paper sheet detection sensors (not shown) at

each portion of the base machine feed path **20** for detecting the conveyance state of the paper sheet **S** while detecting and counting the paper sheet **S**. These paper sheet detection sensors detect, as the conveyance state of the paper sheet **S**, whether the paper sheet **S** is conveyed normally without oblique feeding, near-feeding, or multi-feeding, or whether there is abnormal conveyance in which there is oblique feeding, near-feeding, or multi-feeding, and detect a jam state of the paper sheet **S**. That is, the paper sheet detection sensor detects the presence or absence of conveyance abnormalities including abnormal conveyance and occurrence of jams.

(28) The paper sheet detection sensor is, for example, a light-transmissive sensor whose detection unit has a light-emitting unit and a light-receiving unit that allow the paper sheet **S** being conveyed on the base machine feed path **20** to pass therebetween. The paper sheet detection sensor detects that there is a paper sheet **S** at the position of the paper sheet detection sensor when the amount of light received by the light-receiving unit drops below a first predetermined value while the light-emitting unit is emitting light. At this time, the paper sheet detection sensor determines that the paper sheet **S** is being multi-fed if the amount of light transmitted through the paper sheet **S** is equal to or less than a second predetermined value that is smaller than the first predetermined value, and detects that the paper sheet **S** is not being multi-fed if greater than the second predetermined value.

(29) The paper sheet detection sensor has a left and right pair of detection units when viewed in the conveyance direction of the paper sheet **S**, and when the deviation in the detection timing of one paper sheet **S** by these detection units is equal to or greater than to a predetermined time, the paper sheet detection sensor detects that the paper sheet **S** is skewed, and if less than the predetermined time, detects that the paper sheet **S** is not skewed.

(30) When the detection interval between the previously detected paper sheet **S** and the next detected paper sheet **S** is equal to or shorter than a predetermined time, the paper sheet detection sensor detects a near feed for these paper sheets **S**, and when the predetermined time is exceeded, detects no near feed for these paper sheets **S**.

(31) Further, when the paper sheet detection sensor continuously detects the paper sheet **S** for a predetermined time or longer, it detects that the paper sheet **S** is stuck at the position of the paper sheet detection sensor, that is, that a jam has occurred. Further, if the paper sheet detection sensor does not detect the paper sheet **S** even after a predetermined time has elapsed from the timing at which the paper sheet **S** should be detected, the paper sheet detection sensor detects that the paper sheet **S** has become stuck upstream of the position of the paper sheet detection sensor, that is, that a jam has occurred.

(32) A predetermined one of these paper sheet detection sensors is also used as a timing sensor for switching the sorting portions **30**, **34** to **36**.

(33) The base machine **2** is also provided with a master control portion **56** (control portion) that internally controls the loading portion **15**, the rejection portion **16**, the identification conveyance portion **21**, the validator **22**, the rejection conveyance portion **27**, the outward-facing conveyance portion **28**, the sorting portion **30**, the tributary conveyance portions **31** to **33**, the sorting portions **34** to **36**, and the stacking portions **41** to **43**, such as sorting the paper sheet **S** to the rejection portion **16** and the stacking portions **41** to **43** based on the identification result in the validator **22**. The master control portion **56** controls the entire base machine **2** and also, by being communicatively connected via an interface to, among the first expansion unit **3** to the fourth expansion unit **6** shown in FIG. **1**, those units that are interconnected to the base machine **2** to constitute the same paper sheet processing device **1**, outputs control command signals. As shown in FIG. **2**, the base machine **2** is provided with a power supply portion **57** that is connected to an external power supply and supplies power to each part of the base machine **2**.

(34) The first expansion unit **3** to the fourth expansion unit **6** and the terminal pocket **7** shown in FIG. **1** can be adjacently connected in a selective manner to the left side surface **24** of the base machine **2**.

(35) <First Expansion Unit 3>

(36) As shown in FIG. 3, the first expansion unit 3 has an extension feed path 64 that opens to the upper part of the right side surface 61 of the main body, extends horizontally and linearly toward the left side surface 62 opposite to the right side surface 61, and opens to the upper part of the left side surface 62, and a branching feed path 65 that branches off from an intermediate position on the left side surface 62 side of the extension feed path 64 and extends downward. As shown in FIG. 1, when the first expansion unit 3 is connected adjacent to the base machine 2 with the right side surface 61 thereof overlaid on the left side surface 24 of the base machine 2, the extension feed path 64 receives and conveys the paper sheet S from the outward-facing conveyance portion 28 of the base machine 2. In other words, the extension feed path 64 extends from the base machine feed path 20, and the branching feed path 65 branches off from the extension feed path 64.

(37) As shown in FIG. 3, the first expansion unit 3 has a sorting portion 66 that sorts the paper sheet S conveyed from the right side surface 61 to the left side surface 62 along the extension feed path 64 to the branching feed path 65 and more to the left side surface 62 side of the extension feed path 64 than the branching feed path 65.

(38) The branching feed path 65 has a main conveyance portion 70 that receives the paper sheet S from an intermediate position of the extension feed path 64 and conveys it downward, and three tributary conveyance portions 71 to 73 branching off from the main conveyance portion 70. The tributary conveyance portions 71 to 73 are arranged side by side in the vertical direction of the main body, and branch off from the main conveyance portion 70 toward the right side surface 61.

(39) The first expansion unit 3 has, at the position of the tributary conveyance portion 71, a sorting portion 74 that sorts the paper sheet S of the main conveyance portion 70 to the tributary conveyance portion 71 and more to the tributary conveyance portion 72 side of the main conveyance portion 70 than the tributary conveyance portion 71. The first expansion unit 3 has, at the position of the tributary conveyance portion 72, a sorting portion 75 that sorts the paper sheet S of the main conveyance portion 70 to the tributary conveyance portion 72 and more to the tributary conveyance portion 73 side of the main conveyance portion 70 than the tributary conveyance portion 72. The first expansion unit 3 has, at the position of the tributary conveyance portion 73, a sorting portion 76 that sorts the paper sheet S of the main conveyance portion 70 to the tributary conveyance portion 73 and more to the side opposite the extension feed path 64 than the tributary conveyance portion 73 of the main conveyance portion 70.

(40) The first expansion unit 3 is provided with paper sheet detection sensors that detect and count the paper sheet S and detect the conveyance state thereof in each part of the extension feed path 64 and the branching feed path 65. These paper sheet detection sensors are the same as the paper sheet detection sensors of the base machine 2, and detect the presence or absence of abnormal conveyance of the paper sheet S, the occurrence of jamming, and the like. A predetermined one of these paper sheet detection sensors is also used as a timing sensor for switching the sorting portions 66, 74 to 76.

(41) The first expansion unit 3 is provided with an extension feed path driving portion (not shown) for driving the extension feed path 64 and a branching feed path driving portion (not shown) for driving the branching feed path 65 independently. In other words, the extension feed path 64 and the branching feed path 65 can operate independently of each other.

(42) The first expansion unit 3 has a plurality of, specifically, four stacking portions 81 to 84 of the same shape on the right side of the tributary conveyance portions 71 to 73 and the end of the main conveyance portion 70. The stacking portions 81 to 84 are arranged vertically. The stacking portions 81 to 84 each perform a stacking process of receiving and stacking the paper sheet S fed out from the branching feed path 65. Each of the stacking portions 81 to 84 opens to the front surface of the main body of the first expansion unit 3.

(43) The uppermost stacking portion 81 of the stacking portions 81 to 84 is connected to the uppermost tributary conveyance portion 71 of the tributary conveyance portions 71 to 73, the

stacking portion **82** under the stacking portion **81** is connected to the tributary conveyance portion **72** under the tributary conveyance portion **71**, the stacking portion **83** under the stacking portion **82** is connected to the tributary conveyance portion **73** under the tributary conveyance portion **72**, and the lowest stacking portion **84** is connected to the end of the main conveyance portion **70**.

(44) When the paper sheet processing device **1** has the configuration shown in FIG. **1**, the stacking portions **81** to **84** each receive and stack paper sheets **S** that, after being identified by the validator **22** while being conveyed on the base machine feed path **20**, are conveyed by the extension feed path **64** and the branching feed path **65** and fed out from the branching feed path **65**. In other words, the first expansion unit **3** has the plurality of stacking portions **81** to **84** that are connected to the branching feed path **65** and stack the paper sheet **S** after being identified by the validator **22**.

(45) As shown in FIG. **3**, the stacking portion **81** stacks the paper sheet **S** fed out from the tributary conveyance portion **71** so as to be removable to the outside from the front surface of the main body of the first expansion unit **3**. The stacking portion **81** includes a bottom surface **88** inclined slightly downward to the right with respect to the horizontal, a wall plate **89** positioned perpendicularly to the top of the bottom surface **88** and movable to the left and right, and an impeller **90** that is arranged at the left portion of the bottom surface **88** and that causes the paper sheet **S** fed out from the tributary conveyance portion **71** to abut the bottom surface **88** and pushes the paper sheet **S** toward the wall plate **89**. Therefore, the paper sheets **S** fed out to the stacking portion **81** are stacked in the left-right direction, and are taken out to the front from the front surface opening of the main body of the first expansion unit **3**. The wall plate **89** moves left and right according to the stacking amount of the paper sheet **S** in the stacking portion **81**.

(46) The stacking portions **82** to **84** have the same structure as the stacking portion **81**, with the stacking portion **82** stacking the paper sheet **S** fed out from the branch conveyance portion **72** so as to be removable to the outside, the stacking portion **83** stacking the paper sheet **S** fed out from the branch conveyance portion **73** so as to be removable to the outside, and the stacking portion **84** stacking the paper sheet **S** fed out from the end of the main conveyance portion **70** so as to be removable to the outside.

(47) A slave control portion **95** (control unit) is provided inside the first expansion unit **3** as shown in FIG. **3**, and by being communicably connected to the base machine **2** shown in FIG. **2** to receive a control command signal from the master control unit **56** thereof, controls the first expansion unit **3** on the basis of this control command signal. Further, the first expansion unit **3** is provided with a power supply portion **96** that is connected to an external power source and supplies power to each portion of the first expansion unit **3**.

(48) The base machine **2** and the first expansion unit **3** to the fourth expansion unit **6** including a separate first expansion unit **3** can be adjacently connected in a selective manner to the right side surface **61** side of the first expansion unit **3**. Another first expansion unit **3** to the fourth expansion unit **6** including a separate first expansion unit **3** and the terminal pocket **7** can be adjacently connected in a selective manner to the left side surface **62** side of the first expansion unit **3**, which is the side opposite the base machine **2**.

(49) <Second Expansion Unit **4**>

(50) As shown in FIG. **4**, the second expansion unit **4** has an extension feed path **104** that opens to the upper part of the right side surface **101** of the main body, extends horizontally and linearly toward the left side surface **102** opposite to the right side surface **101**, and opens to the upper part of the left side surface **102**, and a branching feed path **105** that branches off from an intermediate position on the left side surface **102** side of the extension feed path **104** and extends downward. As shown in FIG. **1**, when the second expansion unit **4** is connected adjacent to the first expansion unit **3** so that the right side surface **101** thereof overlaps the left side surface **62** of the first expansion unit **3**, for example as shown in FIG. **1**, the extension feed path **104** receives and conveys the paper sheet **S** from the extension feed path **64** of the first expansion unit **3**. In other words, the extension feed path **104** is extended from the base machine feed path **20** via the extension feed path **64**, and

the branching feed path **105** branches off from the extension feed path **104**.

(51) As shown in FIG. **4**, the second expansion unit **4** has a sorting portion **106** that sorts the paper sheet **S** conveyed from the right side surface **101** to the left side surface **102** along the extension feed path **104** to the branching feed path **105** and more to the left side surface **102** side of the extension feed path **104** than the branching feed path **105**.

(52) The branching feed path **105** has a main conveyance portion **110** that receives the paper sheet **S** from an intermediate position of the extension feed path **104** and conveys it downward, and a tributary conveyance portion **111** branching off from the main conveyance portion **110**. The tributary conveyance portion **111** branches off from the main conveyance portion **110** toward the right side surface **101**. The second expansion unit **4** has, at the position of the tributary conveyance portion **111**, a sorting portion **114** that sorts the paper sheet **S** of the main conveyance portion **110** to the tributary conveyance portion **111** and to more to the side opposite the extension feed path **104** than the tributary conveyance portion **111** of the main conveyance portion **110**.

(53) The second expansion unit **4** is provided with paper sheet detection sensors that detect and count the paper sheet **S** and detect the conveyance state thereof in each part of the extension feed path **104** and the branching feed path **105**. These paper sheet detection sensors are the same as the paper sheet detection sensors of the base machine **2**, and detect the presence or absence of abnormal conveyance of the paper sheet **S**, the occurrence of jamming, and the like. A predetermined one of these paper sheet detection sensors is also used as a timing sensor for switching the sorting portions **106**, **114**.

(54) The second expansion unit **4** is provided with an extension feed path driving portion (not shown) for driving the extension feed path **104** and a branching feed path driving portion (not shown) for driving the branching feed path **105** independently. In other words, the extension feed path **104** and the branching feed path **105** can operate independently of each other.

(55) The second expansion unit **4** has a plurality of, specifically, two stacking portions **121**, **122** on the right side of the tributary conveyance portion **111** and the end of the main conveyance portion **110**. The stacking portions **121**, **122** are arranged vertically. The stacking portions **121**, **122** each perform a stacking process of receiving and stacking the paper sheet **S** fed out from the branching feed path **105**, and open to the front surface of the main body of the second expansion unit **4**. The upper stacking portion **121** of the stacking portions **121** and **122** is connected to the branch conveyance portion **111**, while the lower stacking portion **122** is connected to the end of the main conveyance portion **110**.

(56) When the paper sheet processing device **1** has the configuration shown in FIG. **1**, the stacking portions **121**, **122** each receive and stack paper sheets **S** that, after being identified by the validator **22** while being conveyed on the base machine feed path **20**, are conveyed by the extension feed paths **64**, **104** and the branching feed path **105** and fed out from the branching feed path **105**. In other words, the second expansion unit **4** has the plurality of stacking portions **121**, **122** that are connected to the branching feed path **105** and stack the paper sheet **S** after being identified by the validator **22**.

(57) As shown in FIG. **4**, the stacking portion **121** stacks the paper sheet **S** fed out from the tributary conveyance portion **111** so as to be removable to the outside from the front surface of the main body of the second expansion unit **4**. The stacking portion **121** has a bottom plate **126** that is horizontally arranged and which can be raised and lowered, a wall surface **127** which is arranged on the left side of the bottom plate **126** and perpendicular thereto, and an impeller **128** that is arranged at the upper portion of the wall surface **127** and that causes the paper sheet **S** fed out from the tributary conveyance portion **111** to abut the wall surface and fall to the bottom plate **126**. Therefore, the paper sheets **S** fed out to the stacking portion **121** are stacked vertically on the bottom plate **126**, and are taken out to the front from the front surface opening of the second expansion unit **4**. The bottom plate **126** moves up and down according to the stacking amount of the paper sheet **S** in the stacking portion **121**.

(58) The stacking portion **122** has the same structure as the stacking portion **121**, with the stacking portion **122** stacking the paper sheet S fed out from the end of the main conveyance portion **110** so as to be removable to the outside.

(59) A slave control portion **135** (control unit) is provided inside the second expansion unit **4** as shown in FIG. **4**, and by being communicably connected to the base machine **2** shown in FIG. **1** to receive a control command signal from the master control portion **56** thereof, controls the second expansion unit **4** on the basis of this control command signal. Further, the second expansion unit **4** is provided with a power supply portion **136** that is connected to an external power source and supplies power to each portion of the second expansion unit **4**. The respective stackable amounts of paper sheet S of the stacking portions **121** and **122** are larger than the stackable amounts of paper sheet S of the stacking units **81** to **84** of the first expansion unit **3**.

(60) The base machine **2** and the first expansion unit **3** to the fourth expansion unit **6** including a separate second expansion unit **4** can be adjacently connected in a selective manner to the right side surface **101** side of the second expansion unit **4**. The first expansion unit **3** to the fourth expansion unit **6** including a separate second expansion unit **4** and the terminal pocket **7** can be adjacently connected in a selective manner to the left side surface **102** side of the second expansion unit **4**, which is the side opposite the base machine **2**.

(61) <Third Expansion Unit 5>

(62) As shown in FIG. **5**, the third expansion unit **5** has an extension feed path **144** that opens to the upper part of the right side surface **141** of the main body, extends horizontally and linearly toward the left side surface **142** opposite to the right side surface **141**, and opens to the upper part of the left side surface **142**, and a branching feed path **145** that branches off from an intermediate position on the left side surface **142** side of the extension feed path **144** and extends downward. As shown in FIG. **1**, when the third expansion unit **5** is connected adjacent to the second expansion unit **4** so that the right side surface **141** thereof overlaps the left side surface **102** of the second expansion unit **4**, the extension feed path **144** receives and conveys the paper sheet S from the extension feed path **104** of the second expansion unit **4**. In other words, the extension feed path **144** is extended from the base machine feed path **20** via the extension feed paths **64**, **104**, and the branching feed path **145** branches off from the extension feed path **144**.

(63) As shown in FIG. **5**, the third expansion unit **5** has a sorting portion **146** that sorts the paper sheet S conveyed from the right side surface **141** to the left side surface **142** along the extension feed path **144** to the branching feed path **145** and more to the left side surface **142** side of the extension feed path **144** than the branching feed path **145**.

(64) The branching feed path **145** has a main conveyance portion **150** that receives the paper sheet S from an intermediate position of the extension feed path **144** and conveys it downward, and tributary conveyance portions **151**, **152** branching off from the main conveyance portion **150**. The tributary conveyance portions **151**, **152** branch off from the main conveyance portion **150** toward the right side surface **141**. The third expansion unit **5** has, at the position of the tributary conveyance portion **151**, a sorting portion **154** that sorts the paper sheet S of the main conveyance portion **150** to the tributary conveyance portion **151** and more to the tributary conveyance portion **152** side of the main conveyance portion **150** than the tributary conveyance portion **151**. The third expansion unit **5** has, at the position of the tributary conveyance portion **152**, a sorting portion **155** that sorts the paper sheet S of the main conveyance portion **150** to the tributary conveyance portion **152** and more to the side opposite the extension feed path **144** than the tributary conveyance portion **152** of the main conveyance portion **150**.

(65) The third expansion unit **5** is provided with paper sheet detection sensors that detect and count the paper sheet S and detect the conveyance state thereof in each part of the extension feed path **144** and the branching feed path **145**. These paper sheet detection sensors are the same as the paper sheet detection sensors of the base machine **2**, and detect the presence or absence of abnormal conveyance of the paper sheet S, the occurrence of jamming, and the like. A predetermined one of

these paper sheet detection sensors is also used as a timing sensor for switching the sorting portions **146, 154, 155**.

(66) The third expansion unit **5** is provided with an extension feed path driving portion (not shown) for driving the extension feed path **144** and a branching feed path driving portion (not shown) for driving the branching feed path **145** independently. In other words, the extension feed path **144** and the branching feed path **145** can operate independently of each other.

(67) The third expansion unit **5** has a plurality of, specifically, three stacking portions **161** to **163** of the same shape on the right side of the tributary conveyance portions **151, 152** and the end of the main conveyance portion **150**. The stacking portions **161** to **163** are arranged vertically. The stacking portions **161** to **163** each perform a stacking process of receiving and stacking the paper sheet **S** fed out from the branching feed path **145**. Each of the stacking portions **161** to **163** opens to the front surface of the main body of the third expansion unit **5**.

(68) The uppermost stacking portion **161** of the stacking portions **161** to **163** is connected to the upper tributary conveyance portion **151** of the tributary conveying portions **151, 152**, the middle stacking portion **162** is connected to the lower tributary conveyance portion **152**, and the lowermost stacking portion **163** is connected to the end of the main conveyance portion **150**.

(69) When the paper sheet processing device **1** has the configuration shown in FIG. **1**, the stacking portions **161** to **163** each receive and stack the paper sheet **S** that, after being identified by the validator **22** while being conveyed on the base machine feed path **20**, are conveyed by the extension feed paths **64, 104, 144** and the branching feed path **145** and fed out from the branching feed path **145**. In other words, the third expansion unit **5** has the plurality of stacking portions **161** to **163** that are connected to the branching feed path **145** and stack the paper sheet **S** after being identified by the validator **22**.

(70) As shown in FIG. **5**, the stacking portion **161** stacks the paper sheet **S** fed out from the tributary conveyance portion **151** so as to be removable to the outside from the front surface of the main body of the third expansion unit **5**. The stacking portion **161** includes a bottom surface **168** inclined slightly downward to the right with respect to the horizontal, a wall plate **169** positioned perpendicularly to the upper side of the bottom surface **168** and movable to the left and right, and an impeller **170** that is arranged at the left portion of the bottom surface **168**, causes the paper sheet **S** fed out from the tributary conveyance portion **151** to abut the bottom surface **168** and pushes the paper sheet **S** toward the wall plate **169**. Therefore, the paper sheets **S** fed out to the stacking portion **161** are stacked in the left-right direction, and are taken out to the front from the front surface opening of the main body of the third expansion unit **5**. The wall plate **169** moves left and right according to the stacking amount of the paper sheet **S** in the stacking portion **161**.

(71) The stacking portions **162, 163** also have the same structure as the stacking portion **161**, with the stacking portion **162** stacking the paper sheet **S** fed out from the branch conveyance portion **152** so as to be removable to the outside, and the stacking portion **163** stacking the paper sheet **S** fed out from the end of the main conveyance portion **150** so as to be removable to the outside.

(72) A slave control portion **175** (control unit) is provided inside the third expansion unit **5** as shown in FIG. **5**, and by being communicably connected to the base machine **2** shown in FIG. **1** to receive a control command signal from the master control portion **56** thereof, controls the third expansion unit **5** on the basis of this control command signal. Further, the third expansion unit **5** is provided with a power supply portion **176** that is connected to an external power source and supplies power to each portion of the third expansion unit **5**. The respective stackable amounts of paper sheets **S** of the stacking portions **161** to **163** are larger than the stackable amounts of paper sheets **S** of the stacking portions **81** to **84** of the first expansion unit **3**, and smaller than the stackable amounts of paper sheets **S** of the stacking portions **121, 122** of the second expansion unit **4**.

(73) The base machine **2** and the first expansion unit **3** to the fourth expansion unit **6** including a separate third expansion unit **5** can be adjacently connected in a selective manner to the right side

surface **141** side of the third expansion unit **5**. Another first expansion unit **3** to the fourth expansion unit **6** including a separate third expansion unit **5** and the terminal pocket **7** can be adjacently connected in a selective manner to the left side surface **142** side of the third expansion unit **5**, which is the side opposite the base machine **2**.

(74) <Fourth Expansion Unit **6**>

(75) As shown in FIG. **6**, the fourth expansion unit **6** has an extension feed path **184** that opens to the upper part of the right side surface **181** of the main body, extends horizontally and linearly toward the left side surface **182** opposite to the right side surface **181**, and opens to the upper part of the left side surface **182**, and a branching feed path **185** that extends downward from an intermediate position on the left side surface **182** side of the extension feed path **184**. As shown in FIG. **1**, when the fourth expansion unit **6** is connected adjacent to the third expansion unit **5** so that the right side surface **181** thereof overlaps for example the left side surface **142** of the third expansion unit **5**, the extension feed path **184** receives and conveys the paper sheet **S** from the extension feed path **144** of the third expansion unit **5**. In other words, the extension feed path **184** is extended from the base machine feed path **20** via the extension feed paths **64**, **104**, **144**, and the branching feed path **185** branches off from the extension feed path **184**.

(76) As shown in FIG. **6**, the fourth expansion unit **6** has a sorting portion **186** that sorts the paper sheet **S** conveyed from the right side surface **181** to the left side surface **182** along the extension feed path **184** to the branching feed path **185** and more to the left side surface **182** side of the extension feed path **184** than the branching feed path **185**.

(77) The side of the branching feed path **185** opposite the extension feed path **184** is connected to a bundling portion **189** that stacks and binds a predetermined number (**100**) of paper sheets **S**.

(78) The branching feed path **185** has a main conveyance portion **190** that receives the paper sheet **S** from an intermediate position of the extension feed path **184** and conveys it downward, and a tributary conveyance portion **191** branching off from the main conveyance portion **190**. The tributary conveyance portion **191** branches off from the main conveyance portion **190** toward the right side surface **181**. The fourth expansion unit **6** has, at the position of the tributary conveyance portion **191**, a sorting portion **194** that sorts the paper sheet **S** of the main conveyance portion **190** to the tributary conveyance portion **191** and to more to the side opposite the extension feed path **184** than the tributary conveyance portion **191** of the main conveyance portion **190**.

(79) The fourth expansion unit **6** is provided with paper sheet detection sensors that detect and count the paper sheet **S** and detect the conveyance state thereof in each part of the extension feed path **184** and the branching feed path **185**. These paper sheet detection sensors are the same as the paper sheet detection sensors of the base machine **2**, and detect the presence or absence of abnormal conveyance of the paper sheet **S**, the occurrence of jamming, and the like. A predetermined one of these paper sheet detection sensors is also used as a timing sensor for switching the sorting portions **186**, **194**.

(80) The fourth expansion unit **6** is provided with an extension feed path driving portion (not shown) for driving the extension feed path **184** and a branching feed path driving portion (not shown) for driving the branching feed path **185** independently. In other words, the extension feed path **184** and the branching feed path **185** can operate independently of each other.

(81) The bundling portion **189** has a plurality of, specifically, two bonding stackers **201**, **202** (stacking portions) on the right side of the tributary conveyance portion **191** and the end of the main conveyance portion **190**. The bonding stackers **201**, **202** are arranged vertically. The bonding stackers **201**, **202** each perform an alignment and stacking process of receiving the paper sheet **S** fed out from the branching feed path **185** and aligning and stacking a predetermined binding unit sheet number (for example, 100 sheets), horizontally oriented, in the vertical direction. The upper bonding stacker **201** of the bonding stackers **201** and **202** is connected to the tributary conveyance portion **191**, while the lower bonding stacker **202** is connected to the end of the main conveyance portion **190**.

(82) When the paper sheet processing device **1** has the configuration shown in FIG. **1**, the bonding stackers **201**, **202** each receive and stack paper sheets **S** that, after being identified by the validator **22** while being conveyed on the base machine feed path **20**, are conveyed by the extension feed paths **64**, **104**, **144**, **184** and the branching feed path **185** and fed out from the branching feed path **185**. In other words, the fourth expansion unit **6** has the plurality of bonding stackers **201**, **202** that are connected to the branching feed path **185** and stack the paper sheet **S** after being identified by the validator **22**.

(83) As shown in FIG. **6**, the bundling portion **189** has a descending conveyance portion **204** for loading the stacked paper sheet **S** fed rightward from the bonding stackers **201** and **202** in a horizontal state and conveying the paper sheet **S** vertically downward. The bundling portion **189** also has, below the bonding stacker **202**, a sealing portion **205** that collects a predetermined number of loose paper sheets **S** into a bundling unit to manufacture a small bundle of paper sheets integrated with bundling tape by bundling the stacked paper sheets **S** conveyed down by the descending conveyance portion **204** and sent out from the descending conveyance portion **204** in a horizontal state to the left side of the machine.

(84) The sealing portion **205** creates the small bundles of paper sheets one bundle at a time, having gripping members **207**, **208** that sandwich from top and bottom the stacked paper sheets **S** set leftward in the main body from the descending conveyance portion **204** and bend them into a V shape, and creates the small bundle of paper sheets by wrapping bundling tape around the stacked paper sheets **S** held by the gripping members **207**, **208**, gluing and cutting the bundling tape, and then pulling the gripping members **207**, **208** away from the bundling tape.

(85) Further, the bundling portion **189** has a drawer-type storage chamber **211** below the sealing portion **205** for receiving the small bundles of paper sheets produced by the sealing portion **205**. The storage chamber **211** can be pulled out from the front surface of the fourth expansion unit **6**. That is, the storage chamber **211** accepts small bundles of paper sheets from the sealing portion **205** in a state of being pushed into the main body, and by afterward being pulled out from the front surface, the small bundles of paper sheets can be removed from the machine. A door **216** is provided on the front of the main body of the fourth expansion unit **6** at the location of the bonding stackers **201**, **202** to allow fractional paper sheets of the bonding stackers **201**, **202** to be removed from the machine.

(86) A slave control portion **225** (control unit) is provided inside the fourth expansion unit **6** as shown in FIG. **6**, and by being communicably connected to the base machine **2** shown in FIG. **1** to receive a control command signal from the master control portion **56** thereof, controls the fourth expansion unit **6** on the basis of this control command signal. Further, the fourth expansion unit **6** is provided with a power supply portion **226** that is connected to an external power source and supplies power to each portion of the fourth expansion unit **6**.

(87) The base machine **4** and the first expansion unit **3** to the fourth expansion unit **6** including a separate fourth expansion unit **6** can be adjacently connected in a selective manner to the right side surface **181** side of the fourth expansion unit **6**. Another first expansion unit **3** to the fourth expansion unit **6** including a separate fourth expansion unit **6** and the terminal pocket **7** can be adjacently connected in a selective manner to the left side surface **182** side of the fourth expansion unit **6**, which is the side opposite the base machine **2**.

(88) <Terminal Pocket 7>

(89) As shown in FIG. **7**, the terminal pocket **7** has an opening portion **230** at the top of right side surface **231**. The terminal pocket **7** is a storage pocket that receives the paper sheet **S** fed out from the opening portion **230** and stacks the paper sheet **S** in the vertical direction on the bottom surface **235** in a horizontal orientation. The terminal pocket **7** has an opening/closing door **232** on the front surface for allowing the paper sheet **S** housed in the terminal pocket **7** to be taken out to the outside.

(90) The base machine **2** and the first expansion unit **3** to the fourth extension unit **6** can be

connected in a selective manner to the right side surface **231** side of the terminal pocket **7**. When the base machine **2** is adjacently connected to the right side surface **231** side of the terminal pocket **7**, the opening portion **230** of the terminal pocket **7** receives and stores the paper sheet **S** conveyed by the outward-facing conveyance portion **28** of the base machine **2** shown in FIG. **2** at the conveyance terminal position in the paper sheet processing device **1**.

(91) When any one of the first expansion unit **3** to the fourth expansion unit **6** is adjacently connected to the right side surface **231** of the terminal pocket **7**, the paper sheet **S** conveyed on the extension feed path of the expansion unit adjacent to the terminal pocket **7** among the extension feed paths **64**, **104**, **144**, **185** shown in FIGS. **3** to **6** is received and stored through the opening portion **230** at the conveyance terminal position of the paper sheet processing device **1**.

(92) <Overall Configuration of Paper Sheet Processing Device **1**>

(93) The paper sheet processing device **1** is manufactured by preparing the required number of the required types from the multiple types of the first expansion unit **3** to fourth expansion unit **6**, sequentially connecting them to the base machine **2** in a set order, and connecting the terminal pocket **7** at the position farthest from the base machine **2**. That is, various combinations of the paper sheet processing device **1** are possible, such as making the paper sheet processing device **1** by connecting only one of the first expansion unit **3** to the fourth expansion unit **6** selected and set adjacent to the base machine **2** and connecting the terminal pocket **7** thereto, making the paper sheet processing device **1** by connecting a plurality of units of different types selected and set among the first expansion unit **3** to the fourth expansion unit **6** to the base machine **2** and connecting the terminal pocket **7** thereto, making the paper sheet processing device **1** by connecting a plurality of units of the same type among the first expansion unit **3** to the fourth expansion unit **6** to the base machine **2** and further connecting one or more units of different types, and connecting the terminal pocket **7** thereto. That is, one paper sheet processing device **1** is manufactured by combining one base machine **2** with any number of the first expansion unit **3** to the fourth expansion unit **6**, together with the terminal pocket **7**. Of course, in any combination, when a plurality of units out of the first expansion unit **3** to the fourth expansion unit **6** are connected to the base machine **2**, the arrangement order of the expansion units can also be selected and set. The paper sheet processing device **1** can be configured without the terminal pocket **7**.

(94) When the paper sheet processing device **1** includes at least one of the expansion units **3** through **6**, in any combination, the master control portion **56** of the base machine **2** is connected in a communicable manner to the slave control portions among the connected slave control portions **95**, **135**, **175**, **225** of the expansion units **3** through **6** via communication cables (not shown), and outputs control command signals to the slave control portions of the connected units. When the paper sheet processing device **1** includes at least one of the first expansion unit **3** to the fourth expansion unit **6**, in any combination, the power supply portion of the connected one of the power supply portions **96**, **136**, **176**, **226** of the expansion units **3** through **6** is turned on and off in conjunction with the power supply portion **57** of the base machine **2**.

(95) For example, as shown in FIG. **1**, one first expansion unit **3** is connected to the base machine **2** by a coupling mechanism (not shown), with the right side surface **61** of the first expansion unit **3** making contact with the left side surface **24** of the base machine **2**, and the outward-facing conveyance portion **28** and the extension feed path **64** arranged so that the paper sheet **S** can be delivered. One second expansion unit **4** is connected to this first expansion unit **3** by a coupling mechanism (not shown), with the right side surface **101** of the second expansion unit **4** making contact with the left side surface **62** of the first expansion unit **3**, and the extension feed path **64** and the extension feed path **104** arranged so that the paper sheet **S** can be delivered. One third expansion unit **5** is connected to this second expansion unit **4** by a coupling mechanism (not

shown), with the right side surface **141** of the third expansion unit **5** making contact with the left side surface **102** of the second expansion unit **4**, and the extension feed path **104** and the extension feed path **144** arranged so that the paper sheet **S** can be delivered. One fourth expansion unit **6** is connected to this third expansion unit **5** by a coupling mechanism (not shown), with the right side surface **181** of the fourth expansion unit **6** making contact with the left side surface **142** of the third expansion unit **5**, and the extension feed path **144** and the extension feed path **184** arranged so that the paper sheet **S** can be delivered. One terminal pocket **7** is connected to this fourth expansion unit **6** by a coupling mechanism (not shown), with the right side surface **231** of the terminal pocket **7** making contact with the left side surface **182** of the fourth expansion unit **6**, so that a paper sheet **S** can be passed from the extension feed path **184** to the terminal pocket **7**. In this manner, a single paper sheet processing device **1** is manufactured. In this case, the terminal pocket **7** receives and stacks the paper sheet **S** from the opening at the end of the extension feed path **184**.

(96) In this configuration, in the paper sheet processing device **1**, the master control portion **56** of the base machine **2** controls the base machine **2** and also the slave control portion **95** of the first expansion unit **3**, the slave control portion **95** of the second expansion unit **4**, the slave control portion **175** of the third expansion unit **5**, and the slave control portion **225** of the fourth expansion unit **6** by outputting control command signals to each.

(97) Unit station address **#1** is preset for the first expansion unit **3**, unit station address **#2** for the second expansion unit **4**, unit station address **#3** for the third expansion unit **5**, and unit station address **#4** for the fourth expansion unit **6**.

(98) When performing necessary communication with, for example, the slave control portion **95** of the first expansion unit **3**, the master control portion **56** of the base machine **2** transmits a control command signal specifying the unit station address **#1**. When the control command signal specifying unit station address **#1** is received in this manner, only the slave control portion **95** of the first expansion unit **3** recognizes the reception of a control command signal addressed to its own station. Conversely, when the slave control portion **95** of the first expansion unit **3** makes a required communication to the master control portion **56** of the base machine **2**, the slave control portion **95** does so by sending a signal with the unit station address **#1** appended. Upon receiving a signal specifying unit station address **#1**, only the master control portion **56** of the base machine **2** recognizes the reception of the signal from the slave control portion **95** of the first expansion unit **3**.

(99) This mechanism is similar between the master control portion **56** of the base machine **2** and the slave control portion **135** of the second expansion unit **4**, between the master control portion **56** of the base machine **2** and the slave control portion **175** of the third expansion unit **5**, and between the master control portion **56** of the base machine **2** and the slave control portion **225** of the fourth expansion unit **6**. This is the mechanism by which such an interface is established between the master control portion **56** and the multiple slave control portions **95**, **135**, **175**, and **225**.

(100) In the paper sheet processing device **1**, the master control portion **56** of the base unit **2** sets, based on a setting input to the operation display portion **55**, whether the destination by classification for the classified paper sheet **S** is any of the stacking portions **41** to **43** of the base machine **2**, the stacking portions **81** to **84** of the first expansion unit **3**, the stacking portions **121**, **122** of the second expansion unit **4**, the stacking portions **161** to **163** of the third expansion unit **5**, the bundling portion **189** of the fourth expansion unit **6** and the terminal pocket **7**.

(101) The master control portion **56** causes the paper sheet **S** to be fed one by one to the identification conveyance portion **21** by the input unit **15**, conveyed by the identification conveyance portion **21**, and identified by the validator **22**. The destination of each paper sheet **S** is determined from the identification result of the validator **22**. The master control portion **56** causes the paper sheet **S** other than the paper sheet **S** successfully identified by the identifying portion unit **22** as count targets to be guided from the identification conveyance portion **21** to the rejection conveyance portion **27** by the sorting portion **30** shown in FIG. 2, and fed out from the terminal of the rejection conveyance portion **27** to the rejection portion **16**. The master control portion **56**

guides the paper sheet S of the type successfully identified from the identification result of the validator **22** as a counting object and to be conveyed to the stacking portion **41** from the identification conveyance portion **21** to the rejection conveyance portion **27** by the sorting portion **30**, and causes the paper sheet S to be sorted from the rejection conveyance portion **27** to the tributary conveyance portion **31** by the sorting portion **34** and fed out to the stacking portion **41**. The master control portion **56** guides the paper sheet S of the type successfully identified from the identification result of the validator **22** as a counting object and to be conveyed to the stacking portion **42** from the identification conveyance portion **21** to the rejection conveyance portion **27** by the sorting portion **30**, and causes the paper sheet S to be sorted from the rejection conveyance portion **27** to the tributary conveyance portion **32** by the sorting portion **35** and fed out to the stacking portion **42**. The master control portion **56** guides the paper sheet S of the type successfully identified from the identification result of the validator **22** as a counting object and to be conveyed to the stacking portion **43** from the identification conveyance portion **21** to the rejection conveyance portion **27** by the sorting portion **30**, and causes the paper sheet S to be sorted from the rejection conveyance portion **27** to the tributary conveyance portion **33** by the sorting portion **36** and fed out to the stacking portion **43**.

(102) The master control portion **56** guides the paper sheet S of the type successfully identified from the identification result of the validator **22** as a counting object and to be conveyed to any of the stacking portions **81** to **84** of the first expansion unit **3**, the stacking portions **121** and **122** of the second expansion unit **4**, the stacking portions **161** to **163** of the third expansion unit **5**, and the bundling portion **189** of the fourth expansion unit **6** from the identification conveyance portion **21** to the outward-facing conveyance portion **28** by the sorting portion **30**, and conveys the paper sheet S by the outward-facing conveyance portion **28** to the first expansion unit **3** shown in FIG. **1**. At the same time, the master control portion **56** outputs a control command signal including the identification information of the conveyance destination for this paper sheet S to the slave control portion **95** of the first expansion unit **3**, the slave control portion **135** of the second expansion unit **4**, the slave control portion **175** of the third expansion unit **5**, and the slave control portion **225** of the fourth expansion unit **6**.

(103) The slave control portion **95** of the first expansion unit **3** controls the first expansion unit **3** based on the control command signal received from the master control portion **56**. For example, upon detecting the paper sheet S by a predetermined paper sheet detection sensor (not shown) provided more to the right side surface **61** side than the sorting portion **66** in the extension feed path **64** shown in FIG. **3**, when the control command signal for the paper sheet S contains a designation of any of the stacking portions **81** to **84** as the conveyance destination, the extension feed path **64**, the branching feed path **65** and the sorting portions **66**, **74** to **76** are controlled to send this paper sheet S to the designated one of the stacking portions **81** to **84**. On the other hand, when the control command signal for the paper sheet S detected by this predetermined paper sheet detection sensor does not contain designation of any of the stacking portions **81** to **84** as the conveyance destination, the extension feed path **64** and the sorting portion **66** are controlled to send this paper sheet S to the extension feed path **104** of the second extension unit **4** shown in FIG. **1**.

(104) Therefore, the extension feed path **64** of the first expansion unit **3** accepts the paper sheet S that has been successfully identified as a counting object by the validator **22** of the base machine **2** and conveyed by the outward-facing conveyance portion **28**, and conveys the paper sheet S toward the outside of the machine on the side of the first expansion unit **3** on the opposite side of the base machine **2**. Also, the branching feed path **65** of the first expansion unit **3** branches off from the extension feed path **64** to convey the paper sheet S. The stacking portions **81** to **84** of the first expansion unit **3** classify and stack the paper sheet S conveyed by the branching feed path **65**.

(105) The slave control portion **135** of the second expansion unit **4** controls the second expansion unit **4** based on the control command signal received from the master control portion **56**. For example, upon detecting the paper sheet S by a predetermined paper sheet detection sensor (not

shown) provided more to the right side surface **101** side than the sorting portion **106** in the extension feed path **104** shown in FIG. 4, when the control command signal for the paper sheet S contains a designation of either one of the stacking portions **121**, **122** as the conveyance destination, the extension feed path **104**, the branching feed path **105** and the sorting portions **106**, **114** are controlled to send this paper sheet S to the designated one of the stacking portions **121**, **122**. On the other hand, when the control command signal for the paper sheet S detected by this predetermined paper sheet detection sensor does not contain a designation of either of the stacking portions **121**, **122** as the conveyance destination, the extension feed path **104** and the sorting portion **106** are controlled to send this paper sheet S to the extension feed path **144** of the third expansion unit **5** shown in FIG. 1.

(106) Therefore, the extension feed path **104** of the second expansion unit **4** accepts the paper sheet S that has been successfully identified as a counting object by the validator **22** of the base machine **2** and conveyed by the outward-facing conveyance portion **28**, and conveys the paper sheet S toward the outside of the machine on the side of the second expansion unit **4** on the opposite side of the base machine **2**. Also, the branching feed path **105** of the second expansion unit **4** branches off from the extension feed path **104** to convey the paper sheet S. The stacking portions **121**, **122** of the second expansion portion **4** stack the paper sheet S conveyed by the branching feed path **105**.

(107) The slave control portion **175** of the third expansion unit **5** controls the third expansion unit **5** based on the control command signal received from the master control portion **56**. For example, upon detecting the paper sheet S by a predetermined paper sheet detection sensor (not shown) provided more to the right side surface **141** side than the sorting portion **146** in the extension feed path **144** shown in FIG. 5, when the control command signal for the paper sheet S contains a designation of any one of the stacking portions **161** to **163** as the conveyance destination, the extension feed path **144**, the branching feed path **145** and the sorting portions **146**, **154**, **155** are controlled to send this paper sheet S to the designated one of the stacking portions **161** to **163**. On the other hand, when the control command signal for the paper sheet S detected by this predetermined paper sheet detection sensor does not contain a designation of any of the stacking portions **161** to **163** as the conveyance destination, the extension feed path **144** and the sorting portion **146** are controlled to send this paper sheet S to the extension feed path **184** of the fourth expansion unit **6** shown in FIG. 1.

(108) Therefore, the extension feed path **144** of the third expansion unit **3** accepts the paper sheet S that has been successfully identified as a counting object by the validator **22** of the base machine **3** and conveyed by the outward-facing conveyance portion **28**, and conveys the paper sheet S toward the outside of the machine on the side of the third expansion unit **5** on the opposite side of the base machine **2**. Also, the branching feed path **145** of the third expansion unit **5** branches off from the extension feed path **144** to convey the paper sheet S. The stacking portions **161** to **163** of the third expansion portion **5** classify and stack the paper sheet S conveyed by the branching feed path **145**.

(109) The slave control portion **225** of the fourth expansion unit **6** controls the fourth expansion unit **6** based on the control command signal received from the master control portion **56**. For example, upon detecting the paper sheet S by a predetermined paper sheet detection sensor provided more to the right side surface **181** side than the sorting portion **186** in the extension feed path **184** shown in FIG. 6, when the control command signal for the paper sheet S contains a designation of the bundling portion **189** as the conveyance destination, the extension feed path **184**, the branching feed path **185** and the sorting portions **186**, **194** are controlled to send this paper sheet S to the then-currently stacking one among the bonding stackers **201**, **202**. On the other hand, when the control command signal for the paper sheet S detected by this predetermined paper sheet detection sensor does not contain a designation of the bundling portion **189** as the conveyance destination, the extension feed path **184** and the sorting portion **186** are controlled to send this paper sheet S to the terminal pocket **7** shown in FIG. 1.

(110) The slave control unit **225** of the fourth extension unit **6** feeds the paper sheet S to one of the

bonding stackers **201**, **202** shown in FIG. **6**, and when the number of stacked sheets in one thereof reaches the unit number of sheets for bundling, the paper sheets **S** are fed into the other one, and the stacked paper sheets **S** of the bundling unit number are conveyed by the descending conveyance portion **204**, bundled by the sealing portion **205**, and fed into the storage chamber **211**.

(111) Therefore, the extension feed path **184** of the fourth expansion unit **6** accepts the paper sheet **S** that has been successfully identified as a counting object by the validator **22** of the base machine **2** shown in FIG. **1** and conveyed by the outward-facing conveyance portion **28**, and conveys the paper sheet **S** toward the outside of the machine on the side of the fourth expansion unit **6** on the opposite side of the base machine **2**. Also, the branching feed path **185** of the fourth expansion unit **6** branches off from the extension feed path **184** to convey the paper sheet **S**. The bundling portion **189** of the fourth expansion portion **6** bundles the paper sheet **S** conveyed by the branching feed path **185**.

(112) <Control Contents: Normal>

(113) The master control unit **56** of the base machine **2** takes in the paper sheet **S** fed into the loading portion **15** one by one, and conveys the paper sheet **S** that has been successfully identified as a counting object by the validator **22** to, for example, the predetermined stacking portions **41** to **43** to be stacked, in accordance with the counting mode. Also, the paper sheet **S** other than the paper sheet **S** successfully identified as a counting object is conveyed to the rejection portion **16** to be stacked.

(114) A counting mode is a mode pertaining to operation that has been prepared in advance in plurality and stored in the storage portion of the master control portion **56**. A mode that, for example, among the paper sheet **S** successfully identified by the validator **22** as a counting object, sorts and stacks by denomination paper sheet **S** of 10,000-yen, 5,000-yen, and 1,000-yen notes in the three stacking portions **41** to **43** of the base machine **2**, and stacks other paper sheet **S** in the rejection portion **16** is set as a first counting mode. In the first counting mode, for example, 10,000-yen notes are set to be stacked in the stacking portion **41**, 5,000-yen notes are stacked in the stacking portion **42**, and 1,000-yen notes are stacked in the stacking portion **43**, respectively.

(115) A mode that, for example, among the paper sheet **S** successfully identified by the validator **22** as a counting object, sorts and stacks by denomination paper sheet **S** of 10,000-yen, 5,000-yen, 2,000-yen, and 1,000-yen notes in the four stacking portions **81** to **84** of the first expansion unit **3** and stacks other paper sheets **S** in the rejection portion **16** is set as a second counting mode. In the second counting mode, for example, 10,000-yen notes are set to be stacked in the stacking portion **81**, 5,000-yen notes are stacked in the stacking portion **82**, 2,000-yen notes are stacked in the stacking portion **83**, and 1,000-yen notes are stacked in the stacking portion **84**, respectively.

(116) For example, as an operation for counting a large number of paper sheets **S**, a mode that, for example, among the paper sheet **S** successfully identified by the validator **22** as a counting object, sorts and stacks by denomination paper sheets **S** of 10,000-yen notes in the two stacking portions **121** and **122** of the second expansion unit **4**, 5,000-yen bills, 2,000-yen, and 1,000-yen notes in the three stacking portions **161** to **163** of the third expansion portion **5**, and stacks other paper sheets **S** in the rejection portion **16** is set as a third counting mode. In the third counting mode, for example, 5,000-yen notes are set to be stacked in the stacking portion **161**, 2,000-yen notes are stacked in the stacking portion **162**, and 1,000-yen notes are stacked in the stacking portion **163**, respectively. In addition, in the third counting mode, 10,000-yen notes are stacked first in the stacking portion **121** of the two stacking portions **121** and **122**, and when the stacking portion **121** is full, the 10,000-yen notes are stacked in the stacking portion **122**.

(117) A mode that, for example, after stacking by denomination 10,000-yen and 1,000-yen notes, among the paper sheet **S** successfully identified by the validator **22** as a counting object, in the two bonding stackers **201**, **202** of the fourth expansion unit **6**, performs sealing with the sealing portion **205** located thereunder and stores the sealed small bundles in the storage chamber **211** is set as the fourth counting mode.

(118) As described above, the operator can select the desired counting mode on the operation display portion **55**, and then feed the paper sheet **S** into the loading portion **15** for counting.

(119) A count sensor (not shown) is provided at each entrance of the stacking portions **41** to **43**, **81** to **84**, **121**, **122**, **161** to **163** and the bonding stackers **201** and **202**. The master control portion **56** of the base machine **2** counts the number of paper sheets **S** that have passed through the count sensors, and confirms and detects the number of sheets stacked in each of the stacking portions **41** to **43**. The slave control portion **95** of the first expansion unit **3** counts the number of paper sheets **S** that have passed through the count sensors, and confirms and detects the number of sheets stacked in each of the stacking portions **81** to **84**. Similarly, the slave control portion **135** of the second expansion unit **4** counts the number of paper sheets **S** that have passed through the count sensors, and confirms and detects the number of sheets stacked in each of the stacking portions **121** and **122**, the slave control portion **175** of the third expansion unit **5** counts the number of paper sheets **S** that have passed through the count sensors, and confirms and detects the number of sheets stacked in each of the stacking portions **161** to **163** and the slave control portion **225** of the fourth expansion unit **6** counts the number of paper sheets **S** that have passed through the count sensors, and confirms and detects the number of sheets counted in each of the bonding stackers **201** and **202**.

(120) The stacking portions **41** to **43**, **81** to **84**, **121**, **122**, **161** to **163** and the bonding stackers **201** and **202** are each provided with a stacking portion residual sensor (not shown), with the slave control portions **95**, **135**, **175**, **225** of the first expansion units **3** to the fourth expansion unit **6** capable of detecting whether or not a paper sheet **S** remains in the stacking portions **41** to **43**, **81** to **84**, **121**, **122**, **161** to **163**, and the bonding stackers **201** and **202**, respectively, depending on whether the corresponding stacking portion residual sensor detects a paper sheet **S**. The information detected by the slave control portions **95**, **135**, **175**, and **225** is transmitted to the master control portion **56** of the base machine **2** through the above-described interface, and is centrally managed by the master control portion **56**.

(121) (Second Counting Mode: Normal)

(122) Next, in the second counting mode, control of the master control portion **56** of the base machine **2** and the slave control portions **95**, **135**, **175**, **225** of the first expansion unit **3** to the fourth expansion unit **6** when counting the paper sheet **S** will be described in detail.

(123) First, when an operator loads the paper sheet **S** into the loading portion **15** and presses the “count” button (not shown) on the operation display portion **55**, the master control portion **56** recognizes this and notifies the plurality of slave control portions **95**, **135**, **175** and **225** by transmitting a control command signal including a command indicating that counting will now be performed in the second counting mode and to begin the counting. In response, all of the slave control portions **95**, **135**, **175** and **225** recognize that they will start counting in the second counting mode.

(124) Then, the slave control unit **95** starts driving the extension feed path **64** and the branching feed path **65**, and notifies the master control portion **56** of this fact by transmitting a signal including that effect. The slave control unit **135** starts driving the extension feed path **104** and the branching feed path **105**, and notifies the master control portion **56** of this fact by transmitting a signal including that effect. The slave control unit **175** starts driving the extension feed path **144** and the branching feed path **145**, and notifies the master control portion **56** of this fact by transmitting a signal including that effect. The slave control unit **225** starts driving the extension feed path **184** and the branching feed path **185**, and notifies the master control portion **56** of this fact by transmitting a signal including that effect.

(125) Subsequently, the master control portion **56** drives the loading portion **15** and the base machine feed path **20** of the base machine **2**, causing the paper sheet **S** in the loading portion **15** to be taken into the machine one by one to be conveyed. Then, the master control portion **56** causes the validator **22** to identify the paper sheet **S** while being conveyed by the identification

conveyance portion **21** of the base machine feed path **20**. If the first paper sheet S identified by the validator **22** is, for example, a 10,000-yen note that has been successfully identified as a counting object, denomination information of the 10,000-yen note is associated with the information on this first paper sheet S and stored, and conveyance destination information to the effect that the conveyance destination thereof is the stacking portion **81** of the first expansion unit **3** is also associated and stored.

(126) If the second paper sheet S next identified by the validator **22** is, for example, a 5,000-yen note that has been successfully identified as a counting object, the master control portion **56** stores in association with the information on this second paper sheet S denomination information of the 5,000-yen note, and also stores and associates conveyance destination information to the effect that the conveyance destination thereof is the stacking portion **82** of the first expansion unit **3**.

(127) If the third paper sheet S next identified by the validator **22** is, for example, a 2,000-yen note that has been successfully identified as a counting object, the master control portion **56** stores in association with the information on this third paper sheet S denomination information of the 2,000-yen note, and also stores and associates conveyance destination information to the effect that the conveyance destination thereof is the stacking portion **83** of the first expansion unit **3**.

(128) If the fourth paper sheet S next identified by the validator **22** is, for example, a 1,000-yen note that has been successfully identified as a counting object, the master control portion **56** stores in association with the information on this fourth paper sheet S denomination information of the 1,000-yen note, and also stores and associates conveyance destination information to the effect that the conveyance destination thereof is the stacking portion **84** of the first expansion unit **3**.

(129) As described above, the master control portion **56**, based on the identification results in the validator **22** for each paper sheet S taken in from the loading portion **15**, for each paper sheet S that is successfully identified as a counting object, performs the process of associating the denomination information and destination information with the information of the paper sheet S so identified as described above, while transmitting and notifying control command signals including the results of this association process to all the multiple slave control portions **95**, **135**, **175**, **225**.

(130) That is, the master control portion **56** transmits to and notifies all the multiple slave control portions **95**, **135**, **175**, **225** of the results of the association processing each time a paper sheet S taken in from loading portion **15** is successfully identified by validator **22** as a counting object, such that the first paper sheet S identified by the validator **22** is a 10,000-yen note and the conveyance destination thereof is the stacking portion **81** of the first expansion unit **3**, the second paper sheet S next identified by the validator **22** is a 5,000-yen note and the conveyance destination thereof is the stacking portion **82** of the first expansion unit **3**, the third paper sheet S identified by the validator **22** is a 2,000-yen note and the conveyance destination thereof is the stacking portion **83** of the first expansion unit **3**, and the fourth paper sheet S identified by the validator **22** is a 1,000-yen note and the conveyance destination thereof is the stacking portion **84** of the first expansion unit **3**.

(131) In addition, the master control portion **56**, while conveying on the base machine feed path **20** the paper sheet S that has been successfully identified as a counting object by the validator **22**, conveys the paper sheet to the extension feed path **64** of the first expansion unit **3**, with the sorting direction of the sorting portion **30** made the direction to direct the paper sheet S from the identification conveyance portion **21** to the outward-facing conveyance portion **28**. In this way, the paper sheets S that have been successfully identified as a counting object by the validator **22** are transferred from the base machine **2** to the first expansion unit **3**.

(132) The paper sheets S not correctly identified as counting objects by the validator **22** are conveyed to the rejection portion **16** and stacked, with the sorting direction of the sorting portion **30** made the direction to direct the paper sheet S from the identification conveyance portion **21** to the rejection conveyance portion **27**.

(133) When the paper sheet S loaded into the loading portion **15** runs out, if there is a paper sheet S

that has been stacked in the rejection portion **16**, the operator reloads the paper sheet S into the loading portion **15** and presses the “count” button (not shown) on the operation display portion **55**. Then, the paper sheet S loaded into the loading portion **15** is processed in the same manner as described above, and if the paper sheet S is successfully identified by the validator **22** as a counting object, the master control portion **56** causes this paper sheet S to be conveyed to the first expansion unit **3**.

(134) The slave control portion **95** of the first extension unit **3** detects the paper sheet S transferred from the outward-facing conveyance portion **28** of the base machine **2** to the extension feed path **64** with a paper sheet detection sensor (not shown) provided more to the outward-facing conveyance portion **28** side of the extension feed path **64** than the sorting portion **66**. At that time, the slave control portion **95** recognizes, based on the association information transmitted from the master control portion **56** described above, that the first paper sheet S conveyed on the extension feed path **64** is a 10,000-yen note and that its conveyance destination is the stacking portion **81** of the first expansion unit **3**. Therefore, the slave control portion **95** causes the paper sheet S to be conveyed along the main conveyance portion **70** and the tributary conveyance portion **71** of the branching feed path **65** by the sorting portions **66** and **74** and to be stacked in the stacking portion **81**.

(135) The slave control portion **95** recognizes that the second paper sheet S conveyed on the extension feed path **64** is a 5,000-yen note and that the conveyance destination thereof is the stacking portion **82** of the first expansion unit **3**. Therefore, the slave control portion **95** causes the paper sheet S to be conveyed along the main conveyance portion **70** and the tributary conveyance portion **72** of the branching feed path **65** by the sorting portions **66**, **74**, **75** and to be stacked in the stacking portion **82**.

(136) The slave control portion **95** recognizes that the third paper sheet S conveyed on the extension feed path **64** is a 2,000-yen note and that the conveyance destination thereof is the stacking portion **83** of the first expansion unit **3**. Therefore, the slave control portion **95** causes the paper sheet S to be conveyed along the main conveyance portion **70** and the tributary conveyance portion **73** of the branching feed path **65** by the sorting portions **66**, **74** to **76** and to be stacked in the stacking portion **83**.

(137) The slave control portion **95** recognizes that the fourth paper sheet S conveyed on the extension feed path **64** is a 1,000-yen note and that the conveyance destination thereof is the stacking portion **84** of the first expansion unit **3**. Therefore, the slave control portion **95** causes the paper sheet S to be conveyed along the main conveyance portion **70** of the branching feed path **65** by the sorting portions **66**, **74** to **76** and to be stacked in the stacking portion **84**.

(138) As described above, the slave control portion **95** of the first expansion unit **3** performs a stacking operation to stack all of the paper sheet S received from the base machine **2** in a predetermined stacking portion among the stacking portions **81** to **84** based on the association information communicated from the master control portion **56**, respectively.

(139) When the stacking operation for all the paper sheet S received from the base machine **2** is completed, the slave control portion **95** of the first expansion unit **3** notifies the master control portion **56** by sending an end notification signal to that effect. In response, the master control portion **56** sends a control command signal including a “counting end stop” command to each of the multiple slave control portions **95**, **135**, **175**, and **225** to notify them. Then the slave control portion **95** stops the conveyance operation of the extension feed path **64** and the branching feed path **65**, the slave control portion **135** stops the conveyance operation of the extension feed path **104** and the branching feed path **105**, the slave control portion **175** stops the conveyance operation of the extension feed path **144** and the branching feed path **145**, and the slave control portion **225** stops the conveyance operation of the extension feed path **184** and the branching feed path **185**, respectively.

(140) Subsequently, the master control portion **56** displays on the operation display portion **55** the fact that the counting process has been successfully completed, and on the basis of the

identification result of the validator **22**, displays for the paper sheet **S** successfully identified as a counting target, a detailed description such as the quantity and amount of money for each denomination as well as the total amount of money by this counting process. When the operator confirms the display and presses the “done” button (not shown) on the operation display portion **55**, the master control portion **56** confirms the counting transaction process performed in this second counting mode.

(141) (Third Counting Mode: Normal)

(142) Next, in the third counting mode, control of the master control portion **56** of the base machine **2** and the slave control portions **95**, **135**, **175**, **225** of the first expansion unit **3** to the fourth expansion unit **6** during counting of the paper sheets **S** will be described in detail.

(143) First, when an operator loads the paper sheet **S** into the loading portion **15** and presses the “count” button (not shown) on the operation display portion **55**, the master control portion **56** recognizes this and notifies the plurality of slave control portions **95**, **135**, **175** and **225** by transmitting a control command signal including a command indicating that counting will now be performed in the third counting mode and to begin the counting. In response, all of the slave control portions **95**, **135**, **175** and **225** recognize that they will start counting in the third counting mode.

(144) Then, the slave control unit **95** starts driving the extension feed path **64** and the branching feed path **65**, and notifies the master control portion **56** of this fact by transmitting a signal including that effect. The slave control unit **135** starts driving the extension feed path **104** and the branching feed path **105**, and notifies the master control portion **56** of this fact by transmitting a signal including that effect. The slave control unit **175** starts driving the extension feed path **144** and the branching feed path **145**, and notifies the master control portion **56** of this fact by transmitting a signal including that effect. The slave control unit **225** starts driving the extension feed path **184** and the branching feed path **185**, and notifies the master control portion **56** of this fact by transmitting a signal including that effect.

(145) Subsequently, the master control portion **56** drives the loading portion **15** and the base machine feed path **20** of the base machine **2**, causing the paper sheet **S** in the loading portion **15** to be taken into the machine one by one and conveyed. Then, the master control portion **56** causes the validator **22** to identify the paper sheet **S** while being conveyed by the identification conveyance portion **21** of the base machine feed path **20**. If the first paper sheet **S** identified by the validator **22** is, for example, a 10,000-yen note that has been successfully identified as a counting object, denomination information of the 10,000-yen note is associated with the information on this first paper sheet **S** and stored, and conveyance destination information to the effect that the conveyance destination thereof are the two stacking portions **121** and **122** of the second expansion unit **4** is also associated and stored.

(146) If the second paper sheet **S** next identified by the validator **22** is, for example, a 5,000-yen note that has been successfully identified as a counting object, the master control portion **56** stores in association with the information on this second paper sheet **S** denomination information of the 5,000-yen note, and also stores and associates conveyance destination information to the effect that the conveyance destination thereof is the stacking portion **161** of the third expansion unit **5**.

(147) If the third paper sheet **S** next identified by the validator **22** is, for example, a 2,000-yen note that has been successfully identified as a counting object, the master control portion **56** stores in association with the information on this third paper sheet **S** denomination information of the 2,000-yen note, and also stores and associates conveyance destination information to the effect that the conveyance destination thereof is the stacking portion **162** of the third expansion unit **5**.

(148) If the fourth paper sheet **S** next identified by the validator **22** is, for example, a 1,000-yen note that has been successfully identified as a counting object, the master control portion **56** stores in association with the information on this fourth paper sheet **S** denomination information of the 1,000-yen note, and also stores and associates conveyance destination information to the effect that

the conveyance destination thereof is the stacking portion **163** of the third expansion unit **5**.

(149) As described above, the master control portion **56**, based on the identification results in the validator **22** for each paper sheet **S** taken in from the loading portion **15**, for each paper sheet **S** that is successfully identified as a counting object, performs the process of associating the denomination information and destination information with the information of the paper sheet **S** so identified as described above, while transmitting and notifying control command signals including the results of this association process to all the multiple slave control portions **95**, **135**, **175**, **225**.

(150) That is, the master control portion **56** transmits to and notifies all the multiple slave control portions **95**, **135**, **175**, **225** the results of the association processing each time a paper sheet **S** taken in from loading portion **15** is successfully identified by validator **22** as a counting object, such that the first paper sheet **S** identified by the validator **22** is a 10,000-yen note and the conveyance destination thereof are the two stacking portions **121** and **122** of the second expansion unit **4**, that the second paper sheet **S** next identified by the validator **22** is a 5,000-yen note and the conveyance destination thereof is the stacking portion **161** of the third expansion unit **5**, that the third paper sheet **S** next identified by the validator **22** is a 2,000-yen note and the conveyance destination thereof is the stacking portion **162** of the third expansion unit **5**, and that the fourth paper sheet **S** next identified by the validator **22** is a 1,000-yen note and the conveyance destination thereof is the stacking portion **163** of the third expansion unit **5**.

(151) In addition, the master control portion **56**, while conveying on the base machine feed path **20** the paper sheet **S** that has been successfully identified as a counting object by the validator **22**, conveys the paper sheet to the extension feed path **64** of the first expansion unit **3**, with the sorting direction of the sorting portion **30** made the direction to direct the paper sheet **S** from the identification conveyance portion **21** to the outward-facing conveyance portion **28**. In this way, the paper sheets **S** that have been successfully identified as a counting object by the validator **22** are transferred from the base machine **2** to the first expansion unit **3**.

(152) The paper sheets **S** not correctly identified as counting objects by the validator **22** are conveyed to the rejection portion **16** and stacked, with the sorting direction of the sorting portion **30** made the direction to direct the paper sheet **S** from the identification conveyance portion **21** to the rejection conveyance portion **27**.

(153) When the paper sheet **S** loaded into the loading portion **15** runs out, if there is a paper sheet **S** that has been stacked in the rejection portion **16**, the operator reloads the paper sheet **S** into the loading portion **15** and presses the “count” button (not shown) on the operation display portion **55**. Then, the paper sheet **S** loaded into the loading portion **15** is processed in the same manner as described above, and if the paper sheet **S** is successfully identified by the validator **22** as a counting object, the master control portion **56** causes this paper sheet **S** to be conveyed to the first expansion unit **3**.

(154) The slave control portion **95** of the first extension unit **3** detects the paper sheet **S** transferred from the outward-facing conveyance portion **28** of the base machine **2** to the extension feed path **64** with a paper sheet detection sensor (not shown) provided more to the outward-facing conveyance portion **28** side of the extension feed path **64** than the sorting portion **66**. At that time, the slave control portion **95** recognizes, based on the association information transmitted from the master control portion **56** described above, that the conveyance destination of all the paper sheet **S** conveyed on the extension feed path **64** is the second expansion unit **4** or the third expansion unit **5** from the first to the last sheet. The slave control portion **95** has all the paper sheet **S** conveyed as they are on the extension feed path **64**, to be sequentially conveyed to the extension feed path **104** of the second expansion unit **4**.

(155) The slave control portion **135** of the second expansion unit **4** detects the paper sheet **S** transferred from the extension feed path **64** to the extension feed path **104** of the first expansion unit **3** with a paper sheet detection sensor (not shown) provided more to the extension feed path **64** side of the extension feed path **104** than the sorting portion **106**. At that time, the slave control

portion **135** recognizes, based on the association information transmitted from the master control portion **56** described above, that the first paper sheet S conveyed on the extension feed path **104** is a 10,000-yen note and that the conveyance destination thereof are the stacking portions **121** and **122** of the second expansion unit **4**. Therefore, the slave control portion **135** causes the paper sheet S to be conveyed along the main conveyance portion **110** and the tributary conveyance portion **111** of the branching feed path **105** by the sorting portions **106** and **114** and to be stacked in the stacking portion **121** on the upper stage. When the upper-stage stacking portion **121** subsequently becomes full due to the paper sheet S being conveyed to the upper-stage stacking portion **121**, the slave control portion **135** causes the 10,000-yen note paper sheet S from that point to be conveyed along the main conveyance portion **110** of the branching feed path **105** by the sorting portions **106** and **114** and to be stacked in the lower-stage stacking portion **122**.

(156) On the other hand, the slave control portion **135** has all the paper sheets S other than 10,000-yen notes conveyed as they are on the extension feed path **104**, to be sequentially conveyed to the extension feed path **144** of the third expansion unit **5**.

(157) The slave control portion **175** of the third expansion unit **5** detects the paper sheet S transferred from the extension feed path **104** of the second expansion unit **4** to the extension feed path **144** with a paper sheet detection sensor (not shown) provided more to the extension feed path **104** side of the extension feed path **144** than the sorting portion **146**. At that time, the slave control portion **175** recognizes, based on the association information transmitted from the master control portion **56** described above, that the first paper sheet S conveyed on the extension feed path **144** is a 5,000-yen note and that the conveyance destination thereof is the stacking portions **161** of the third expansion unit **5**. Therefore, the slave control portion **175** causes the paper sheet S to be conveyed along the main conveyance portion **150** and the tributary conveyance portion **151** of the branching feed path **145** by the sorting portions **146** and **154** and to be stacked in the stacking portion **161**.

(158) The slave control portion **175** recognizes that the second paper sheet S conveyed on the extension feed path **144** is a 2,000-yen note and that the conveyance destination thereof is the stacking portion **162** of the third expansion unit **5**. Therefore, the slave control portion **175** causes the paper sheet S to be conveyed along the main conveyance portion **150** and the tributary conveyance portion **152** of the branching feed path **145** by the sorting portions **146**, **154**, **155** and to be stacked in the stacking portion **162**.

(159) The slave control portion **175** recognizes that the third paper sheet S conveyed on the extension feed path **144** is a 1,000-yen note and that the conveyance destination thereof is the stacking portion **163** of the third expansion unit **5**. Therefore, the slave control portion **175** causes the paper sheet S to be conveyed along the main conveyance portion **150** of the branching feed path **145** by the sorting portions **146**, **154**, **155** and to be stacked in the stacking portion **163**.

(160) As described above, the slave control portion **135** of the second expansion unit **4** and the slave control portion **175** of the third expansion unit **5** perform a stacking operation to stack all of the paper sheet S transferred from the base machine **2** in a predetermined stacking portion among the stacking portions **121**, **122**, **161** to **163** based on the association information transmitted from the master control portion **56** to each.

(161) When the stacking operation for all the paper sheet S transferred from the base machine **2** via the first expansion unit **3** is completed, the slave control portion **135** of the second expansion unit **4** and the slave control portion **175** of the third expansion unit **5** notify the master control portion **56** by sending an end notification signal to that effect. In response, the master control portion **56** sends a control command signal including a “counting end stop” command to each of the multiple slave control portions **95**, **135**, **175**, and **225** to notify them. Then the slave control portion **95** stops the conveyance operation of the extension feed path **64** and the branching feed path **65**, the slave control portion **135** stops the conveyance operation of the extension feed path **104** and the branching feed path **105**, the slave control portion **175** stops the conveyance operation of the

extension feed path **144** and the branching feed path **145**, and the slave control portion **225** stops the conveyance operation of the extension feed path **184** and the branching feed path **185**, respectively.

(162) Subsequently, the master control portion **56** displays on the operation display portion **55** the fact that the counting process has been successfully completed, and on the basis of the identification result of the validator **22**, displays for the paper sheet S successfully identified as a counting target, a detailed description such as the quantity and amount of money for each denomination as well as the total amount of money by this counting process. When the operator confirms the display and presses the “done” button (not shown) on the operation display portion **55**, the master control portion **56** confirms the counting transaction process performed in this third counting mode.

(163) <Control Contents: When an Abnormality Occurs>

(164) The normal counting operation was described above, but an abnormality may occur during the counting operation. Details of the control in that case will be described below.

(165) There are two main types of abnormalities: those that require an emergency stop (immediate stop by applying a brake to the feed path) because no further conveyance operation of the target feed path can be performed, and those that require a normal stop (stop after the paper sheet on the feed path has been conveyed) so that the conveyor operation itself of the feed path can be performed.

(166) Abnormalities that require an emergency stop include abnormalities such as an abnormality in which a cover (not shown) of each feed path is opened during the conveyance operation, jamming of the paper sheet S, an abnormal motor rotation of the drive unit, and an operation switching failure in the sorting portion. On the other hand, abnormalities that can be handled by a normal stop include the occurrence of a conveyance abnormality such as double feeding and skewing of the paper sheet S, detection of an abnormality in a sensor such as the paper sheet detection sensor and the stacking portion full detection sensor, and defective intake of paper sheet S at the loading portion **15**. Another factor pertaining to the occurrence of an abnormality is the location where the abnormality occurred. The master control portion **56** and the slave control portions **95**, **135**, **175**, **225** are designed to perform control after an abnormality occurs based on these two factors.

(167) (First Abnormality in Second Counting Mode)

(168) The first abnormality in the second counting mode is a conveyance abnormality that has occurred inside the base machine **2** and requires an “emergency stop”.

(169) As the first abnormality in the second counting mode, details of the control in the case of, for example, a paper sheet jam abnormality requiring an “emergency stop” occurring at the validator **22** during the counting operation in the second counting mode will be described below.

(170) If the validator **22** detects, during normal counting operation in the second counting mode described above, the occurrence of a paper sheet S jam in which a paper sheet S identified by the validator **22** of the base machine **2** becomes stuck in the validator **22**, the master control portion **56** first immediately stops driving of the loading portion **15** of the base machine **2** to stop the taking-in operation of a paper sheet S by the loading portion **15**. At the same time, the master control portion **56** also immediately stops the conveyance operation of the base machine feed path **20**. At that time, if there is a possibility that the paper sheet S being conveyed on the base machine feed path **20** will stop at the position of the sorting portion **30** in the upper left part of the base machine **2**, the master control portion **56**, by stopping the base machine feed path **20** not immediately but after a certain amount of time has elapsed, so that the paper sheet S being conveyed may pass beyond the sorting portion **30** to the extension feed path **64** of the first expansion portion **3**.

(171) After the conveyance operation of the base machine transport path **20** is stopped, the master control portion **56** performs a notification by transmitting to all of the multiple slave control portions **95**, **135**, **175**, **225** a control command signal including a command that an abnormality

requiring an “emergency stop” has occurred in the base machine **2** and that the extension transfer paths **64, 104, 144, 184** and the branching feed paths **65, 105, 145, 185** should undergo a “normal finalizing stop”.

(172) Upon receiving the above control command signal, if there are paper sheets **S** being conveyed on the extension feed path **64** and the branching feed path **65** of the first expansion unit **3**, the slave control portion **95** of the first expansion unit **3** causes a stacking operation to be performed to stack the paper sheet **S** in the predetermined stacking portion among the stacking portions **81** to **84** based on the association information transmitted from the master control portion **56** while continuing the conveyance operation of that paper sheet **S**. In other words, when an abnormality occurs at the base machine **2**, a paper sheet **S** that has already been delivered to the first expansion unit **3** in a normal state are stacked in the corresponding stacking portions among the stacking portions **81** to **84** in the same manner as under normal conditions.

(173) In other words, during the occurrence of the first abnormality in the second counting mode, the slave control portion **95** continues the conveyance operation of both the extension feed path **64** and the branching feed path **65** without immediate stoppage when a paper sheet jam abnormality has occurred requiring an “emergency stop” in the validator **22**, depending on the location and type of the conveyance abnormality that occurred, and conveys the paper sheet **S** of the extension feed path **64** and the branching feed path **65** to the predetermined stacking portion of any of the stacking portions **81** to **84** for stacking.

(174) In the case where the base machine **2** continues to perform the conveyance operation for some predetermined time in order to prevent the paper sheet **S** from stopping at the position of the sorting portion **30**, the paper sheet **S** that has passed over the sorting portion **30** to be transferred to the extension feed path **64** after the abnormality occurrence and just prior to the predetermined time elapsing is detected with a predetermined paper sheet detection sensor more on the outward-facing conveyance portion **28** side of the extension feed path **64** than the branching feed path **65**. Then, the slave control portion **95** of the first expansion unit **3**, regarding this paper sheet **S** as the last paper sheet **S** to be continuously conveyed in the first expansion unit **3**, performs the stacking operation to perform stacking to the corresponding stacking portion among the stacking portions **81** to **84** until this last paper sheet **S**.

(175) Then, when this stacking operation is completed, the slave control portion **95** of the first expansion unit **3** notifies the master control portion **56** by transmitting a signal to that effect. In response, the master control portion **56** sends a control command signal including a “counting end stop” command to each of the multiple slave control portions **95, 135, 175, 225** to notify them. Then, the slave control portions **95, 135, 175, 225** stop the conveyance operations of the extension feed paths **64, 104, 144, 184** and the branching feed paths **65, 105, 145, 185**.

(176) Subsequently, the master control portion **56** then finalizes the count of the paper sheets **S** that have been successfully conveyed to the stacking portions **81** to **84**. Note that the master control portion **56** considers all the paper sheet **S** other than the paper sheet **S** normally conveyed to the stacking portions **81** to **84** to be unfinalized.

(177) Then, as shown in FIG. **8**, the master control portion **56** causes the operation display portion **55** to indicate that a paper sheet **S** jam abnormality has occurred in the validator **22**, the error code thereof, and the base machine feed path **20** and validator **22** (the bold lines in FIG. **8**), where paper sheets **S** remain and require removal, as detected based on the detection results of the paper sheet detection sensors (not shown) in each portion.

(178) At the same time, the master control portion **56** causes the operation display portion **55** to indicate that the counting of the counted and conveyed paper sheet **S** already stacked in the stacking portions **81** to **84** has been finalized, and details of the quantity and amount by denomination and the total monetary amount of the paper sheet **S** that has been counted and finalized in this manner.

(179) Then, the operator recognizes this fact and starts the abnormality recovery process. First, the

side cover (not shown) of the base machine **2** is opened to expose the validator **22** and the base machine feed path **20**, the jammed paper sheet S remaining in the validator **22** is removed, and if there is a paper sheet S stuck elsewhere in the base machine feed path **20**, that paper sheet S is also removed, and then the side cover (not shown) of the base machine **2** is closed. Then, as shown in FIG. **9**, the master control unit **56** causes the operation display portion **55** to display the fact that the jam abnormality has been resolved, a display prompting the user to press the “reset” button on the operation display portion **55**, and the “reset” button. Seeing this, the operator presses the “reset” button on the operation display portion **55**.

(180) Then, the master control portion **56** performs a reset operation to drive the base machine feed path **20** for a predetermined time to convey the paper sheet S in the base machine feed path **20** to the rejection portion **16**, with the sorting portion **30** oriented to convey the paper sheet S from the identification conveyance portion **21** to the rejection conveyance portion **27**. Then, if there is any paper sheet S that was forgotten to be removed in the base machine feed path **20**, this reset operation will convey those paper sheets S to the rejection portion **16**. When there is a paper sheet S that was conveyed to the rejection portion **16**, the master control portion **56**, as shown in FIG. **10**, causes the operation display portion **55** to display an indication guiding the user to remove the paper sheet S in the rejection portion **16** and indicate the location of the rejection portion **16** (the bold frame portion in FIG. **10**).

(181) When the residual detection sensor (not shown in the figure) of the rejection portion **16** detects that the operator has removed the paper sheet S from the rejection portion **16**, the master control portion **56** displays on the operation display portion **55** that the reset process, that is, the abnormality recovery process, is complete, as shown in FIG. **11**, and also displays on the operation display portion **55** a guide to the effect that the paper sheet S removed from the rejection portion **16** can be loaded into the loading portion **15** for continued counting, and a “continued counting” button.

(182) Thus, even if a paper sheet S jam occurs in the validator **22** during normal counting operation in the second counting mode, the paper sheet S downstream of the jam, which is being conveyed in the first expansion unit **3**, is stacked in the corresponding stacking portion among the stacking portions **81** to **84** based on the association information transmitted from the master control portion **56** for each. Therefore, the number of paper sheets S stacked in the stacking portions **81** to **84** can be counted and finalized.

(183) In the above, during the abnormality recovery process, the operator can remove the paper sheet S in the validator **22** and the base machine feed path **20** of the base machine **2** where the abnormality occurred, and also remove the paper sheet S in the rejection portion **16** if there are any there. That is, the operator does not need to access the stacking portions **81** to **84**, the extension feed path **64** and the branching feed path **65** of the first expansion unit **3**, and does not need to remove the paper sheet S from the stacking portions **81** to **84**, the extension feed path **64** and the branching feed path **65**. Therefore, it is possible to reduce the burden on the operator involved in the operation of the abnormality recovery process.

(184) (Second Abnormality in Second Counting Mode)

(185) The second abnormality in the second counting mode is a conveyance abnormality that has occurred inside the base machine **2** and can be handled by a “normal stop”.

(186) As the second abnormality in the second counting mode, details of the control will be described below for the case of, for example, a paper sheet S intake failure occurring in the loading portion **15** during the counting operation in the second counting mode and that can be resolved with a “normal stop”.

(187) If, during the normal counting operation in the second counting mode described above, an intake failure occurs for a paper sheet S to be taken in by loading portion **15** of the base machine **2**, and the paper sheet S becomes stuck in the loading portion **15**, the master control portion **56** first immediately stops the driving of the loading portion **15** to stop the operation of taking in the paper

sheet S. On the other hand, the master control portion **56** does not stop the conveyance operation of the base machine feed path **20** but allows the conveyance operation to continue, and so all the paper sheets S successfully identified as counting targets by the validator **22** among the paper sheets S being conveyed on the base machine feed path **20** are passed to the first expansion unit **3**. On the other hand, the paper sheets S not successfully identified as counting objects by the validator **22** are conveyed to the rejection portion **16**.

(188) The master control portion **56** notifies all of the multiple slave control portions **95**, **135**, **175**, **225** that an abnormality requiring a “normal stop” has occurred in the base machine **2**, and transmits thereto a control command signal including a command to perform a “normal finalizing stop” on the extension transfer paths **64**, **104**, **144**, **184** and the branching feed paths **65**, **105**, **145**, **185**. In response, the slave control portion **95** of the first expansion unit **3** continues to perform the conveyance operation for the paper sheets S being conveyed on the extension feed path **64** and the branching feed path **65** of the first expansion unit **3**, including the paper sheets S accepted from the base unit **2** even after the occurrence of the abnormality, and while doing so causes a stacking operation to be performed that stacks the paper sheet S in the corresponding stacking portion among the stacking portions **81** to **84**, based on the association information transmitted from the master control portion **56**.

(189) In other words, during the occurrence of the second abnormality in the second counting mode, the slave control portion **95** continues the conveyance operation of all of the base machine feed path **20**, the extension feed path **64** and the branching feed path **65** without immediate stoppage when an intake malfunction of the paper sheet S has occurred that can be resolved with a “normal stop” in the loading portion **15**, depending on the location and type of the conveyance abnormality that occurred, and stacks the paper sheet S of the base machine feed path **20**, the extension feed path **64** and the branching feed path **65** in the predetermined stacking portion of any of the stacking portions **81** to **84**.

(190) Then, when this stacking operation is completed, the slave control portion **95** of the first expansion unit **3** notifies the master control portion **56** by performing a transmission to that effect. In response, the master control portion **56** sends a control command signal including a “counting end stop” command to each of the multiple slave control portions **95**, **135**, **175**, **225** to notify them. Then, the plurality of slave control portions **95**, **135**, **175**, **225** stop the conveyance operations of the extension feed paths **64**, **104**, **144**, **184** and the branching feed paths **65**, **105**, **145**, **185**.

(191) Subsequently, the master control portion **56** then finalizes the count of the paper sheets S that have been successfully conveyed to the stacking portions **81** to **84**. Note that the master control portion **56** considers all the paper sheets S other than the paper sheets S normally conveyed to the stacking portions **81** to **84** to be unfinalized.

(192) The master control portion **56** then causes the operation display portion **55** to perform a display according to that shown in FIGS. **8** through **11** during the first abnormality in the second counting mode. That is, the master control portion **56** first causes the operation display portion **55** to display an error code to the effect that an intake malfunction abnormality has occurred in the loading portion **15**. At the same time, the master control portion **56** also causes the operation display portion **55** to display that the counting of the counted and conveyed paper sheet S already stacked in the stacking portions **81** to **84** has been finalized, and details of the quantity and amount by denomination and the total monetary amount of the paper sheet S that has been counted and finalized in this manner.

(193) Then, the operator recognizes this fact and starts the abnormality recovery process. The operator first opens the front cover of the base machine **2** (not shown) to expose the intake portion **15** and removes paper sheets S that are stuck in the intake portion **15**. The operator then closes the front cover of the base machine **2** (not shown) and presses the “reset” button (not shown) on the operation display portion **55**.

(194) Thus, even if a paper sheet S intake malfunction has occurred in the intake portion **15** during

the normal counting operation in the second counting mode, a stacking operation is performed to stack in the corresponding stacking portion among the stacking portions **81** to **84** based on the association information transmitted from the master control portion **56** the paper sheet **S** being conveyed in the base machine feed path **20** of the base machine **2**, and the extension feed path **64** and the branching feed path **65** of the first expansion unit **3**, at least on the downstream side. Therefore, the number of paper sheets **S** stacked in the stacking portions **81** to **84** can be counted and finalized.

(195) In the above, during the abnormality recovery process, the operator should remove the paper sheet **S** in the loading portion **15** of the base machine **2** where the abnormality occurred, and also remove the paper sheet **S** in the rejection portion **16** if there are any there. That is, the operator does not need to access the base machine feed path **20** of the base machine **2**, and the stacking portions **81** to **84**, the extension feed path **64** and the branching feed path **65** of the first expansion unit **3**, and does not need to remove the paper sheet **S** from the base machine feed path **20**, the stacking portions **81** to **84**, the extension feed path **64** and the branching feed path **65**. Therefore, it is possible to reduce the burden on the operator involved in the operation of the abnormality recovery process.

(196) (Third Abnormality in Second Counting Mode)

(197) The third abnormality in the second counting mode is a conveyance abnormality requiring an “emergency stop” that occurred in the extension feed path **64** of the first expansion unit **3**.

(198) As the third abnormality in the second counting mode, details of the control in the case of, for example, a paper sheet jam abnormality requiring an “emergency stop” occurring in, for example, the extension feed path **64** of the first expansion unit **3** during the counting operation in the second counting mode will be described below.

(199) During the normal counting operation in the above-described second counting mode, suppose a paper sheet jam abnormality requiring an “emergency stop” has occurred more on the outward-facing conveyance portion **28** side of the extension feed path **64** of the first expansion unit **3** than the sorting portion **66**. Then, first, the slave control portion **95** of the first expansion unit **3** detects this, and notifies the master control portion **56** by sending a signal to the effect that a paper sheet jam abnormality requiring an “emergency stop” has occurred on the outward-facing conveyance portion **28** side of the extension feed path **64** of the first expansion unit **3** rather than the sorting portion **66**.

(200) In response, the master control portion **56** notifies all of the multiple slave control portion **95**, **135**, **175**, **225** to the effect that a paper sheet jam abnormality requiring an “emergency stop” has occurred on the outward-facing conveyance portion **28** side of the extension feed path **64** of the first expansion unit **3** rather than the sorting portion **66**, and transmits to all of the slave control portions **95**, **135**, **175**, **225** of the first expansion unit **3** to the fourth expansion unit **6** a control command signal including a command to the effect that the extension feed paths **64**, **104**, **144**, **184** are to undergo an “emergency stop”, and transmits a control command signal including a command to the effect that the branching feed paths **65**, **105**, **145**, **185** are to undergo an “normal finalizing stop”. The master control portion **56** immediately stops the drive of the loading portion **15** of the base machine **2** to stop the taking-in operation of a paper sheet **S** by the loading portion **15**.

(201) If the master control portion **56** determines that, with regard to the conveyance operation of the base machine feed path **20**, there is a risk of a paper sheet **S** on the base machine feed path **20** being passed downstream to the first expansion unit **3** as a result of the switching operation of the sorting unit **30** not occurring in time, the master control portion **56** immediately stops the conveyance operation of the base machine feed path **20**. In contrast, upon determining that all the paper sheet **S** in the base machine feed path **20** can be conveyed to the rejection portion **16** normally due to the switching of the sorting portion **30** happening in time, the master control portion **56** switches the conveyance destination to the rejection portion **16** and conveys all the paper sheet **S** on the base machine feed path **20** to the rejection portion **16** to be stacked.

(202) Upon receiving the above control command signal from the master control portion **56**, the slave control portion **95** of the first expansion unit **3** immediately stops the conveyance operation of the extension feed path **64** and continues the conveyance operation of the branching feed path **65** to carry out the conveyance operation. As a result, the first expansion unit **3** performs a stacking operation to stack the paper sheet **S** being conveyed on the branching feed path **65** in the corresponding stacking portion among the stacking portions **81** to **84**, based on the association information transmitted from the master control portion **56**.

(203) In other words, during the occurrence of the third abnormality in the second counting mode, the slave control portion **95** continues the conveyance operation of the branching feed path **65**, which is one of the extension feed path **64** and the branching feed path **65**, without immediate stoppage when a paper sheet jam abnormality has occurred requiring an “emergency stop” in the extension feed path **64** of the first expansion unit **3**, depending on the location and type of the conveyance abnormality that occurred, and conveys the paper sheet **S** of the branching feed path **65** to the predetermined stacking portion of any of the stacking portions **81** to **84**.

(204) Then, when this stacking operation is completed, the slave control portion **95** of the first expansion unit **3** notifies the master control portion **56** by performing a transmission to that effect. In response, the master control portion **56** sends a control command signal including a “counting end stop” command to each of the multiple slave control portions **95**, **135**, **175**, **225** to notify them. Then, the plurality of slave control portions **95**, **135**, **175**, **225** stop the conveyance operations of the extension feed paths **64**, **104**, **144**, **184** and the branching feed paths **65**, **105**, **145**, **185**.

(205) Subsequently, the master control portion **56** then finalizes the count of the paper sheets **S** that have been successfully conveyed to the stacking portions **81** to **84**. Note that the master control portion **56** considers all the paper sheets **S** other than the paper sheets **S** normally conveyed to the stacking portions **81** to **84** to be unfinalized.

(206) The master control portion **56** causes the operation display portion **55** to perform a display according to that shown in FIGS. **8** through **11** during the first abnormality in the second counting mode. That is, the master control portion **56** first causes the operation display portion **55** to indicate that a paper sheet jam abnormality has occurred in the extension feed path **64** of the first expansion unit **3**, the error code thereof, and the extension feed path **64** where paper sheets **S** remain and require removal, as detected based on the detection results of the paper sheet detection sensors (not shown) in each portion. At this time, if the base machine feed path **20** of the base machine **2** also corresponds to a location where the paper sheets **S** remain and need to be removed, the base machine feed path **20** is also displayed.

(207) At the same time, the master control portion **56** causes the operation display portion **55** to also indicate that the counting of the counted and conveyed paper sheet **S** already stacked in the stacking portions **81** to **84** has been finalized, and details of the quantity and amount by denomination and the total monetary amount of the paper sheet **S** that has been counted and finalized in this manner.

(208) Then, the operator recognizes this fact and starts the abnormality recovery process. The operator opens the top cover (not shown) of the first expansion unit **3** to expose the expansion feed path **64**, removes the jammed paper sheet stuck in the extension feed path **64**, and also removes a paper sheet **S** stuck elsewhere in the extension feed path **64**, if any, and closes the top cover. When the base machine feed path **20** of the base machine **2** is displayed on the operation display portion **55** as the location of the paper sheet **S** that needs to be removed, the operator opens the front cover (not shown) of the base machine **2** to expose the base machine feed path **20**, removes the paper sheet **S** stuck in the base machine feed path **20**, and closes this front cover. Then the “reset” button (not shown) on the operation display portion **55** is pressed.

(209) Then, the master control portion **56** switches the sorting portion **30** to the direction conveying the paper sheet **S** from the identification conveyance portion **21** to the rejection conveyance portion **27**, to cause the base machine feed path **20** to convey a paper sheet **S** to the rejection portion **16**.

Then, if there is any paper sheet S that was forgotten to be removed in the base machine feed path **20**, this reset operation will cause those paper sheets S to be conveyed to the rejection portion **16** and stacked. At this time, the master control portion **56** causes the operation display portion **55** to show an indication guiding the user to remove the paper sheet S from the rejection portion **16**.

(210) At the same time, the master control portion **56** outputs a control command signal to the slave control portion **95** of the first expansion unit **3**, instructing it to perform a reset operation. The slave control portion **95** of the first expansion unit **3** then performs a reset operation of the extension feed path **64** and the branching feed path **65**. If there is any paper sheet S that was forgotten to be removed in the extension feed path **64**, the slave control portion **95** switches the conveyance destination so that if there is an empty stacking portion among the stacking portions **81** to **84** where the paper sheet S is yet to be stacked, all the paper sheets S that were forgotten to be removed are collectively conveyed to the empty stacking portion in the stacking portions **81** to **84** as a destination for evacuation. In other words, the paper sheet S that was forgotten to be removed in the extension feed path **64** are conveyed to an empty stacking portion as an evacuation destination among the stacking portions **81** to **84**, rather than to a predetermined conveyance destination based on the association information transmitted from the master control portion **56**.

(211) That is, even if the operator forgets to remove the paper sheet S stuck in the target extension feed path **64** during the abnormality recovery processing operation after a conveyance abnormality has occurred, the slave control portion **95** switches the conveyance destination of such stuck paper sheet S to an empty stacking portion as an evacuation destination among the stacking portions **81** to **84** during the reset operation of the target extension feed path **64** and branching feed path **65**.

(212) If, at the time of this reset operation, there is no empty stacking portion among the stacking portions **81** to **84** in which no paper sheet S is stacked, the slave control portion **95** collectively conveys all the paper sheets S that were forgotten to be removed in the extension feed path **64** to the stacking portion with the lowest stacked number of paper sheets S among the stacking portions **81** to **84** as the evacuation destination.

(213) Note that, for example, a predetermined stacking portion, for example, stacking portion **84**, is set in advance from among the stacking portions **81** to **84** as an evacuation destination for the paper sheet S, and the slave control portion **95** may gather all the paper sheets S that were forgotten to be removed from the extension feed path **64** during the reset operation of the extension feed path **64** and the branching feed path **65**, and have them conveyed to this stacking portion **84** for accumulation.

(214) After the reset operation is completed, the slave control portion **95** of the first expansion unit **3** notifies the master control portion **56** by transmitting information on the stacking portion that is the evacuation destination when paper sheets S that were forgotten to be removed from the extension feed path **64** are evacuated to one of the stacking portions **81** to **84**. In response, the master control portion **56** causes the operation display portion **55** to show a guidance display on removing the paper sheet S in the stacking portion that is the evacuation destination. In this case, the master control portion **56** assumes that all paper sheets S in the stacking portion of this evacuation destination are unfinalized. The paper sheets S removed from the stacking portion that is the evacuation destination are reloaded into the loading portion **15**, and are identified and counted again in a continuation process after the abnormality recovery process.

(215) Thus, even if a paper sheet jam abnormality has occurred in the extension feed path **64** of the first expansion unit **3**, a stacking operation is performed to stack in a predetermined stacking portion among the stacking portions **81** to **84** based on the association information transmitted from the master control portion **56** the paper sheet S being conveyed in the branching feed path **65** of the first expansion unit **3**, at least on the downstream side. Therefore, it is possible to count and finalize the paper sheet S that has been stacked in a stacking portion which is not the evacuation destination of the paper sheet S that was forgotten to be removed, among the stacking portions **81** to **84**.

(216) In addition, even if there are paper sheets S that have been forgotten to be removed on the

extension feed path **64** during the abnormality recovery process by the operator, if there is an empty stacking portion among the stacking portions **81** to **84**, the paper sheets **S** are collectively conveyed and stacked in that stacking portion, and if there is no empty stacking portion, the paper sheets are collectively conveyed and stacked in the stacking portion with the least number of paper sheets among the stacking portions **81** to **84**, or to a predetermined stacking portion among the stacking portions **81** to **84**. Therefore, the operator does not need to access stacking portions other than the evacuation destination among the stacking portions **81** to **84** and the branching feed path **65**, and so it is no longer necessary to remove a paper sheet **S** from stacking portions other than the evacuation destination and the branching feed path **65**. Accordingly, it is possible to reduce the burden on the operator involved in the operation of the abnormality recovery process.

(217) Any one stacking portion among the stacking portions **121** and **122** of the second expansion unit **4**, the stacking portions **161** to **163** of the third expansion unit **5**, and the terminal pocket **7**, for example, the stacking portion **121**, may serve as the evacuation destination of the paper sheets **S** when there are paper sheets **S** that were forgotten to be removed from the extension feed path **64** during the abnormality recovery process.

(218) (Fourth Abnormality in Second Counting Mode)

(219) The fourth abnormality in the second counting mode is a conveyance abnormality, resolved by a “normal stop”, that occurred in the extension feed path **64** of the first expansion unit **3**.

(220) As the fourth abnormality in the second counting mode, details of the control in the case of, for example, a paper sheet double-feed abnormality requiring a “normal stop” occurring in, for example, the extension feed path **64** of the first expansion unit **3** during the counting operation in the second counting mode will be described below.

(221) During the normal counting operation in the above-described second counting mode, suppose a paper sheet double-feed abnormality requiring a “normal stop” has occurred more on the outward-facing conveyance portion **28** side of the extension feed path **64** of the first expansion unit **3** than the sorting portion **66**. Then, first, the slave control portion **95** of the first expansion unit **3** detects this, and notifies the master control portion **56** by sending a signal to the effect that a paper sheet double-feed abnormality requiring a “normal stop” has occurred on the outward-facing conveyance portion **28** side of the extension feed path **64** of the first expansion unit **3** rather than the sorting portion **66**.

(222) Here, the paper sheet double-feed abnormality is an abnormality in which two or three paper sheets **S** are conveyed in an overlapping manner, which may result in incorrect feeding to the designated conveyance destination based on the association information transmitted from the master control portion **56**. Here, since it is not possible to determine whether two sheets or three sheets are overlapped in the double feeding, there is a possibility that paper sheet **S** subsequently conveyed may not be correctly conveyed to the predetermined conveyance destination based on the association information transmitted from the master control portion **56**. However, since there is no abnormality such as the extension feed path **64** becoming clogged with jammed paper sheet, the conveyance operation itself can continue.

(223) Upon receiving notification that a paper sheet double-feed abnormality requiring a “normal stop” has occurred on the outward-facing conveyance portion **28** side of the extension feed path **64** rather than the sorting portion **66**, the master control portion **56** notifies all of the slave control portions **95**, **135**, **175**, **225** by transmitting a signal to the effect that a paper sheet double-feed abnormality requiring a “normal stop” has occurred on the outward-facing conveyance portion **28** side of the extension feed path **64** of the first expansion unit **3** rather than the sorting portion **66**.

(224) The master control portion **56** transmits to all of the multiple slave control portions **95**, **135**, **175**, **225** of the second expansion unit **4** to the fourth expansion unit **6** a control command signal including a command to perform a “normal non-finalizing stop” on the extension feed paths **104**, **144**, **184**, and transmits a control command signal including a command to perform a “normal finalizing stop” on the branching feed paths **105**, **145**, **185**.

(225) The master control portion **56** immediately stops the drive of the loading portion **15** of the base machine **2** to stop the taking-in operation of a paper sheet **S** by the loading portion **15**. In the event of determining that, with regard to the conveyance operation of the base machine feed path **20**, there is a risk of a paper sheet **S** on the base machine feed path **20** being passed downstream to the first expansion unit **3** as a result of the switching operation of the sorting unit **30** not occurring in time, the master control portion **56** immediately stops the conveyance operation of the base machine feed path **20**. In contrast, in the event of determining that all the paper sheet **S** in the base machine feed path **20** can be conveyed to the rejection portion **16** normally due to the switching of the sorting portion **30** happening in time, the master control portion **56** has all the paper sheet **S** on the base machine feed path **20** conveyed to the rejection portion **16** to be stacked.

(226) The slave control portion **95** of the first expansion unit **3**, which receives the above control command signal from the master control portion **56**, continues the conveyance operation of the branching feed path **65** and performs a stacking operation that causes the paper sheet **S** being conveyed on the branching feed path **65** to be stacked in the corresponding stacking portion among the stacking portions **81** to **84**, based on the association information transmitted from the master control portion **56**.

(227) Although the conveyance operation of the extension feed path **64** is continued, for a paper sheet **S** being conveyed on the extension feed path **64** including the paper sheet **S** involved in the paper sheet double-feed abnormality, the slave control portion **95** switches the conveyance destination so that if there is an empty stacking portion among the stacking portions **81** to **84** where the paper sheet **S** is yet to be stacked, all the paper sheets **S** are collectively conveyed to this empty stacking portion as an evacuation destination.

(228) That is, when a conveyance abnormality requiring “normal stop” occurs in the extension feed path **64**, the slave control portion **95** switches the conveyance destination so that the paper sheets **S** being conveyed in the extension feed path **64** are conveyed to an empty stacking portion as an evacuation destination where paper sheets **S** are not stacked among the stacking portions **81** to **84**. Note that even when a conveyance abnormality requiring “normal stop” occurs in the branching feed path **65**, the slave control portion **95** switches the conveyance destination so that the paper sheets **S** being conveyed in the branching feed path **65** and the extension feed path **64** are conveyed to an empty stacking portion as an evacuation destination where paper sheets **S** are not stacked among the stacking portions **81** to **84**.

(229) If there is no empty stacking portion where paper sheets **S** are yet to be stacked among the stacking portions **81** to **84**, the slave control portion **95** collectively conveys all the paper sheets **S** being conveyed in the extension feed path **64** to the stacking portion with the lowest stacked number of paper sheets **S** among the stacking portions **81** to **84** as the evacuation destination. In other words, when a conveyance abnormality requiring a “normal stop” has occurred, if an empty stacking portion where paper sheets **S** are yet to be stacked does not exist among the stacking portions **81** to **84**, the slave control portion **95** switches the conveyance destination so that the paper sheet **S** being conveyed in the extension feed path **64** or in the branching feed path **65** is conveyed to the stacking portion with the lowest number of stacked sheets among the stacking portions **81** to **84** serving as the evacuation destination.

(230) In this case, when a conveyance abnormality requiring a “normal stop” occurs in the extension feed path **64** and there are no empty stacking portions among the stacking portions **81** to **84**, the slave control portion **95** switches the conveyance destination so that paper sheets **S** being conveyed on the extension feed path **64** are conveyed to the stacking portion with the lowest number of stacked sheets among the stacking portions **81** to **84**. When a conveyance abnormality requiring a “normal stop” occurs in the branching feed path **65** and there are no empty stacking portions among the stacking portions **81** to **84**, the slave control portion **95** switches the conveyance destination so that paper sheets **S** being conveyed on the branching feed path **65** and the extension feed path **64** are conveyed to the stacking portion with the lowest number of stacked

sheets among the stacking portions **81** to **84**.

(231) For example, a predetermined evacuation destination stacking portion, for example, the stacking portion **84**, may be set as the evacuation destination of the paper sheet among the stacking portions **81** to **84**, and the slave control portion **95** may cause all the paper sheet to be collectively conveyed to the stacking portion **84** to be stacked. In other words, the slave control portion **95** may switch the conveyance destination so that when a conveyance abnormality occurs, the paper sheet S being conveyed in the extension feed path **64** or the paper sheet S being conveyed in the branching feed path **65** is conveyed to a predetermined stacking portion that has been preset among the stacking portions **81** to **84**.

(232) In this case, when a conveyance abnormality requiring a “normal stop” occurs in the extension feed path **64**, the slave control portion **95** switches the conveyance destination so that paper sheet S being conveyed on the extension feed path **64** is conveyed to the predetermined stacking portion among the stacking portions **81** to **84**, for example, stacking portion **84**. When a conveyance abnormality requiring a “normal stop” has occurred in the branching feed path **65**, the slave control portion **95** switches the conveyance destination so that the paper sheet S being conveyed on the branching feed path **65** and the paper sheet S being conveyed on the extension feed path **64** is conveyed to the predetermined stacking portion among the stacking portions **81** to **84**, for example, stacking portion **84**.

(233) In other words, during the occurrence of the fourth abnormality in the second counting mode, the slave control portion **95** continues the conveyance operation of both the extension feed path **64** and the branching feed path **65** without immediate stoppage when an abnormality has occurred requiring a “normal stop” in the extension feed path **64** of the first expansion unit **3**, depending on the location and type of the conveyance abnormality that occurred, and conveys the paper sheet S of the extension feed path **64** and the branching feed path **65** to the predetermined stacking portion of any of the stacking portions **81** to **84**.

(234) Then, when this stacking operation is completed, the slave control portion **95** of the first expansion unit **3** notifies the master control portion **56** by performing a transmission to that effect. In response, the master control portion **56** sends a control command signal including a “counting end stop” command to each of the multiple slave control portions **95**, **135**, **175**, **225** to notify them. Then, the plurality of slave control portions **95**, **135**, **175**, **225** stop the conveyance operations of the extension feed paths **64**, **104**, **144**, **184** and the branching feed paths **65**, **105**, **145**, **185**.

(235) Subsequently, the master control portion **56** then finalizes the count of the paper sheet S normally conveyed to the stacking portion other than the stacking portion serving as the evacuation destination for the paper sheet S in the extension feed path **64** of the stacking portions **81** to **84**. Note that the master control portion **56** considers all the paper sheet S other than the paper sheet S normally conveyed to the stacking portion other than the stacking portion serving as the evacuation destination for the paper sheet S in the extension feed path **64** of the stacking portions **81** to **84** to be unfinalized.

(236) The master control portion **56** then causes the operation display portion **55** to perform a display according to that shown in FIGS. **8** through **11** during the first abnormality in the second counting mode. In other words, the master control portion **56** causes the operation display portion **55** to indicate, first, that a paper sheet double-feed abnormality requiring a “normal stop” has occurred in the extension feed path **64** and that the unfinalized paper sheet S being conveyed has been stacked in a stacking portion of the evacuation destination among the stacking portions **81** to **84** as described above, that there is a need to remove the stacked paper sheet S including the paper sheet S in question from this evacuation destination stacking portion, the error code thereof, and the location in the base machine feed path **20** where a paper sheet S remains and needs to be removed, having been detected based on the detection results of the paper sheet detection sensors (not shown) in each portion.

(237) At the same time, the master control portion **56** also causes the operation display portion **55**

to indicate that the counting of the counted and conveyed paper sheet S already stacked in the stacking portions **81** to **84** has been finalized except for the paper sheet S in the aforementioned evacuation destination stacking portion, and details of the quantity and amount by denomination and the total monetary amount of the paper sheet S that has been counted and finalized in this manner.

(238) Then, the operator recognizes this fact and starts the abnormality recovery process. The operator removes the paper sheet S from the evacuation destination stacking portion among the stacking portions **81** to **84**, guided by the operation display portion **55**. When the base machine feed path **20** of the base machine **2** is displayed on the operation display portion **55** as the location of the paper sheet S that needs to be removed, the operator opens the front cover (not shown) of the base machine **2** to expose the base machine feed path **20**, and removes the stuck paper sheet S. Moreover, when there is guidance to remove the paper sheet S in the rejection portion **16**, the paper sheet S in the rejection portion **16** is also removed.

(239) When the operator presses the “reset” button (not shown), the master control portion **56** performs a reset operation in the base machine **2** to switch the sorting portion **30** to the direction of conveying a paper sheet S from the identification conveyance portion **21** to the rejection conveyance portion **27**, to cause the paper sheet S stuck in the base machine feed path **20** to be conveyed to the rejection portion **16** in the base machine feed path **20**. Thereby, if there is any paper sheet S stuck in the base machine feed path **20**, this will cause all the stuck paper sheet S to be conveyed to the rejection portion **16** and stacked. At this time, the master control portion **56** causes the operation display portion **55** to display guidance on removing the paper sheet S from the rejection portion **16**.

(240) At the same time, the master control portion **56** outputs a control command signal to the slave control portion **95** of the first expansion unit **3**, instructing it to perform a reset operation. The slave control portion **95** of the first expansion unit **3** then performs a reset operation of the extension feed path **64** and the branching feed path **65**.

(241) As described above, even if a double-feed abnormality has occurred in the extension feed path **64** of the first expansion unit **3**, the paper sheet S being conveyed in the branching feed path **65** of the first expansion unit **3**, on the downstream side of the abnormality, is stacked in a predetermined stacking portion among the stacking portions **81** to **84** based on the association information transmitted from the master control portion **56**. Therefore, the counting of the number of paper sheets S stacked in a stacking portions other than the evacuation destination of the paper sheets S of the extension feed path **64**, among the stacking portions **81** to **84**, can be regarded as finalized.

(242) In addition, even when a paper sheet double-feed abnormality occurs in the extension feed path **64**, the conveying operation of the extension feed path **64** is continued without being stopped. If there is an empty stacking portion among the stacking portions **81** to **84** to serve as an evacuation destination, the paper sheets S of the extension feed path **64** are collectively conveyed and stacked in this stacking portion, and if there is no empty stacking portion, the paper sheet S of the extension feed path **64** is collectively conveyed and stacked in the stacking portion with the least number of stacked paper sheet S among the stacking portions **81** to **84** as the evacuation destination.

Alternatively, the paper sheets S on the extension feed path **64** are collectively conveyed and stacked in a predetermined stacking portion, for example, the stacking portion **84**, which is a predetermined evacuation destination set in advance as an evacuation destination among the stacking portions **81** to **84**. Therefore, the operator does not need to access stacking portions other than the evacuation destination, the extension feed path **64** and the branching feed path **65**, and so it is no longer necessary to remove the paper sheet S. Accordingly, it is possible to reduce the burden on the operator involved in the operation of the abnormality recovery process.

(243) When a conveyance abnormality requiring a “normal stop” has occurred in the branching feed path **64**, any one stacking portion among the stacking portions **121** and **122** of the second

expansion unit **4**, the stacking portions **161** to **163** of the third expansion unit **5**, and the terminal pocket **7**, for example, the stacking portion **121**, may serve as the evacuation destination of the paper sheet **S** being conveyed on the extension feed path **63**.

(244) (First Abnormality in Third Counting Mode)

(245) The first abnormality in the third counting mode is a conveyance abnormality requiring an “emergency stop” that occurred in the extension feed path **104** of the second expansion unit **4**.

(246) As the first abnormality in the third counting mode, details of the control in the case of, for example, a paper sheet jam abnormality requiring an “emergency stop” occurring in, for example, the extension feed path **104** of the second expansion unit **4** during the counting operation in the third counting mode will be described below.

(247) During the normal counting operation in the above-described third counting mode described above, suppose a paper sheet jam abnormality requiring an “emergency stop” has occurred more on the extension feed path **64** side of the extension feed path **104** in the second expansion unit **4** than the sorting portion **106**. Then, first, the slave control portion **135** of the second expansion unit **4** detects this, and notifies the master control portion **56** by transmitting a signal to the effect that a paper sheet jam abnormality requiring an “emergency stop” has occurred more on the extension feed path **64** side of the extension feed path **104** in the second expansion unit **4** than the sorting portion **106**.

(248) In response, the master control portion **56** notifies all of the slave control portions **95**, **125**, **175**, **225** by transmitting thereto a signal to the effect that a paper sheet jam abnormality requiring an “emergency stop” has occurred more on the extension feed path **64** side of the extension feed path **104** in the second expansion unit **4** than the sorting portion **106**.

(249) At the same time, the master control portion **56** transmits to the slave control portion **135** of the second expansion unit **4** a control command signal including a command to perform an “emergency stop” on the extension feed path **104**, and transmits a control command signal including a command to perform a “normal finalizing stop” on the branching feed path **105**.

(250) The master control portion **56** notifies the slave control portion **95** of the first expansion unit **3** by transmitting thereto a control command signal including a command to perform a “normal non-finalizing stop” on the extension feed path **64**, and transmitting thereto a control command signal including a command to perform a “normal finalizing stop” on the branching feed path **65**.

(251) In addition, the master control portion **56** transmits to the slave control portions **175** and **225** of the third expansion unit **5** and fourth expansion unit **6** a control command signal that includes a command to perform a “normal finalizing stop” on the extension feed paths **144** and **184** and the branching feed paths **145** and **185**.

(252) The master control portion **56** immediately stops the drive of the loading portion **15** of the base machine **2** to stop the taking-in operation of a paper sheet **S** by the loading portion **15**. Note that, if a determination is made that, with regard to the conveyance operation of the base machine feed path **20**, there is a risk of a paper sheet **S** on the base machine feed path **20** being passed downstream to the first expansion unit **3** as a result of the switching operation of the sorting unit **30** not occurring in time, the conveyance operation of the base machine feed path **20** is immediately stopped. Conversely, in the case of a determination that all the paper sheet **S** in the base machine feed path **20** can be conveyed to the rejection portion **16** normally due to the switching of the sorting portion **30** happening in time, all the paper sheet **S** on the base machine feed path **20** is conveyed to the rejection portion **16** to be stacked.

(253) The slave control portion **95** of the first expansion unit **3**, which receives the above control command signal from the master control portion **56**, continues the conveyance operation of the extension feed path **64** and the branching feed path **65** to cause the paper sheet **S** being conveyed on the extension feed path **64** to be stacked in a predetermined stacking portion, for example, stacking portion **84**, as an evacuation destination among the stacking portions **81** to **84**, which are all empty.

(254) In other words, during the occurrence of the first abnormality in the third counting mode, the slave control portion **95** continues the conveyance operation of both the extension feed path **64** and the branching feed path **65** without immediate stoppage when a paper sheet jam abnormality has occurred requiring an “emergency stop” in the extension feed path **104** of the second expansion unit **4**, depending on the location and type of the conveyance abnormality that occurred, and conveys the paper sheet S of the extension feed path **64** to the predetermined stacking portion of the evacuation destination of any of the stacking portions **81** to **84**.

(255) The slave control portion **135** of the second expansion unit **4**, which receives the above control command signal from the master control portion **56**, immediately stops the conveyance operation of the extension conveying path **104** while continuing the conveyance operation of the branching feed path **105**, and performs a stacking operation for the paper sheet S being conveyed on the branching feed path **105** to the corresponding stacking portion among the stacking portions **121** and **122**, based on the association information transmitted from the master control portion **56**.

(256) In other words, during the occurrence of the first abnormality in the third counting mode, the slave control portion **135** continues the conveyance operation of the branching feed path **105**, which is one of the extension feed path **104** and the branching feed path **105**, without immediate stoppage when a paper sheet jam abnormality has occurred requiring an “emergency stop” in the extension feed path **104** of the second expansion unit **4**, depending on the location and type of the conveyance abnormality that occurred, and conveys the paper sheet S of the branching feed path **105** to the predetermined stacking portion of either of the stacking portions **121** and **122**.

(257) If there are paper sheets S being conveyed on the extension feed path **144** and the branching conveyance route **145**, the slave control portion **175** of the third expansion unit **5**, which receives the aforementioned control command signal from the master control portion **56**, performs the stacking operation to the corresponding stacking portion among the stacking portions **161** to **163** based on the association information transmitted from the master control portion **56**, while continuing to perform the conveyance operation.

(258) In other words, during the occurrence of the first abnormality in the third counting mode, the slave control portion **175** continues the conveyance operation of both the extension feed path **144** and the branching feed path **145** without immediate stoppage when a paper sheet jam abnormality has occurred requiring an “emergency stop” in the extension feed path **104** of the second expansion unit **4**, depending on the location and type of the conveyance abnormality that occurred, and conveys the paper sheet S of the extension feed path **144** and the branching feed path **145** to the predetermined stacking portion of any of the stacking portions **161** to **163**.

(259) As described above, when the plurality of the first expansion unit **3** to the third expansion unit **5** are provided, these slave control portions **95**, **135**, **175** will, depending on the location of the conveyance abnormality that has occurred and the type of conveyance abnormality, convey the paper sheet S being conveyed downstream from the extension feed path **104** of the second expansion unit **4**, which is the location of the conveyance abnormality that has occurred, to the predetermined stacking portion among the stacking portions **161** to **163**, and convey the paper sheet S being conveyed upstream of the extension feed path **104** of the second expansion unit **4** to the predetermined stacking portion among the stacking portions **81** to **84** upstream of the extension feed path **104**, which are different from the stacking portions **161** to **163**. In other words, the slave control portions **95**, **135**, **175** individually switch the conveyance destination of the paper sheet S being conveyed downstream from the location of the conveyance abnormality that occurred and the paper sheet S being conveyed upstream from the location of the conveyance abnormality that occurred so as to either convey the paper sheet S to a predetermined stacking portion or to a separate stacking portion, according to the location of the conveyance abnormality that occurred and the type of conveyance abnormality.

(260) When the slave control portions **95**, **135**, **175** of the first expansion unit **3** to the third expansion unit **5** finish their respective processes, they transmit a signal to that effect to the master

control portion **56** to give notification. In response, the master control portion **56** sends a control command signal including a “counting end stop” command to each of the multiple slave control portions **95**, **135**, **175**, **225** to notify them. Then, the plurality of slave control portions **95**, **135**, **175**, **225** stop the conveyance operations of the extension feed paths **64**, **104**, **144**, **184** and the branching feed paths **65**, **105**, **145**, **185**.

(261) Subsequently, the master control portion **56** finalizes the counting of the paper sheet **S** normally conveyed to the stacking portions **121**, **122**, **161** to **163**, which are other than the stacking portion serving as the evacuation destination for the paper sheet **S** in the extension feed path **64** of the stacking portions **81** to **84**. Note that the master control portion **56** considers all the paper sheet **S** other than the paper sheet **S** normally conveyed to the stacking portions **121**, **122**, **161** to **163** to be unfinalized.

(262) The master control portion **56** then causes the operation display portion **55** to perform a display according to that shown in FIGS. **8** through **11** during the first abnormality in the second counting mode. That is, the master control portion **56** first causes the operation display portion **55** to indicate that a paper sheet jam abnormality has occurred in the extension feed path **104** of the second expansion unit **4**, the error code thereof, and the location where the paper sheets **S** remain and require removal, as detected based on the detection results of the paper sheet detection sensors (not shown) in each portion.

(263) At the same time, the master control portion **56** causes the operation display portion **55** to also indicate that the counting of the counted and conveyed paper sheet **S** already stacked in the stacking portions **121** and **122** and the stacking portions **161** to **163** has been finalized, and details of the quantity and amount by denomination and the total monetary amount of the paper sheet **S** whose counting has been finalized in this manner.

(264) Then, the operator recognizes this fact and starts the abnormality recovery process. The operator opens the top cover (not shown) of the second expansion unit **4** to expose the expansion feed path **104**, removes the jammed paper sheet **S** stuck in the extension feed path **104**, and also removes a paper sheet **S** stuck elsewhere in the extension feed path **104**, if any, and closes this top cover. When the base machine feed path **20** of the base machine **2** is displayed on the operation display portion **55** as a location where the paper sheet **S** needs to be removed, the operator opens the front cover (not shown) of the base machine **2** to expose the base machine feed path **20**, removes the stuck paper sheet **S**, and then closes this front cover. When the operation display portion **55** displays that the paper sheet **S** is being rejected by the rejection portion **16**, the operator removes the paper sheet **S** from the rejection portion **16**. Further, when the operation display portion **55** displays that the paper sheets **S** are being stacked in the stacking portion as the evacuation destination among the stacking portions **81** to **84**, the operator removes the paper sheet **S** from this stacking portion. Then the “reset” button (not shown) on the operation display portion **55** is pressed.

(265) Then, the master control portion **56** performs a reset operation in the base machine **2** that switches the sorting portion **30** to the direction of conveying the paper sheet **S** from the identification conveyance portion **21** to the rejection conveyance portion **27**, to cause the paper sheet **S** of the identification conveyance portion **21** to be conveyed to the rejection portion **16**. Then, if there is any paper sheet **S** that was forgotten to be removed in the base machine feed path **20**, this reset operation will cause those paper sheets **S** to be conveyed to the rejection portion **16** and stacked. At this time, the master control portion **56** causes the operation display portion **55** to display guidance on the removal of the paper sheet **S** from the rejection portion **16**.

(266) At the same time, the master control portion **56** outputs a control command signal to the slave control portion **95** of the first expansion unit **3**, instructing it to perform a reset operation. The slave control portion **95** of the first expansion unit **3** then performs a reset operation of the extension feed path **64** and the branching feed path **65**.

(267) At the same time, the master control portion **56** outputs a control command signal to the slave

control portion **175** of the third expansion unit **5**, instructing it to perform a reset operation. The slave control portion **175** of the third expansion unit **5** then performs a reset operation of the extension feed path **144** and the branching feed path **145**.

(268) At the same time, the master control portion **56** outputs a control command signal to the slave control portion **135** of the second expansion unit **4**, instructing it to perform a reset operation. At this time, if there is a paper sheet **S** stuck in the extension feed path **104**, the slave control portion **175** switches the conveyance destination so that if there is an empty stacking portion among the stacking portions **121** and **122** where the paper sheet **S** is yet to be stacked, all the paper sheets **S** in question are collectively conveyed to the empty stacking portion in the stacking portions **121** and **122** as an evacuation destination.

(269) That is, even if the operator forgets to remove the paper sheet **S** stuck in the target extension feed path **104** during the abnormality recovery processing operation after a conveyance abnormality has occurred, the slave control portion **135** switches the conveyance destination of such paper sheet **S** stuck in the extension feed path **104** to an empty stacking portion as an evacuation destination among the stacking portions **121** and **122** during the reset operation of the extension feed path **104** and branching feed path **105**.

(270) Note that, at this time, if there is no empty stacking portion among the stacking portions **121** and **122** in which no paper sheet **S** is stacked, the slave control portion **135** switches the conveyance destination so as to collectively convey all the paper sheet **S** stuck in the extension feed path **104** to the stacking portion with the lowest stacked number of paper sheet **S** among the stacking portions **121** and **122** as the evacuation destination.

(271) Moreover, for example, as a conveyance change destination of a paper sheet **S** stuck in the extension feed path **104**, the slave control portion **175** may set a predetermined stacking portion, for example, the stacking portion **122**, in advance as the evacuation destination among stacking portions **121** and **122**, and then collectively convey all the paper sheet in question to this stacking portion **122** to be stacked.

(272) After the reset operation is completed, the slave control portion **135** of the second expansion unit **4** notifies the master control portion **56** by transmitting information on the stacking portion that is the evacuation destination of a paper sheet **S** stuck in the extension feed path **104** among the stacking portions **121** and **122**. In response, the master control portion **56** causes the operation display portion **55** to show a guidance display on removing the paper sheet **S** in the stacking portion. In this case, the master control portion **56** determines all the paper sheets **S** in the stacking portion, of the stack portions **121** and **122**, that is the evacuation destination for the paper sheets **S** that were stuck in the extension feed path **104** to be unfinalized.

(273) Thus, even if a paper sheet jam abnormality has occurred in the extension feed path **104** of the second expansion unit **4**, the paper sheets **S** being conveyed in the branching feed path **105** of the second expansion unit **4** and the extension feed path **144** and the branching feed path **145** of the third expansion unit **5**, on the downstream side of the abnormality, are stacked in the corresponding stacking portion of the stacking portions **121** and **122** and the stacking portions **161** to **163**, on the basis of the association information transmitted from the master control portion **56**. Therefore, the counting of the paper sheet **S** stacked in the stacking portion other than the aforementioned evacuation destination of the stacking portions **121** and **122** of the second expansion unit **4** and the stacking portions **161** to **163** of the third expansion unit **5** can be regarded as finalized.

(274) In addition, even if there are paper sheets **S** that have been forgotten to be removed on the extension feed path **104** during the abnormality recovery process by the operator, if there is an empty stacking portion among the stacking portions **121** and **122**, the paper sheets **S** are collectively conveyed and stacked in that stacking portion as an evacuation destination, and if there is no empty stacking portion, the paper sheets are collectively conveyed and stacked in the stacking portion with the least number of paper sheets among the stacking portions **121** and **122**, or to a predetermined stacking portion among the stacking portions **121** and **122** as an evacuation

destination. For this reason, the operator does not need to access stacking portions other than the evacuation destination of the paper sheet S forgotten to be removed, and the branching feed path **105**, and so it is no longer necessary to remove the paper sheet S. Accordingly, it is possible to reduce the burden on the operator involved in the operation of the abnormality recovery process. (275) Moreover, even if there is paper sheet S that has been forgotten to be removed on the extension feed path **104**, during the reset operation the paper sheet S in question will be conveyed to the aforementioned stacking portion of the evacuation destination among the stacking portions **121** and **122** to be stacked. From this point as well, it is possible to reduce the burden on the operator involved in the operation of the abnormality recovery process.

(276) If there is paper sheet S that has been forgotten to be removed from the extension feed path **104** during the abnormality recovery process performed by the operator, the terminal pocket **7** may be used as the evacuation destination.

(277) (Second Abnormality in Third Counting Mode)

(278) The second abnormality in the third counting mode is a conveyance abnormality, resolved by a “normal stop”, that occurred in the extension feed path **104** of the second expansion unit **4**.

(279) As the second abnormality in the third counting mode, details of the control in the case of, for example, a paper sheet double-feed abnormality requiring a “normal stop” occurring in, for example, the extension feed path **104** of the second expansion unit **4** during the counting operation in the third counting mode will be described below.

(280) During the normal counting operation in the above-described third counting mode described above, suppose a paper sheet double-feed abnormality requiring a “normal stop” has occurred more on the extension feed path **64** side of the extension feed path **104** in the second expansion unit **4** than the sorting portion **106**. Then, first, the slave control portion **135** of the second expansion unit **4** detects this, and notifies the master control portion **56** by transmitting a signal to the effect that a paper sheet double-feed abnormality requiring a “normal stop” has occurred more on the extension conveyance portion **64** side of the extension feed path **104** in the second expansion unit **4** than the sorting portion **106**.

(281) In response, the master control portion **56** notifies all of the slave control portions **95**, **125**, **175**, **225** by transmitting thereto a signal to the effect that a paper sheet double-feed abnormality requiring a “normal stop” has occurred more on the extension conveyance portion **64** side of the extension feed path **104** in the second expansion unit **4** than the sorting portion **106**.

(282) In addition, the master control portion **56** transmits to the slave control portion **95** of the first expansion unit **3**, the slave control portion **135** of the second expansion unit **4**, and the slave control portion **175** of the third expansion unit **5** a control command signal that includes a command to perform a “normal finalizing stop” of the branching feed paths **65**, **105**, **145**.

(283) In addition, the master control portion **56** notifies the slave control portion **95** of the first expansion unit **3** and the slave control portion **135** of the second expansion unit **4** by transmitting thereto a control command signal that includes a command to make the extension feed paths **64** and **104** perform a “normal non-finalizing stop”.

(284) In addition, the master control portion **56** notifies the slave control portion **175** of the third expansion unit **5** downstream of the second expansion unit **4** by transmitting thereto a control command signal that includes a command to perform a “normal finalizing stop” of the extension feed path **144**.

(285) The master control portion **56** immediately stops the drive of the loading portion **15** of the base machine **2** to stop the taking-in operation of a paper sheet S by the loading portion **15**. Note that, if a determination is made that, with regard to the conveyance operation of the base machine feed path **20**, there is a risk of a paper sheet S on the base machine feed path **20** being passed downstream to the first expansion unit **3** as a result of the switching operation of the sorting unit **30** not occurring in time, the conveyance operation of the base machine feed path **20** is immediately stopped. Conversely, in the case of a determination that all the paper sheet S in the base machine

feed path **20** can be conveyed to the rejection portion **16** normally due to the switching of the sorting portion **30** happening in time, all the paper sheet S of the base machine feed path **20** is conveyed to the rejection portion **16** to be stacked.

(286) Upon receiving the above control command signal from the master control portion **56**, the slave control portion **95** of the first expansion unit **3** continues the conveyance operation of the extension feed path **64** to perform the conveyance operation and transfer the paper sheet S to the extension feed path **104** of the second expansion unit **4** on the downstream side.

(287) If there are paper sheets S being conveyed on the branching feed path **105**, the slave control portion **135** of the second expansion unit **4**, which receives the aforementioned control command signal from the master control portion **56**, stacks the paper sheets to the predetermined stacking portion among the stacking portions **121** and **122** based on the association information transmitted from the master control portion **56**, while continuing to perform the conveyance operation.

(288) Although the conveyance operation of the extension feed path **104** is continued, since the paper sheets S being conveyed on the extension feed path **104** including the paper sheets S involved in the paper sheet double-feed abnormality and the paper sheets S transferred from the extension feed path **64** are not finalized, if there is an empty stacking portion among the stacking portions **121** and **122** where the paper sheet S is yet to be stacked, all the paper sheets S are collectively conveyed to this empty stacking portion of the stacking portions **121** and **122** as an evacuation destination.

(289) That is, when a conveyance abnormality requiring a “normal stop” occurs in the extension feed path **104**, the slave control portions **95** and **135**, in order to evacuate the paper sheet S being conveyed in the extension feed paths **64** and **104**, switch the conveyance destinations so that the paper sheets S are conveyed to an empty stacking portion where paper sheets S are not stacked among the stacking portions **121** and **122**. Note that when a conveyance abnormality requiring a “normal stop” occurs in the branching feed path **105**, the slave control portions **95** and **135**, in order to evacuate the paper sheet S being conveyed in the extension feed paths **64** and **104** and the branching feed path **105**, switch the conveyance destinations so as to convey the paper sheet S to an empty stacking portion where paper sheets S are not stacked among the stacking portions **121** and **122**.

(290) At this time, if there is no empty stacking portion as an evacuation destination in the stacking portions **121** and **122**, the slave control portion **135** collectively conveys the paper sheet S being conveyed on the extension feed paths **64** and **104** to the stacking portion with the least number of paper sheets S stacked among the stacking portions **121** and **122** serving as the evacuation destination.

(291) In other words, when a conveyance abnormality requiring a “normal stop” has occurred, if an empty stacking portion where paper sheets S are yet to be stacked does not exist among the stacking portions **121** and **122**, the slave control portion **135** switches the conveyance destination so that the paper sheets S being conveyed in the extension feed path **104** or the paper sheets being conveyed in the branching feed path **105** are conveyed to the stacking portion with the lowest number of stacked sheets among the stacking portions **121** and **122**.

(292) Specifically, when a conveyance abnormality requiring a “normal stop” occurs in the extension feed path **104** and there are no empty stacking portions among the stacking portions **121** and **122**, the slave control portion **135** switches the conveyance destination so that paper sheets S being conveyed on the extension feed path **104** are conveyed to the stacking portion with the lowest number of stacked sheets among the stacking portions **121** and **122**. When a conveyance abnormality requiring a “normal stop” occurs in the branching feed path **105** and there are no empty stacking portions among the stacking portions **121** and **122**, the slave control portion **135** switches the conveyance destination so that paper sheets S being conveyed on the branching feed path **105** and the extension feed path **104** are conveyed to the stacking portion with the lowest number of stacked sheets among the stacking portions **121** and **122**.

(293) Alternatively, as the evacuation destination for a paper sheet S when a conveyance abnormality requiring a “normal stop” occurs for example in the extension feed path **104** and the branching feed path **105**, a predetermined conveyance destination stacking portion, for example, the stacking portion **122**, may be set among the stacking portions **121** and **122**, and the paper sheet S in question may be collectively conveyed to the stacking portion **122** to be stacked.

(294) In other words, when a conveyance abnormality requiring a “normal stop” occurs in the extension feed path **104** or the branching feed path **105**, the slave control portion **135** may switch the conveyance destination so as to convey the paper sheet S being conveyed in the extension feed path **104** or the paper sheet S being conveyed in the branching feed path **105** to a predetermined stacking portion that has been set in advance among the stacking portions **121** and **122**.

(295) Specifically, when a conveyance abnormality requiring a “normal stop” occurs in the extension feed path **104**, the slave control portion **135** switches the conveyance destination so that paper sheet S being conveyed on the extension feed path **104** is conveyed to the stacking portion of a predetermined conveyance destination among the stacking portions **121** and **122**, for example, stacking portion **122**. When a conveyance abnormality requiring a “normal stop” occurs in the branching feed path **105**, the slave control portion **135** switches the conveyance destination so as to convey the paper sheet S being conveyed on the branching feed path **105** and the extension feed path **104** to the stacking portion of a predetermined conveyance destination among the stacking portions **121** and **122**, for example, stacking portion **122**.

(296) As described above, during the occurrence of the second abnormality in the third counting mode, the slave control portions **95** and **135** continue the conveyance operation of all of the extension feed paths **64** and **104** and the branching feed path **105** without immediate stoppage when an abnormality has occurred requiring a “normal stop” in the extension feed path **104** of the second expansion unit **4**, depending on the location and type of the conveyance abnormality that occurred, and conveys the paper sheet S of the extension feed paths **64** and **104** and the branching feed path **105** to the evacuation destination stacking portion of either of the stacking portions **121** and **122**.

(297) If there are paper sheets S being conveyed on the extension feed path **144** and the branching conveyance route **145**, the slave control portion **175** of the third expansion unit **5**, which receives the aforementioned control command signal from the master control portion **56**, performs the stacking operation to the corresponding stacking portion among the stacking portions **161** to **163** based on the association information transmitted from the master control portion **56**, while continuing to perform the conveyance operation.

(298) In other words, during the occurrence of a conveyance abnormality, the slave control portion **175** continues the conveyance operation of both the extension feed path **144** and the branching feed path **145** without immediate stoppage when an abnormality has occurred requiring a “normal stop” in the extension feed path **104** of the second expansion unit **4** on the upstream side, depending on the location and type of the conveyance abnormality that occurred, and conveys the paper sheet S of the extension feed path **144** and the branching feed path **145** to the predetermined stacking portion of any of the stacking portions **161** to **163**.

(299) As described above, when the plurality of the first expansion unit **3** to the third expansion unit **5** are provided, the slave control portions **95**, **135**, **175** will, depending on the location of the conveyance abnormality that has occurred and the type of conveyance abnormality, convey the paper sheet S being conveyed downstream from the extension feed path **104** of the second expansion unit **4**, which is the location of the conveyance abnormality that has occurred, to the predetermined stacking portion among the stacking portions **161** to **163**, and convey the paper sheet S being conveyed upstream of the extension feed path **104** of the second expansion unit **4** to the predetermined stacking portion among the stacking portions **121** and **122** separate from the stacking portions **161** to **163**. In other words, the slave control portions **95**, **135**, **175** individually switch the conveyance destination of the paper sheet S being conveyed downstream from the

location of the conveyance abnormality that occurred and the paper sheet S being conveyed upstream from the location of the conveyance abnormality that occurred so as to either convey the paper sheet S to a predetermined stacking portion or to a separate stacking portion, according to the location of the conveyance abnormality that occurred and the type of conveyance abnormality.

(300) Then, when these stacking operations are completed, the slave control portions **95, 135, 175** of the first expansion unit **3** to the third expansion unit **5** notify the master control portion **56** by performing a transmission to that effect. In response, the master control portion **56** transmits a control command signal including a “counting end stop” command to each of the multiple slave control portions **95, 135, 175, 225**. Then, the plurality of slave control portions **95, 135, 175, 225** stop the conveyance operations of the extension feed paths **64, 104, 144, 184** and the branching feed paths **65, 105, 145, 185**.

(301) Subsequently, the master control portion **56** finalizes the count of the paper sheet S normally conveyed to the stacking portions other than the aforementioned evacuation destination stacking portion of the stacking portions **121** and **122**, and the stacking portions **161** to **163**. Note that the master control portion **56** considers all the paper sheet S other than the paper sheet S normally conveyed to the stacking portions other than the aforementioned evacuation destination stacking portion of the stacking portions **121** and **122**, and the stacking portions **161** to **163** to be unfinalized.

(302) The master control portion **56** then causes the operation display portion **55** to perform a display according to that shown in FIGS. **8** through **11** during the first abnormality in the second counting mode. In other words, the master control portion **56** causes the operation display portion **55** to indicate, first, that a paper sheet double-feed abnormality requiring a “normal stop” has occurred in the extension feed path **104** and that the unfinalized paper sheet S being conveyed have been stacked in an evacuation destination stacking portion (for example, the stacking portion **122**) as described above, that there is a need to remove the paper sheet S from this stacking portion, the error code thereof, and the location where paper sheets S remain and need to be removed, having been detected based on the detection results of the paper sheet detection sensors (not shown) in each portion.

(303) At the same time, the master control portion **56** causes the operation display portion **55** to indicate that the counting of the counted and conveyed paper sheet S stacked in the stacking portions **161** to **163** and the counting of the paper sheet S of the stacking portion other than the aforementioned evacuation destination stacking portion (for example stacking portion **121**) among the stacking portions **121** and **122** has been finalized, and details of the quantity and amount by denomination and the total monetary amount of the paper sheet S that has been counted and finalized in this manner.

(304) Then, the operator recognizes this fact and starts the abnormality recovery process. The operator removes the paper sheet S from the aforementioned evacuation destination stacking portion of the stacking sections **121** and **122** of the second expansion unit **4** for which guidance was given by the operation display section **55**, and also removes the paper sheet S from the rejection portion **16** when the operation display portion **55** guides the operator to remove the paper sheet S from the rejection portion **16**. When the base machine feed path **20** of the base machine **2** is displayed on the operation display portion **55** as a location where the paper sheet S needs to be removed, the operator opens the front cover (not shown) of the base machine **2** to expose the base machine feed path **20**, removes the stuck paper sheet S, and then closes this front cover.

(305) Then, when the operator presses a “reset” button (not shown) on the operation display portion **55**, the master control portion **56** performs a reset operation that switches sorting portion **30** to the direction of conveying the paper sheet S from the identification conveyance portion **21** to the rejection conveyance portion **27**, to cause the paper sheet S of the identification conveyance portion **21** to be conveyed to the rejection portion **16**. At this time, if there is any paper sheet S stuck in the base machine feed path **20**, this will cause all the paper sheet S to be conveyed to the rejection

portion **16** and stacked. At this time, the master control portion **56** causes the operation display portion **55** to display guidance to guide the operator on removing the paper sheet **S** from the rejection portion **16**.

(306) At the same time, the master control portion **56** outputs a control command signal to the slave control portion **95** of the first expansion unit **3**, instructing it to perform a reset operation. The slave control portion **95** of the first expansion unit **3** then performs a reset operation of the extension feed path **64** and the branching feed path **65**.

(307) At the same time, the master control portion **56** outputs a control command signal to the slave control portion **135** of the second expansion unit **4**, instructing it to perform a reset operation. The slave control portion **135** of the second expansion unit **4** then performs a reset operation of the extension feed path **104** and the branching feed path **105**.

(308) At the same time, the master control portion **56** outputs a control command signal to the slave control portion **175** of the third expansion unit **5**, instructing it to perform a reset operation. The slave control portion **175** of the third expansion unit **5** then performs a reset operation of the extension feed path **144** and the branching feed path **145**.

(309) Thus, even if a paper sheet double-feed abnormality has occurred in the extension feed path **104** of the second expansion unit **4**, the paper sheets **S** being conveyed in the branching feed path **105** of the second expansion unit **4** and the extension feed path **144** and the branching feed path **145** of the third expansion unit **5**, on the downstream side of the abnormality, are stacked in the predetermined stacking portion of the stacking portions **121** and **122** and the stacking portions **161** to **163**, on the basis of the association information transmitted from the master control portion **56**. Therefore, the counting of the paper sheet **S** stacked in the stacking portions **161** to **163** of the third expansion unit **5** and the stacking portion other than the evacuation destination of the stacking portions **121** and **122** of the second expansion unit **4** can be regarded as finalized. Accordingly, it is possible to reduce the operation burden on the operator involved in the abnormality recovery process.

(310) When a conveyance abnormality requiring a “normal stop” has occurred in the extension feed path **104**, the evacuation destination of the paper sheet **S** being conveyed in the extension feed paths **64** and **104** may be the terminal pocket **7**, and so the paper sheet **S** may be conveyed to the terminal pocket **7**.

(311) (Third Abnormality in Third Counting Mode)

(312) The third abnormality in the third counting mode is a conveyance abnormality requiring an “emergency stop” that occurred in the extension feed path **105** of the second expansion unit **4**.

(313) As the third abnormality in the third counting mode, details of the control in the case of a paper sheet jam abnormality requiring an “emergency stop” occurring in the branching feed path **105** of the second expansion unit **4** during the counting operation in the third counting mode will be described below.

(314) During the normal counting operation in the above-described third counting mode described above, suppose a paper sheet jam abnormality requiring an “emergency stop” has occurred more on the stacking portion **122** side of the main conveyance portion **110** of the branching feed path **105** than the sorting portion **114** in the second expansion unit **4**. Then, first, the slave control portion **135** of the second expansion unit **4** detects this, and notifies the master control portion **56** by transmitting a signal to the effect that a paper sheet jam abnormality requiring an “emergency stop” has occurred more on the stacking portion **122** side of the main conveyance portion **110** of the branching feed path **105** than the sorting portion **114** in the second expansion unit **4**.

(315) In response, the master control portion **56** notifies all of the slave control portions **95**, **125**, **175**, **225** by transmitting thereto a signal to the effect that a paper sheet jam abnormality requiring an “emergency stop” has occurred more on the stacking portion **122** side of the main conveyance portion **110** of the branching feed path **105** than the sorting portion **114** in the second expansion unit **4**.

(316) The master control portion **56** transmits to the slave control portion **135** of the second expansion unit **4** a control command signal including a command to perform an “normal non-finalizing stop” on the extension feed path **104**, and transmits thereto a control command signal including a command to perform a “emergency stop” on the branching feed path **105**.

(317) The master control portion **56** transmits to the slave control portion **95** of the first expansion unit **3** a control command signal including a command to perform a “normal non-finalizing stop” on the extension feed path **64**, and transmits thereto a control command signal including a command to perform a “normal finalizing stop” on the branching feed path **65**.

(318) The master control portion **56** transmits to the slave control portions **175** and **225** of the third expansion unit **5** and fourth expansion unit **6** a control command signal that includes a command to perform a “normal finalizing stop” on the extension feed paths **144** and **184** and the branching feed paths **145** and **185**.

(319) The master control portion **56** immediately stops the drive of the loading portion **15** of the base machine **2** to stop the taking-in operation of a paper sheet **S** by the loading portion **15**. Note that, if a determination is made that, with regard to the conveyance operation of the base machine feed path **20**, there is a risk of a paper sheet **S** on the base machine feed path **20** being passed downstream to the first expansion unit **3** as a result of the switching operation of the sorting unit **30** not occurring in time, the master control portion **56** immediately stops the conveyance operation of the base machine feed path **20**. Conversely, in the case of a determination that all the paper sheet **S** in the base machine feed path **20** can be conveyed to the rejection portion **16** normally due to the switching of the sorting portion **30** happening in time, the master control portion **56** has all the paper sheet **S** of the base machine feed path **20** conveyed to the rejection portion **16** to be stacked.

(320) The slave control portion **95** of the first expansion unit **3**, which receives the above control command signal from the master control portion **56**, switches the conveyance destination of the paper sheet **S** being conveyed on the extension feed path **64** to an evacuation destination stacking portion determined in advance among any of the empty stacking portions **81** to **84** of the first expansion unit **3** via the branching feed path **65**, and causes the paper sheet **S** to be stacked at this stacking portion.

(321) In other words, during the occurrence of the third abnormality in the third counting mode, the slave control portion **95** continues the conveyance operation of both the extension feed path **64** and the branching feed path **65** without immediate stoppage when an abnormality has occurred requiring an “emergency stop” in the branching feed path **105** of the second expansion unit **4**, depending on the location and type of the conveyance abnormality that occurred, and conveys the paper sheet **S** of the extension feed path **64** to one of the stacking portions **81** to **84** serving as the predetermined evacuation destination.

(322) Upon receiving the above control command signal from the master control portion **56**, the slave control portion **135** of the second expansion unit **4** immediately stops the conveyance operation of the branching feed path **105**. Further, the paper sheet **S** being conveyed on the extension feed path **104** cannot be conveyed to either of the stacking portions **121** and **122** of the second expansion unit **4**. Therefore, the paper sheets **S** being conveyed on the extension feed path **104** are stacked at the evacuation destination stacking portion among the stacking portions **161** to **163** of the third expansion unit **5** via the extension feed path **144** and the branching feed path **145**. At this time, if there is an empty stacking portion among the stacking portions **161** to **163**, this stacking portion shall be the evacuation destination, and if there is no empty stacking portion among the stacking portions **161** to **163**, the stacking portion with the least number of sheets shall be the evacuation destination. Alternatively, a predetermined stacking portion among the stacking portions **161** to **163**, for example, the stacking portion **163**, shall be the evacuation destination.

(323) In other words, during the occurrence of a conveyance abnormality, the slave control portion **95** continues the conveyance operation of all of the extension feed paths **104** and **144** and the branching feed path **145** without immediate stoppage when an abnormality requiring an

“emergency stop” has occurred in the branching feed path **105** of the second expansion unit **4**, depending on the location and type of the conveyance abnormality that occurred, and conveys the paper sheet **S** of the extension feed path **104** to the stacking portion serving as the predetermined evacuation destination among the stacking portions **161** to **163**.

(324) If there are paper sheets **S** being conveyed on the extension feed path **144** and the branching conveyance route **145**, the slave control portion **175** of the third expansion unit **5**, which receives the aforementioned control command signal from the master control portion **56**, stacks the paper sheets **S** to the predetermined stacking portion among the stacking portions **161** to **163** based on the association information transmitted from the master control portion **56**, while continuing to perform the conveyance operation.

(325) In other words, during the occurrence of a conveyance abnormality, the slave control portion **175** continues the conveyance operation of both the extension feed path **144** and the branching feed path **145** without immediate stoppage when an abnormality requiring an “emergency stop” has occurred in the branching feed path **105** of the second expansion unit **4**, depending on the location and type of the conveyance abnormality that occurred, and conveys the paper sheet **S** of the extension feed path **144** and the branching feed path **145** to the predetermined stacking portion of any of the stacking portions **161** to **163**.

(326) When the slave control portions **95**, **135**, **175** of the first expansion unit **3** to the third expansion unit **5** finish their respective processes, they transmit a signal to that effect to the master control portion **56** to give notification.

(327) In response, the master control portion **56** transmits a control command signal including a “counting end stop” command to each of the multiple slave control portions **95**, **135**, **175**, **225**. Then, the plurality of slave control portions **95**, **135**, **175**, **225** stop the conveyance operations of the extension feed paths **64**, **104**, **144**, **184** and the branching feed paths **65**, **105**, **145**, **185**.

(328) Subsequently, the master control portion **56** then finalizes the counting of the paper sheet **S** normally conveyed to the stacking portions **121** and **122** and the stacking portions other than the evacuation destination stacking portion of the paper sheet **S** of the extension feed paths **64** and **104** among the stacking portions **161** to **163**. Note that the master control portion **56** considers all the paper sheet **S** other than the paper sheet **S** normally conveyed to the stacking portions **121** and **122** and the stacking portions other than the evacuation destination stacking portion of the stacking portions **161** to **163** to be unfinalized.

(329) The master control portion **56** then causes the operation display portion **55** to perform a display according to that shown in FIGS. **8** through **11** during the first abnormality in the second counting mode. That is, the master control portion **56** first causes the operation display portion **55** to indicate that a paper sheet jam abnormality has occurred in the branching feed path **105** of the second expansion unit **4**, the error code thereof, and the location where the paper sheet **S** remains and requires removal, as detected based on the detection results of the paper sheet detection sensors (not shown) in each portion.

(330) At the same time, the master control portion **56** also causes the operation display portion **55** to indicate that the counting of the counted and conveyed paper sheet **S** stacked in the stacking portions **121** and **122** and the paper sheet **S** stacked in the stacking portions **161** to **163** except for the paper sheet **S** of the evacuation destination stacking portion that received paper sheet **S** requiring removal has been finalized, and details of the quantity and amount by denomination and the total monetary amount of the paper sheet **S** that has been counted and finalized in this manner.

(331) Then, the operator recognizes this fact and starts the abnormality recovery process. The operator opens the front cover (not shown) of the second expansion unit **4** to expose the branching feed path **105**, removes the jammed paper sheet **S** stuck in the branching feed path **105**, and also removes a paper sheet **S** stuck elsewhere in the branching feed path **105**, if any, and closes the front cover. When the base machine feed path **20** of the base machine **2** is displayed on the operation display portion **55** as the location where the paper sheets **S** need to be removed, the operator opens

the front cover (not shown) of the base machine **2** to expose the base machine feed path **20**, removes the paper sheets **S** stuck in the base machine feed path **20**, and closes the front cover. When the operation display portion **55** displays that there is a paper sheet **S** rejected by the rejection portion **16**, the operator takes out the paper sheet **S** from the rejection portion **16**. Moreover, when the operation display portion **55** displays that there is a paper sheet **S** that has been evacuated to the predetermined stacking portion serving as the evacuation destination among the stacking portions **81** to **84**, the operator removes the paper sheet **S** from this stacking portion. Moreover, when the operation display portion **55** displays that there is a paper sheet **S** that has been evacuated to the predetermined stacking portion serving as the evacuation destination among the stacking portions **161** to **163**, the operator removes the paper sheet **S** from this stacking portion. Then the “reset” button (not shown) on the operation display portion **55** is pressed.

(332) Then, the master control portion **56** performs a reset operation that switches the sorting portion **30** to the direction of conveying the paper sheet **S** from the identification conveyance portion **21** to the rejection conveyance portion **27**, to cause the paper sheet **S** of the base machine feed path **20** to be conveyed to the rejection portion **16**. Then, if there is any paper sheet **S** that was forgotten to be removed in the base machine feed path **20**, this reset operation will cause those paper sheets **S** to be conveyed to the rejection portion **16** and stacked. At this time, the master control portion **56** causes the operation display portion **55** to display guidance on the removal of the paper sheet **S** from the rejection portion **16**.

(333) At the same time, the master control portion **56** outputs a control command signal to the slave control portion **95** of the first expansion unit **3**, instructing it to perform a reset operation. The slave control portion **95** of the first expansion unit **3** then performs a reset operation of the extension feed path **64** and the branching feed path **65**.

(334) At the same time, the master control portion **56** outputs a control command signal to the slave control portion **175** of the third expansion unit **5**, instructing it to perform a reset operation. The slave control portion **175** of the third expansion unit **5** then performs a reset operation of the extension feed path **144** and the branching feed path **145**.

(335) At the same time, the master control portion **56** outputs a control command signal to the slave control portion **135** of the second expansion unit **4**, instructing it to perform a reset operation. At this time, if there are any paper sheets **S** stuck in the branching feed path **105**, all the paper sheets **S** are conveyed to a predetermined evacuation destination among the stacking portions **121** and **122**, for example, the stacking portion **122**, for stacking. At this time, the master control portion **56** causes the operation display portion **55** to display guidance to guide the operator on removing the paper sheet **S** from the evacuation destination stacking portion **122**. In this case, the master control portion **56** assumes that all paper sheets **S** in this evacuation destination stacking portion **122** are unfinalized.

(336) As described above, even if a paper sheet jam abnormality occurs on the branching feed path **105** of the second expansion unit **4**, the paper sheets **S** being conveyed on the extension feed path **144** and the branching transport path **145** of the third expansion unit **5**, on the downstream side, are stacked in a predetermined stacking portion of the stacking portions **161** to **163** based on the association information transmitted from the master control portion **56**. Therefore, among the stacking portions **161** to **163** of the third expansion unit **5**, the counting can be regarded as finalized except for the paper sheet **S** of the evacuation destination stacking portion from which the paper sheet **S** needs to be removed.

(337) Moreover, if there is a paper sheet **S** that has been forgotten to be removed from the branching feed path **105** during the abnormality recovery processing by the operator, during the reset operation the paper sheet in question will be conveyed to the stacking portion of the evacuation destination among the stacking portions **121** and **122** to be stacked. Therefore, it is possible to reduce the burden on the operator involved in the operation of the abnormality recovery process.

(338) When a conveyance abnormality requiring an “emergency stop” occurs in the branching feed path **105**, the paper sheets S being conveyed in the extension feed paths **64** and **104** may be conveyed to any of the corresponding stacking portions **161** to **163** if the conveyance destination based on the identification result of the validator **22** is one of the stacking portions **161** to **163**, or to the terminal pocket **7** as an evacuation destination if the conveyance destination based on the identification result of the validator **22** is the stacking portions **121** and **122**. In this case, counting of the paper sheet S transported to any of the corresponding stacking portions **161** to **163** will be finalized, while the counting of the paper sheet S conveyed to the terminal pocket **7** will not be finalized.

(339) (Fourth Abnormality in Third Counting Mode)

(340) The fourth abnormality in the third counting mode is a conveyance abnormality requiring an “emergency stop” that occurred in the branching feed path **145** of the third expansion unit **5**.

(341) As the fourth abnormality in the third counting mode, details of the control in the case of a paper sheet jam abnormality requiring an “emergency stop” occurring in the branching feed path **145** of the third expansion unit **5** during the counting operation in the third counting mode will be described below.

(342) During the normal counting operation in the third counting mode described above, suppose a paper sheet jam abnormality requiring an “emergency stop” has occurred between the sorting portion **154** and the sorting portion **155** in the main conveyance portion **150** of the branching feed path **145** of the third expansion unit **5**. Then, the slave control portion **175** of the third expansion unit **5** detects this, and notifies the master control portion **56** by transmitting a signal to the effect that a paper sheet jam abnormality requiring an “emergency stop” has occurred between the sorting portion **154** and the sorting portion **155** in the main conveyance portion **150** of the branching feed path **145** of the third expansion unit **5**.

(343) In response, the master control portion **56** notifies all of the slave control portions **95**, **125**, **175**, **225** by transmitting thereto a signal to the effect that a paper sheet jam abnormality requiring an “emergency stop” has occurred between the sorting portion **154** and the sorting portion **155** in the main conveyance portion **150** of the branching feed path **145** of the third expansion unit **5**.

(344) The master control portion **56** transmits to the slave control portion **175** of the third expansion unit **5** a control command signal including a command to perform an “normal non-finalizing stop” on the extension feed path **144**, and transmits thereto a control command signal including a command to perform an “emergency stop” on the branching feed path **145**.

(345) The master control portion **56** transmits to the slave control portions **95** and **135** of the first expansion unit **3** and the second expansion unit **4** a control command signal including a command to perform a “normal non-finalizing stop” on the extension feed paths **64** and **104**, and transmits thereto a control command signal including a command to perform a “normal finalizing stop” on the branching feed paths **65** and **105**.

(346) The master control portion **56** transmits to the slave control portion **225** of the fourth expansion unit **6** a control command signal that includes a command to perform a “normal finalizing stop” on the extension feed path **184** and the branching feed path **185**.

(347) The master control portion **56** immediately stops the drive of the loading portion **15** of the base machine **2** to stop the taking-in operation of a paper sheet S by the loading portion **15**. Note that, if a determination is made that, with regard to the conveyance operation of the base machine feed path **20**, there is a risk of a paper sheet S on the base machine feed path **20** being passed downstream to the first expansion unit **3** as a result of the switching operation of the sorting unit **30** not occurring in time, the conveyance operation of the base machine feed path **20** is immediately stopped. Conversely, in the case of a determination that all the paper sheet S in the base machine feed path **20** can be conveyed to the rejection portion **16** normally due to the switching of the sorting portion **30** happening in time, all the paper sheet S of the base machine feed path **20** is conveyed to the rejection portion **16** to be stacked.

(348) The slave control portion **95** of the first expansion unit **3**, which has received the above control command signal from the master control portion **56**, when there is a paper sheet **S** being conveyed on the extension feed path **64**, switches the conveyance destination of the paper sheet **S** to an evacuation destination stacking portion determined in advance among the empty stacking portions **81** to **84** of the first expansion unit **3** and causes the paper sheet **S** to be stacked in this stacking portion.

(349) In other words, during the occurrence of the fourth abnormality in the third counting mode, the slave control portion **95** continues the conveyance operation of both the extension feed path **64** and the branching feed path **65** on the upstream side without immediate stoppage when an abnormality has occurred requiring an “emergency stop” in the branching feed path **145** of the third expansion unit **5**, depending on the location and type of the conveyance abnormality that occurred, and conveys the paper sheet **S** of the extension feed path **64** to any of the stacking portions **81** to **84** serving as the predetermined evacuation destination.

(350) If there are paper sheets **S** being conveyed on the branching feed path **105**, the slave control portion **135** of the second expansion unit **4**, which receives the aforementioned control command signal from the master control portion **56**, stacks the paper sheets **S** to the predetermined stacking portion among the stacking portions **121** and **122** based on the association information transmitted from the master control portion **56**, while continuing to perform the conveyance. If there are paper sheets **S** being conveyed in the extension feed path **104**, the paper sheet **S** are stacked in the evacuation destination stacking portion of either of the stacking portions **121** and **122** of the second expansion unit **4**. At this time, if there is an empty stacking portion among the stacking portions **121** and **122**, this stacking portion shall be the evacuation destination, and if there is no empty stacking portion among the stacking portions **121** and **122**, the stacking portion with the least number of sheets shall be the evacuation destination. Alternatively, a preset stacking portion among the stacking portions **121** and **122**, for example, the stacking portion **122**, shall be the evacuation destination.

(351) In other words, during the occurrence of a conveyance abnormality, the slave control portion **135** continues the conveyance operation of both the extension feed path **104** and the branching feed path **105** without immediate stoppage when an abnormality requiring an “emergency stop” has occurred in the branching feed path **145** of the third expansion unit **5**, depending on the location and type of the conveyance abnormality that occurred, and conveys the paper sheet **S** of the extension feed path **104** and the branching feed path **105** to the predetermined stacking portion of either of the stacking portions **121** and **122**.

(352) Upon receiving the above control command signal from the master control portion **56**, the slave control portion **175** of the third expansion unit **5** immediately stops the conveyance operation of the branching feed path **145**. If any paper sheets **S** are being conveyed in the extension feed path **144**, the slave control portion **175** allows the conveying operation to continue as it is, and has the paper sheet **S** conveyed to the terminal pocket **7** via the extension feed path **184** for stacking.

(353) In other words, during the occurrence of a conveyance abnormality, the slave control portions **175** and **225** continue the conveyance operation of the extension feed path **144**, which is one of the extension feed path **144** and the branching feed path **145**, without immediate stoppage when an abnormality requiring an “emergency stop” has occurred in the branching feed path **145** of the third expansion unit **5**, depending on the location and type of the conveyance abnormality that occurred, and convey the paper sheet **S** of the extension feed path **144** to the terminal pocket **7** as the stacking portion via the extension feed path **184**. In other words, the slave control portions **175** and **225** switch the conveyance destination so that the paper sheet **S** being conveyed on the extension feed path **144** is conveyed to the terminal pocket **7** if the location of the conveyance abnormality that has occurred is not in the extension feed path **144**.

(354) The slave control portions **95**, **135**, **175** of the first expansion unit **4** to the fourth expansion unit **6**, upon finishing their respective processes, notify the master control portion **56** by

transmitting a signal thereto to that effect. In response, the master control portion **56** transmits a control command signal including a “counting end stop” command to each of the multiple slave control portions **95, 135, 175, 225**. Then, the plurality of slave control portions **95, 135, 175, 225** stop the conveyance operations of the extension feed paths **64, 104, 144, 184** and the branching feed paths **65, 105, 145, 185**.

(355) Subsequently, the master control portion **56** finalizes the counting of the paper sheet **S** normally conveyed to the stacking portions **161** to **163** and the stacking portion other than the stacking portion serving as the evacuation destination of the paper sheet **S** of the extension feed path **104** among the stacking portions **121** and **122**. Note that the master control portion **56** considers all the paper sheet **S** other than the paper sheet **S** normally conveyed to the stacking portions **161** to **163** and the stacking portion other than the stacking portion serving as the evacuation destination among the stacking portions **121** and **122** to be unfinalized.

(356) The master control portion **56** causes the operation display portion **55** to perform a display according to that shown in FIGS. **8** through **11** during the first abnormality in the second counting mode. That is, the master control portion **56** first causes the operation display portion **55** to indicate that a paper sheet jam abnormality has occurred in the branching feed path **145** of the third expansion unit **5**, the error code thereof, and the location where the paper sheets **S** remain and require removal, as detected based on the detection results of the paper sheet detection sensors (not shown) in each portion.

(357) At the same time, the master control portion **56** also causes the operation display portion **55** to indicate that the counting of the counted and conveyed paper sheet **S** already stacked in the stacking portions **161** to **163** and the paper sheet **S** excluding the paper sheet **S** of the evacuation destination stacking portion of the paper sheet **S** among the stacking portions **121** and **122** has been finalized, and details of the quantity and amount by denomination and the total monetary amount of the paper sheet **S** that has been counted and finalized in this manner.

(358) Then, the operator recognizes this fact and starts the abnormality recovery process. The operator opens the front cover (not shown) of the third expansion unit **5** to expose the branching feed path **145**, removes the jammed paper sheet **S** stuck in the branching feed path **145**, and also removes a paper sheet **S** stuck elsewhere in the branching feed path **145**, if any, and closes the front cover. When the base machine feed path **20** of the base machine **2** is displayed on the operation display portion **55** as a location where the paper sheet **S** needs to be removed, the operator opens the front cover (not shown) of the base machine **2** to expose the base machine feed path **20**, removes the stuck paper sheet **S**, and then closes the front cover. When the operation display portion **55** displays that there is a paper sheet **S** rejected by the rejection portion **16**, the operator takes out the paper sheet **S** from the rejection portion **16**. Moreover, when the operation display portion **55** displays that there is a paper sheet **S** that has been evacuated to the predetermined stacking portion serving as the evacuation destination among the stacking portions **81** to **84**, the operator removes the paper sheet **S** from this stacking portion. Moreover, when the operation display portion **55** displays that there is a paper sheet **S** that has been evacuated to the stacking portion serving as the evacuation destination among the stacking portions **121** and **122**, the operator removes the paper sheet **S** from this stacking portion. Moreover, when the operation display portion **55** displays that there is a paper sheet **S** that has been evacuated to the terminal pocket **7** serving as the evacuation destination, the operator removes the paper sheet **S** from the terminal pocket **7**. Then the operator presses the “reset” button (not shown) on the operation display portion **55**.

(359) Then, the master control portion **56** performs a reset operation that switches the sorting portion **30** to the direction of conveying the paper sheet **S** from the identification conveyance portion **21** to the rejection conveyance portion **27**, to cause the paper sheet **S** of the base machine feed path **20** to be conveyed to the rejection portion **16**. Then, if there is any paper sheet **S** that was forgotten to be removed in the base machine feed path **20**, this reset operation will cause those paper sheets **S** to be conveyed to the rejection portion **16** and stacked. At this time, the master

control portion **56** causes the operation display portion **55** to display guidance on the removal of the paper sheet **S** from the rejection portion **16**.

(360) At the same time, the master control portion **56** outputs a control command signal to the slave control portion **95** of the first expansion unit **3**, instructing it to perform a reset operation. The slave control portion **95** of the first expansion unit **3** then performs a reset operation of the extension feed path **64** and the branching feed path **65**.

(361) At the same time, the master control portion **56** outputs a control command signal to the slave control portion **135** of the second expansion unit **4**, instructing it to perform a reset operation. The slave control portion **135** of the second expansion unit **4** then performs a reset operation of the extension feed path **104** and the branching feed path **105**.

(362) At the same time, the master control portion **56** outputs a control command signal to the slave control portion **175** of the third expansion unit **5**, instructing it to perform a reset operation. The slave control portion **175** of the third expansion unit **5** then performs a reset operation of the extension feed path **144** and the branching feed path **145**.

(363) At this time, if there are any paper sheets **S** stuck in the branching feed path **145** of the third expansion unit **5**, all the stuck paper sheets **S** are conveyed to a predetermined evacuation destination stacking portion among the stacking portions **161** to **163**, for example, the stacking portion **163**, for stacking. At this time, the master control portion **56** causes the operation display portion **55** to display guidance to guide the operator on removing the paper sheet **S** from the evacuation destination stacking portion **163**. In this case, the master control portion **56** assumes that all the paper sheet **S** in this evacuation destination stacking portion **163** are unfinalized.

(364) As described above, if a paper sheet jam abnormality has occurred in the branching feed path **145** of the third expansion unit **5**, the counting of the paper sheet **S** stacked in the stacking portions **161** to **163** of the third expansion unit **5** on the downstream side and the paper sheet **S** stacked in the stacking portion other than the evacuation destination stacking portion of the stacking portions **121** and **122** of the second expansion unit **4** on the upstream side can be regarded as finalized.

(365) Moreover, if there is a paper sheet **S** that has been forgotten to be removed from the branching feed path **145** during the abnormality recovery processing by the operator, during the reset operation the paper sheet in question will be conveyed to the aforementioned evacuation destination stacking portion among the stacking portions **161** to **163** to be stacked. Therefore, it is possible to reduce the burden on the operator involved in the operation of the abnormality recovery process.

(366) When a conveyance abnormality requiring an “emergency stop” occurs in the branching feed path **145**, the paper sheets **S** being conveyed in the extension feed paths **64** and **104** may be conveyed to the stacking portions **121** and **122** if the conveyance destinations thereof based on the identification result of the validator **22** are the stacking portions **121** and **122**, or may be conveyed to the terminal pocket **7** as an evacuation destination if the conveyance destinations based on the identification result of the validator **22** are the stacking portions **161** to **163**. In this case, counting of the paper sheet **S** transported to the stacking portions **121** and **122** will be finalized, while the counting of the paper sheet **S** conveyed to the terminal pocket **7** will not be finalized.

(367) During the counting processing in the second counting mode, the slave control portion **95** of the paper sheet processing device **1** described above continues the conveyance operation of both the extension feed path **64** and the branching feed path **65** without immediate stoppage when the first abnormality occurs, which is a paper sheet jam abnormality requiring an “emergency stop” at the identification section **22**, and conveys the paper sheet **S** of the extension feed path **64** and the branching feed path **65** to the predetermined stacking portion of any of the stacking portions **81** to **84**.

(368) During the occurrence of the second abnormality in which an intake malfunction of the paper sheet **S** has occurred in the loading portion **15** that can be resolved with a “normal stop” in the counting processing of the second counting mode, the master control portion **56** and the slave

control portion **95** continue the conveyance operation of all of the base machine feed path **20**, the extension feed path **64** and the branching feed path **65** without immediate stoppage to stack the paper sheet S of the base machine feed path **20**, the extension feed path **64** and the branching feed path **65** in the predetermined stacking portion of any of the stacking portions **81** to **84**.

(369) During the occurrence of the third abnormality in which a paper sheet jam abnormality has occurred requiring an “emergency stop” in the extension feed path **64** of the first expansion unit **3** in the counting processing of the second counting mode, the slave control portion **95** continues the conveyance operation of the branching feed path **65**, which is one of the extension feed path **64** and the branching feed path **65**, without immediate stoppage to convey the paper sheet S of the branching feed path **65** to the predetermined stacking portion of any of the stacking portions **81** to **84**.

(370) During the occurrence of the fourth abnormality in which an abnormality has occurred requiring a “normal stop” in the extension feed path **64** of the first expansion unit **3** in the counting processing of the second counting mode, the slave control portion **95** continues the conveyance operation of both of the extension feed path **64** and the branching feed path **65** without immediate stoppage to convey the paper sheet S of the extension feed path **64** and the branching feed path **65** to the predetermined stacking portion of any of the stacking portions **81** to **84**.

(371) During the occurrence of the first abnormality in which a paper sheet jam abnormality has occurred requiring an “emergency stop” in the extension feed path **104** of the second expansion unit **4** in the counting processing of the third counting mode, the slave control portion **95** continues the conveyance operation of both of the extension feed path **64** and the branching feed path **65** without immediate stoppage to convey the paper sheet S of the extension feed path **64** to the predetermined stacking portion of any of the stacking portions **81** to **84**.

(372) During the occurrence of the first abnormality in which a paper sheet jam abnormality has occurred requiring an “emergency stop” in the extension feed path **104** of the second expansion unit **4** in the counting processing of the third counting mode, the slave control portion **135** continues the conveyance operation of the branching feed path **105**, which is one of the extension feed path **104** and the branching feed path **105**, without immediate stoppage to convey the paper sheet S of the branching feed path **105** to the predetermined stacking portion of either of the stacking portions **121** and **122**.

(373) During the occurrence of the first abnormality in which a paper sheet jam abnormality has occurred requiring an “emergency stop” in the extension feed path **104** of the second expansion unit **4** in the counting processing of the third counting mode, the slave control portion **175** continues the conveyance operation of both of the extension feed path **144** and the branching feed path **145** without immediate stoppage to convey the paper sheet S of the extension feed path **144** and the branching feed path **145** to the predetermined stacking portion of any of the stacking portions **161** to **163**.

(374) During the occurrence of the second abnormality in which an abnormality has occurred requiring a “normal stop” in the extension feed path **104** of the second expansion unit **4** in the counting processing of the third counting mode, the slave control portions **95** and **135** continue the conveyance operation of all of the extension feed paths **64** and **104** and the branching feed path **105** without immediate stoppage to convey the paper sheet S of the extension feed paths **64** and **104** and the branching feed path **105** to the predetermined stacking portion of either of the stacking portions **121** and **122**.

(375) During the occurrence of the second abnormality in which an abnormality has occurred requiring a “normal stop” in the extension feed path **104** of the second expansion unit **4** in the counting processing of the third counting mode, the slave control portion **175** continues the conveyance operation of both of the extension feed path **144** and the branching feed path **145** without immediate stoppage to convey the paper sheet S of the extension feed path **144** and the branching feed path **145** to the predetermined stacking portion of any of the stacking portions **161**

to **163**.

(376) During the occurrence of the third abnormality in which an abnormality has occurred requiring an “emergency stop” in the branching feed path **105** of the second expansion unit **4** in the counting processing of the third counting mode, the slave control portion **95** continues the conveyance operation of both of the extension feed path **64** and the branching feed path **65** without immediate stoppage to convey the paper sheet S of the extension feed path **64** to the predetermined stacking portion of any of the stacking portions **81** to **84**.

(377) During the occurrence of the third abnormality in which an abnormality has occurred requiring an “emergency stop” in the branching feed path **105** of the second expansion unit **4** in the counting processing of the third counting mode, the slave control portions **135** and **175** continue the conveyance operation of all of the extension feed paths **104** and **144** and the branching feed path **145** without immediate stoppage to convey the paper sheet S of the extension feed paths **104** and **144** and the branching feed path **145** to the predetermined stacking portion of any of the stacking portions **161** to **163**.

(378) During the occurrence of the third abnormality in which an abnormality has occurred requiring an “emergency stop” in the branching feed path **105** of the second expansion unit **4** in the counting processing of the third counting mode, the slave control portion **175** continues the conveyance operation of both of the extension feed path **144** and the branching feed path **145** without immediate stoppage to convey the paper sheet S of the extension feed path **144** and the branching feed path **145** to the predetermined stacking portion of any of the stacking portions **161** to **163**.

(379) During the occurrence of the fourth abnormality in which an abnormality has occurred requiring an “emergency stop” in the branching feed path **145** of the third expansion unit **5** in the counting processing of the third counting mode, the slave control portion **95** continues the conveyance operation of both of the extension feed path **64** and the branching feed path **65** without immediate stoppage to convey the paper sheet S of the extension feed path **64** to the predetermined stacking portion of any of the stacking portions **81** to **84**.

(380) During the occurrence of the fourth abnormality in which an abnormality has occurred requiring an “emergency stop” in the branching feed path **145** of the third expansion unit **5** in the counting processing of the third counting mode, the slave control portion **135** continues the conveyance operation of both of the extension feed path **104** and the branching feed path **105** without immediate stoppage to convey the paper sheet S of the extension feed path **104** and the branching feed path **105** to the predetermined stacking portion of either of the stacking portions **121** and **122**.

(381) During the occurrence of the fourth abnormality in which an abnormality has occurred requiring an “emergency stop” in the branching feed path **145** of the third expansion unit **5** in the counting processing of the third counting mode, the slave control portions **175** and **225** continue the conveyance operation of the extension feed path **144**, which is one of the extension feed path **144** and the branching feed path **145**, without immediate stoppage and convey the paper sheet S of the extension feed path **144** to the terminal pocket **7** as the stacking portion via the extension feed path **184**.

(382) Therefore, during the occurrence of a conveyance abnormality, the paper sheet processing device **1** does not simply immediately stop the conveyance operations of all of the extension feed paths **64**, **104**, **144**, **184** and the branching feed paths **65**, **105**, **145**, **185**, but rather continues as much as possible the conveyance of the paper sheet S being conveyed according to the location of the abnormality that occurred and the type of the abnormality to convey the paper sheet to the predetermined location. This allows the operator to reduce the number of target sheets and target locations of a paper sheet S to be removed from within the paper sheet processing device **1** when the conveyance operations of the extension feed paths **64**, **104**, **144**, **184** and the branching feed paths **65**, **105**, **145**, **185** are finally stopped. Therefore, when a conveyance abnormality occurs, the

work of removing the paper sheet S can be made easier.

(383) During the occurrence of the fourth abnormality in which an abnormality has occurred requiring a “normal stop” in the extension feed path **64** or the branching feed path **65** of the first expansion unit **3** in the counting processing of the second counting mode, the slave control portion **95** switches the conveyance destination so as to convey the paper sheet S being conveyed in the extension feed path **64** or the paper sheet S being conveyed in the branching feed path **65** to an empty stacking portion in which paper sheets S are not stacked among the stacking portions **81** to **84**.

(384) During the occurrence of the second abnormality in which an abnormality has occurred requiring a “normal stop” in the extension feed path **104** or the branching feed path **105** of the second expansion unit **4** in the counting processing of the third counting mode, the slave control portion **135** switches the conveyance destination so as to convey the paper sheet S being conveyed in the extension feed path **104** or the paper sheet S being conveyed in the branching feed path **105** to an empty stacking portion in which paper sheets S are not stacked among the stacking portions **121** and **122**.

(385) Therefore, according to the paper sheet processing unit **1**, when an abnormality occurs and the paper sheet S being conveyed on the extension feed path **64**, the extension feed path **104**, the branching feed path **65** and the branching feed path **105** are stacked in a concentrated manner in an empty stacking portion, the number of paper sheets to be recounted will be reduced. Accordingly, it is possible to reduce the processing load.

(386) During the occurrence of the fourth abnormality in which an abnormality has occurred requiring a “normal stop” in the extension feed path **64** or the branching feed path **65** of the first expansion unit **3** in the counting processing of the second counting mode, the slave control portion **95** switches the conveyance destination so as to convey the paper sheet S being conveyed in the extension feed path **64** or the paper sheet S being conveyed in the branching feed path **65** to the stacking portion with the lowest number of stacked sheets among the stacking portions **81** to **84** when there is no empty stacking portion in which paper sheets S are not stacked among the stacking portions **81** to **84**.

(387) During the occurrence of the fourth abnormality in which an abnormality has occurred requiring a “normal stop” in the extension feed path **104** or the branching feed path **105** of the first expansion unit **4** in the counting processing of the second counting mode, the slave control portion **135** switches the conveyance destination so as to convey the paper sheet S being conveyed in the extension feed path **104** or the paper sheet S being conveyed in the branching feed path **105** to the stacking portion with the lowest number of stacked sheets among the stacking portions **81** to **84** when there is no empty stacking portion in which paper sheets S are not stacked among the stacking portions **81** to **84**.

(388) Therefore, when an abnormality occurs and the paper sheet S being conveyed on the extension feed path **64**, the extension feed path **104**, the branching feed path **65** and the branching feed path **105** are conveyed to and stacked in a concentrated manner in the stacking portion with the lowest number of stacked sheets, the number of paper sheet to be recounted will be reduced. Accordingly, it is possible to reduce the processing load.

(389) During the occurrence of the fourth abnormality in which an abnormality has occurred requiring a “normal stop” in the extension feed path **64** or the branching feed path **65** of the first expansion unit **3** in the counting processing of the second counting mode, the slave control portion **95** may switch the conveyance destination so as to convey the paper sheet S being conveyed in the extension feed path **64** or the paper sheet S being conveyed in the branching feed path **65** to a predetermined stacking portion set in advance among the stacking portions **81** to **84**.

(390) During the occurrence of the second abnormality in which an abnormality has occurred requiring a “normal stop” in the extension feed path **104** or the branching feed path **105** of the second expansion unit **4** in the counting processing of the third counting mode, the slave control

portion **135** may switch the conveyance destination so as to convey the paper sheet S being conveyed in the extension feed path **104** or the paper sheet S being conveyed in the branching feed path **105** to a predetermined stacking portion set in advance among the stacking portions **121** and **122**.

(391) With this configuration, when an abnormality occurs the paper sheet S being conveyed on the extension feed path **64**, the extension feed path **104**, the branching feed path **65** and the branching feed path **105** are stacked in a predetermined stacking portion set in advance as a conveyance destination among the stacking portions **81** to **84**, **121** and **122**. Therefore, the slave control portions **95** and **135** can be easily controlled accordingly. In addition, since the stacking portion to be removed by the operator during the abnormality recovery process is set in advance, the number of places to be processed by the operator is reduced and the stacking locations to be removed are easy to understand.

(392) During the occurrence of the first abnormality in the third counting mode, the slave control portions **95**, **135**, **175** will, depending on the location of the conveyance abnormality that has occurred and the type of conveyance abnormality, convey the paper sheet S being conveyed downstream from the extension feed path **104** of the second expansion unit **4**, which is the location of the conveyance abnormality that has occurred, to the predetermined stacking portion among the stacking portions **121**, **122**, **161** to **163**, and convey the paper sheet S being conveyed upstream of the extension feed path **104** of the second expansion unit **4** to the predetermined stacking portion among the separate stacking portions **81** to **84**.

(393) During the occurrence of the second abnormality in the third counting mode, the slave control portions **95**, **135**, **175** will, depending on the location of the conveyance abnormality that has occurred and the type of conveyance abnormality, convey the paper sheet S being conveyed downstream from the extension feed path **104** of the second expansion unit **4**, which is the location of the conveyance abnormality that has occurred, to the predetermined stacking portion among the stacking portions **161** to **163**, and convey the paper sheet S being conveyed upstream of the extension feed path **104** of the second expansion unit **4** to the predetermined stacking portion among the separate stacking portions **121** and **122**.

(394) Accordingly, in the configuration in which a plurality of the first expansion unit **3** to the third expansion unit **5** are provided, even if the conveyance destinations of the conveyed paper sheet S are distributed among the first expansion unit **3** to the third expansion unit **5**, the number of places to be processed by the operator at the time of abnormality recovery processing is reduced. Accordingly, it is possible to reduce the processing load.

(395) During the occurrence of the fourth abnormality in the third counting mode, if the location of the conveyance abnormality that has occurred is not in the extension feed path **144**, the slave control portions **175** and **225** switch the conveyance destination so that the paper sheet S being conveyed on the extension feed path **144** is conveyed to the terminal pocket **7**.

(396) Accordingly, control of the slave control portions **175** and **225** is facilitated because the termination pocket **7** is positioned as a dedicated rejection portion after an abnormality occurs. In addition, the location of the stacking portion to be pulled out during the abnormality recovery process being the terminal pocket **7** makes it easier for the operator to find.

(397) Even if the operator forgets to remove the paper sheet S stuck in the target extension feed path **64** during the abnormality recovery processing operation after the occurrence of the third abnormality in the second counting mode, the slave control portion **95** switches the conveyance destination of such stuck paper sheet S to the evacuation destination stacking portion among the stacking portions **81** to **84** and conveys the paper sheet S thereto during the reset operation of the target extension feed path **64**.

(398) Even if the operator forgets to remove the paper sheet S stuck in the target extension feed path **104** during the abnormality recovery processing operation after the occurrence of the first abnormality in the third counting mode, the slave control portion **135** switches the conveyance

destination of such stuck paper sheet S to the evacuation destination stacking portion among the stacking portions **121** and **122** and conveys the paper sheet S thereto during the reset operation of the target extension feed path **104**.

(399) Even if the operator forgets to remove the paper sheet S stuck in the target branching feed path **105** during the abnormality recovery processing operation after the occurrence of the third abnormality in the third counting mode, the slave control portion **135** switches the conveyance destination of such stuck paper sheet S to the evacuation destination stacking portion among the stacking portions **121** and **122** and conveys the paper sheet S thereto during the reset operation of the target branching feed path **105**.

(400) Even if the operator forgets to remove the paper sheet S stuck in the target branching feed path **145** during the abnormality recovery processing operation after the occurrence of the fourth abnormality in the third counting mode, the slave control portion **175** switches the conveyance destination of such stuck paper sheet S to the evacuation destination stacking portion among the stacking portions **161** to **163** and conveys the paper sheet S thereto during the reset operation of the target branching feed path **145**.

(401) Accordingly, not only when an abnormality occurs, but also during the reset operation by the abnormality recovery processing operation after the abnormality occurs, the slave control portions **95**, **135**, **175** collectively convey and stack the paper sheet S stuck in the extension feed paths **64**, **104**, **105**, **145** to a predetermined stacking portion, thus reducing the number of processing points for the operator during the abnormality recovery processing. Accordingly, it is possible to reduce the processing load.

(402) During the occurrence of the fourth abnormality in the second counting mode, the master control portion **56**, after finishing conveying the paper sheet S being conveyed in the extension feed path **64** or the paper sheet S being conveyed in the branching feed path **65** to a predetermined stacking portion as the evacuation destination among the stacking portions **81** to **84**, finalizes the counting of the paper sheet S normally conveyed to the stacking portions other than the predetermined stacking portion among the stacking portions **81** to **84**.

(403) During the occurrence of the first abnormality in the third counting mode, the master control portion **56**, after finishing conveying the paper sheet S being conveyed in the extension feed path **64** to a predetermined stacking portion as the evacuation destination among the stacking portions **81** to **84**, finalizes the counting of the paper sheet S normally conveyed to the stacking portions **121**, **122**, **161** to **163** other than this predetermined stacking portion.

(404) During the occurrence of the second abnormality in the third counting mode, the master control portion **56**, after finishing conveying the paper sheet S being conveyed in the extension feed paths **64** and **104** to a predetermined stacking portion as the evacuation destination among the stacking portions **121** and **122**, finalizes the counting of the paper sheet S normally conveyed to the stacking portion other than this predetermined stacking among the stacking portions **121** and **122** and to the stacking portions **161** to **163**.

(405) During the occurrence of the third abnormality in the third counting mode, the master control portion **56**, after finishing conveying the paper sheet S being conveyed in the extension feed path **64** to the predetermined stacking portion as the evacuation destination among the stacking portions **81** to **84** and conveying the paper sheet S being conveyed in the extension feed path **104** to the predetermined stacking portion as the evacuation destination among the stacking portions **161** to **163**, finalizes the counting of the paper sheet S normally conveyed to the stacking portions **121** and **122** and the stacking portions other than the evacuation destination stacking portion among the stacking portions **161** to **163**.

(406) During the occurrence of the fourth abnormality in the third counting mode, the master control portion **56**, after finishing conveying the paper sheet S being conveyed in the extension feed path **64** to the predetermined stacking portion as the evacuation destination among the stacking portions **81** to **84**, conveying the paper sheet S being conveyed in the extension feed path **104** to the

predetermined stacking portion as the evacuation destination among the stacking portions **121** and **122**, and conveying the paper sheet S being conveyed in the extension feed path **144** to the terminal pocket **7** as an evacuation destination, finalizes the counting of the paper sheet S normally conveyed to the stacking portions **161** to **163** and to the stacking portion other than the stacking portion serving as the evacuation destination among the stacking portions **121** and **122**.

(407) Therefore, after the paper sheets S being conveyed are completely conveyed to the predetermined stacking portion serving as the evacuation destination, the counting of the paper sheet S normally conveyed to the stacking portions other than the evacuation destination stacking portions can be finalized. Therefore, the operator only needs to remove the other paper sheet S from the paper sheet processing device **1** for recounting.

(408) In other words, complicated processing by the operator after the occurrence of an abnormality is reduced.

Modifications

(409) During the occurrence of the fourth abnormality in the second counting mode and the occurrence of the second abnormality in the third counting mode described above, that is, when an abnormality requiring a “normal stop” occurs in the extension feed paths **64** and **104**, it was explained that a predetermined conveyance destination stacking portion, such as the stacking portion **84** or the stacking portion **122**, may be set in advance as a changed conveyance destination of the paper sheet S, with the paper sheet S in question being collectively conveyed to and stacked in the stacking portion **84** or stacking portion **122**. In the same way, it was explained that if there are paper sheets S stuck in the extension feed paths **64** and **104** during the reset operation in the abnormality recovery process, the paper sheets S may be conveyed to the stacking portion **84** or **122** for stacking. Alternatively, the paper sheet S may be conveyed and stacked in the terminal pocket **7** arranged at the most downstream portion as a predetermined stacking portion serving as the changed conveyance destination of the paper sheet S.

(410) Further, when an abnormality occurs, the master control portion **56** or the slave control portions **95**, **135**, **175** may switch the conveyance speed of at least any one of the extension feed paths **64**, **104**, **144** and the branching feed paths **65**, **105**, **145** to a lower speed than the conveyance speed during normal counting except when a conveyance abnormality occurs, depending on the location and type of abnormality that has occurred. With such control, after an abnormality occurs, the corresponding one among the extension feed paths **64**, **104**, **144** and branching feed paths **65**, **105**, **145** is conveyed at a low speed, so the paper sheet processing device **1** can prevent detachment of the conveying belts constituting the extension feed paths **64**, **104**, **144** and the branching conveying paths **65**, **105**, **145**, and damage such as tearing of the paper sheet S can be suppressed.

(411) Further, when an abnormality requiring an “emergency stop” occurs, it was described that the slave control portion that is subject to the abnormality occurrence among the slave control portions **95**, **135**, **175** notifies the master control portion **56** to that effect, and stops the feed path requiring an “emergency stop” based on an instruction from the master control portion **56**. However, without being limited thereto, the slave control portion that is the subject of the abnormality occurrence may stop the feed path that requires an “emergency stop” as it is, and notify the master control portion **56** to the effect that an abnormality requiring an “emergency stop” has occurred.

(412) As for control portions, for convenience, the base machine **2** was described as being provided with the master control unit **56**, and each of the first expansion portion **3** to the fourth expansion portion **6** with the slave control portion **95**, **135**, **175**, **225**, respectively, and with regard to the interface, the master control portion **56** and the slave control portions **95**, **135**, **175**, **225** were described as performing one-to-one communication as a result of the corresponding unit station addresses **#1** to **#4** being set in advance for the first expansion unit **3** to the fourth expansion unit **6**, respectively. However, the control is not limited thereto. The control portion may be installed only in the base machine **2** so that said control portion controls all of the base machine **2** and the first

expansion units **3** to the fourth expansion unit **6**.

(413) For convenience, the description was based on the overall configuration of FIG. **1**, but it is not limited thereto, such that the expansion units connected to the base machine **2** may be configured with multiple units of the same type from the first expansion unit **3** to the fourth expansion unit **6**, or only one unit from the first expansion unit **3** to the fourth expansion unit **6** may be connected to the base machine **2**. The paper sheet processing device **1** may be configured with the terminal pocket **7** provided at the opposite end of the base machine **2**, and may be configured without the terminal pocket **7**. In the configuration in which the terminal pocket **7** is provided, there will be more options for the stacking portion to which the paper sheet **S** being conveyed in at least one of the extension feed paths **64**, **104**, **144**, or **184** will be switched in the event of an abnormality.

(414) In the paper sheet processing device **1** of the present embodiment described above, banknotes are described as the medium, being referred to as paper sheets **S**, but the paper sheets **S**, which are the media, were described as being banknotes, but it can be applied to all types of paper sheets **S**, such as gift certificates, checks, promissory notes, securities, and other valuable securities, not limited to banknotes.

INDUSTRIAL APPLICABILITY

(415) According to the paper sheet processing device described above, it becomes possible to easily remove a paper sheet when a conveyance abnormality occurs.

DESCRIPTION OF REFERENCE SYMBOLS

(416) **1** Paper sheet processing device **2** Base machine **3** First expansion unit (expansion unit) **4** Second expansion unit (expansion unit) **5** Third expansion unit (expansion unit) **6** Fourth expansion unit (expansion unit) **7** Terminal pocket (stacking portion, terminal stacking portion) **22** Validator **30** Base machine feed path **56** Master control portion (control portion) **64**, **104**, **144**, **184** Extension feed path **65**, **105**, **145**, **185** Branching feed path **95**, **135**, **175**, **225** Slave control portion (control portion) **81** to **84**, **121**, **122**, **161** to **163** Stacking portion **201**, **202** Bonding stacker (stacking portion) **S** Paper sheet

Claims

1. A paper sheet processing device comprising: a base machine including a base machine feed path that conveys a paper sheet, and a validator that identifies the paper sheet conveyed by the base machine feed path; an expansion unit including an extension feed path that extends from the base machine feed path, a branching feed path that branches from the extension feed path, and a plurality of stacking portions that are connected to the branching feed path and stack the paper sheet identified by the validator; and a control portion that, when a conveyance abnormality occurs, continues conveyance operation of at least one of the extension feed path and the branching feed path without immediately stopping the conveyance operation and conveys the paper sheet to a predetermined stacking portion that is any of the plurality of stacking portions, in accordance with a location where the conveyance abnormality occurred and a type of the conveyance abnormality, wherein the control portion, when the conveyance abnormality occurs and any one of the stacking portions is not empty, switches the conveyance destination so as to convey the paper sheet being conveyed on the extension feed path or the paper sheet being conveyed on the branching feed path to the stacking portion with a lowest number of stacked sheets among the stacking portions.

2. The paper sheet processing device according to claim 1, wherein the control portion, when the conveyance abnormality occurs and a first stacking portion of the stacking portions is empty, switches a conveyance destination so as to convey the paper sheet being conveyed on the extension feed path or the paper sheet being conveyed on the branching feed path to the first stacking portion where no paper sheet is stacked.

3. The paper sheet processing device according to claim 1, wherein the predetermined stacking

portion is set in advance, and the control portion, when the conveyance abnormality occurs, switches a conveyance destination so as to convey the paper sheet being conveyed on the extension feed path or the paper sheet being conveyed on the branching feed path to the predetermined stacking portion.

4. The paper sheet processing device according to claim 1, wherein the expansion unit comprises a plurality of expansion units, and the control portion, in accordance with the location where the conveyance abnormality occurred and the type of the conveyance abnormality, individually switches a conveyance destination of a paper sheet being conveyed downstream of the location where the conveyance abnormality occurred and a paper sheet being conveyed on an upstream side of the location where the conveyance abnormality occurred so that each paper sheet is conveyed to the predetermined stacking portion or to another stacking portion.

5. The paper sheet processing device according to claim 1, further comprising: a terminal stacking portion that receives a paper sheet from an opening portion at the terminal end of the extension feed path and stacks it, wherein the control portion, when the location where the conveyance abnormality occurred is not the extension feed path, switches a conveyance destination so as to convey the paper sheet being conveyed on the extension feed path to the terminal stacking portion.

6. The paper sheet processing device according to claim 1, wherein the control portion switches a first conveyance speed of at least any one of the extension feed path and the branching feed path to a lower speed than a second conveyance speed, in accordance with the location where the conveyance abnormality occurred and the type of the conveyance abnormality, the second conveyance speed being a speed during which no conveyance abnormality occurs.

7. The paper sheet processing device according to claim 1, wherein the control portion, when an operator has forgotten to remove a paper sheet stuck in at least any one of the extension feed path and the branching feed path during an abnormality recovery processing operation after the conveyance abnormality has occurred, switches a conveyance destination of the stuck paper sheet during a reset operation of the at least one of the extension feed path and the branching feed path concerned.

8. The paper sheet processing device according to claim 1, wherein, the control portion, after having completed conveyance of the paper sheet being conveyed on the extension feed path or the paper sheet being conveyed on the branching feed path to the predetermined stacking portion, finalizes counting of a paper sheet normally conveyed to the stacking portion other than the predetermined stacking portion.
